FEDERAL UNIVERSITY OF LAFIA

FACULTY OF SCIENCE

DEPARTMENT OF CHEMISTRY

CHM 121- INTRODUCTORY INORGANIC CHEMISTRY LECTURE NOTES (2 UNITS)

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Course content: Atoms, molecules and chemical reactions, atomic structure, modern electronic theory of atoms, periodic table and periodicity, structure of solids, chemistry of selected metals and non – metals, chemical bonding, intermolecular forces, Radioactivity, chemical equations and stoichiometry.Metals and metallic bonding.

INORGANIC CHEMISTRY: This is a branch of Chemistry that deals with chemical elements and their compounds excluding hydrocarbons and their associated compounds. It can also be defined as the study of the synthesis and behaviour of inorganic and organometallic compounds. Inorganic chemistry covers all chemical compounds except carbon based compounds usually containing C-H bonds which are classified as organic chemistry.

ATOMS, ELEMENTS AND MOLECULES

The Greek philosopher Democritus (470–400 BC) suggested that all matter is composed of tiny, discrete, indivisible particles that he called atoms. His ideas, based entirely on philosophical speculation rather than experimental evidence, were rejected for 2000 years. By the late 1700s, scientists began to realize that the concept of atoms provided an explanation for many experimental observations about the nature of matter.

By the early 1800s, the Law of Conservation of Matter and the Law of Definite Proportions were both accepted as general descriptions of how matter behaves. John Dalton (1766–1844), an English schoolteacher, tried to explain why matter behaves in such systematic ways as those expressed here. In 1808, he published the first "modern" ideas about the existence and nature of atoms. Dalton's explanation summarized and expanded the nebulous concepts of early philosophers and scientists; more importantly, his ideas were based on *reproducible experimental results* of measurements by many scientists. These ideas form the core of **Dalton's Atomic Theory,** one of the highlights in the history of scientific thought. In condensed form, Dalton's ideas may be stated as follows:

1. Elements are composed of extremely small particles, called atoms.
2. All atoms of a given element are identical, having the same size, mass, and chemical properties. The atoms of one element are different from the atoms of all other elements.
3. Compounds are composed of atoms of more than one element. In any compound, the ratio of the numbers of atoms of any two of the elements present is either an integer or a simple fraction.
4. A chemical reaction involves only the separation, combination, or rearrangement of atoms; it does not result in their creation or destruction.

Dalton believed that atoms were solid, indivisible spheres, an idea we now reject. But he showed remarkable insight into the nature of matter and its interactions. Some of his ideas could not be

verified (or refuted) experimentally at the time. They were based on the limited experimental observations of his day. Even with their shortcomings, Dalton's ideas provided a framework that could be modified and expanded by later scientists. Thus John Dalton is often considered to be the father of modern atomic theory.
Figure 1 is a schematic representation of hypotheses 2 and 3. Dalton's concept of an atom was far more detailed and specific than Democritus'. The second hypothesis states that atoms of one element are different from atoms of all other elements. Dalton made no attempt to describe the structure or composition of atoms—he had no idea what an atom is really like. But he did realize that the different properties shown by elements such as hydrogen and oxygen can be explained by assuming that hydrogen atoms are not the same as oxygen atoms.

The third hypothesis suggests that, to form a certain compound, we need not only atoms of the right kinds of elements, but the specific numbers of these atoms as well. This idea is an extension of a law published in 1799 by Joseph Proust, a French chemist. Proust's ***law of definite proportions*** states that *different samples of the same compound always contain its constituent elements in the same proportion by mass.*

Thus, if we were to analyze samples of carbon dioxide gas obtained from different sources, we would find in each sample the same ratio by mass of carbon to oxygen. And be it in any physical form (solid, liquid or gaseous), each form maitain the same ratio by mass of carbon to oxygen. It stands to reason, then, that if the ratio of the masses of different elements in a given compound is fixed; the ratio of the atoms of these elements in the compound also must be constant.

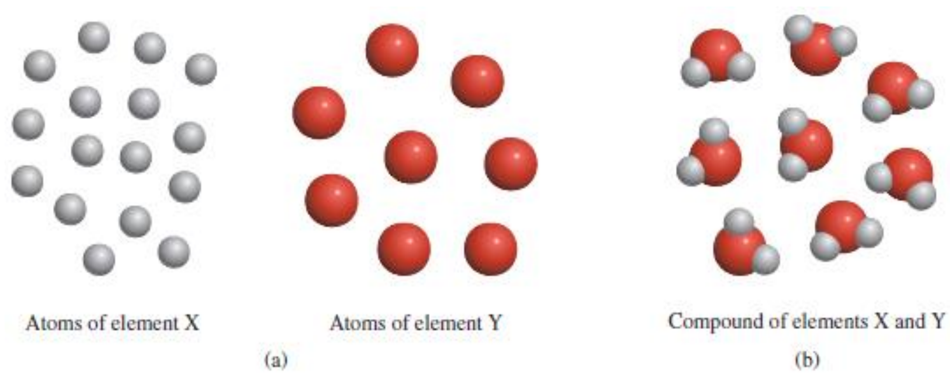


Figure 1.*(a) According to Dalton's atomic theory, atoms of the same element are identical, but atomsof one element are different from atoms of other elements.*

1(b) Compounds formed from atoms of elements X and Y. In this case, the ratio of the atoms of element X to the atoms of element Y is 2:1.

Dalton's third hypothesis also supports another important law, the ***law of multiple proportions.*** According to this law, *if two elements can combine to form more than one compound, the masses of one element that combine with a fixed mass of the other element are in ratios of small whole numbers.* Dalton's theory explains the law of multiple proportions quite simply: The compounds differ in the number of atoms of each kind that combine. For example, carbon forms two stable compounds with oxygen, namely, carbon monoxide and carbon dioxide. Modern measurement techniques indicate that one atom of carbon combines with one atom of oxygen in carbon monoxide and that one atom of carbon combines with two oxygen atoms in carbon dioxide. Thus, the ratio of oxygen in carbon monoxide to oxygen in carbon dioxide is 1:2. This result is consistent with the law of multiple proportions because the mass of an element in a compound is proportional to the number of atoms of the element present (Figure 2). Two Tables below also give more insight on the ***law of multiple proportions.***

Three binary compounds of H and O: H_2O, H_2O_2, H_2O_3

Common Name	%H	%O	Moles H in 100 g of substance	Moles O in 100g	Molar Ratio H/O	Empirical Formula	Molecular Formula
Water	11%	89%	11 mol (11g/1gmol^{-1})	5.6 mol (89g/16 gmol^{-1})	1.95 ~ 2/1	H_2O	H_2O
Hydrogen Peroxide	6.0%	94%	6.0 mol (6g/1 gmol^{-1})	5.9 mol (94g/16 gmol^{-1})	1.01 ~ 1/1	HO	H_2O_2
Hydrogen Trioxide	4.0%	96%	4.0 mol (4g/1 gmol^{-1})	6.0 mol (96g/16 gmol^{-1})	.0.67 ~ 2/3	H_2O_3	H_2O_3

Four binary compounds of N and O: NO, N_2O, NO_2, N_2O_2

Common Name	%N	%O	Moles N in 100 g of substance	Moles O in 100g	Molar Ratio N/O	Empirical Formula	Molecular Formula
Nitric Oxide	47%	53%	3.4 (47g/14gmol^{-1})	3.3 (53g/16 gmol^{-1})	1.03 ~ 1/1	NO	NO
Nitrous Oxide	64%	36%	4.6 (64g/14 gmol^{-1})	2.3 (36g/16 gmol^{-1})	2.0 ~ 2/1	N_2O	N_2O
Nitrogen Dioxide	30%	70%	2.1 (30g/14 gmol^{-1})	4.4 (70g/16 gmol^{-1})	0.48 ~ 1/2	NO_2	NO_2
Dinitrogen Dioxide	47%	53%	3.4 (47g/14 gmol^{-1})	3.3 (53g/16 gmol^{-1})	1.03 ~ 1/1	NO	N_2O_2

Dalton's fourth hypothesis is another way of stating the ***law of conservation of mass,*** which is that *matter can be neither created nor destroyed.* Because matter is made of atoms that are unchanged in a chemical reaction, it follows that mass must be conserved as well. Dalton's brilliant insight into the nature of matter was the main stimulus for the rapid progress of chemistry during the nineteenth century.

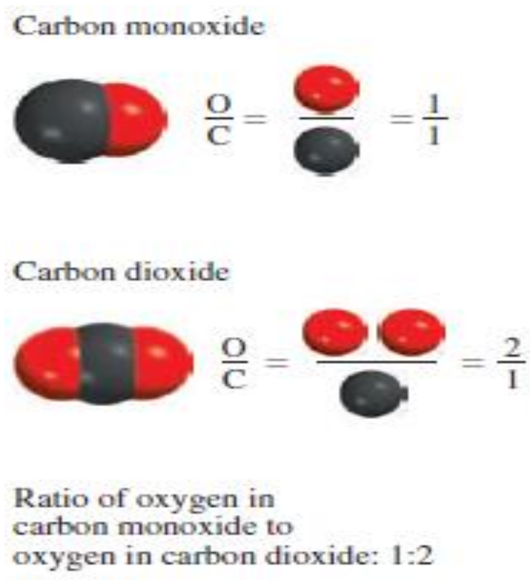


Fig. 2. The illustration of the law of multiple proportions

The smallest particle of an element that maintains its chemical identity through all chemical and physical changes is called an **atom**. Or an atom is the smallest unit quantity of an element that is capable of existence, either alone or in chemical combination with other atoms of the same or another element.

Atoms, and therefore *all* matter, consist principally of three **fundamental particles:** *electrons, protons,* and *neutrons.* These are the basic building blocks of atoms. The masses of protons and neutrons are nearly equal, but the mass of an electron is much smaller. Neutrons carry no charge. The charge on a proton is equal in magnitude, but opposite in sign, to the charge on an electron. Because atoms are electrically neutral, an atom contains equal numbers of electrons and protons.
The **atomic number** (symbol is ***Z***) of an element is defined as the number of protons in the nucleus. In the periodic table, elements are arranged in order of increasing atomic numbers. In a neutral atom the number of protons is equal to the number of electrons, so the atomic number also indicates the number of electrons present in the atom.
The chemical identity of an atom can be determined solely by its atomic number For example, the atomic number of silver is 47.

An **element** is a substance that cannot be decomposed into simpler substances by chemical changes. For instance, neither of the two gases obtained by the electrolysis of water—hydrogen and oxygen—can be further decomposed, so we know that they are elements. As another illustration (Figure below), pure calcium carbonate (a white solid present in limestone and seashells) can be broken down by heating to give another white solid (call it A) and a gas (call it B) in the mass ratio 56.0:44.0. This observation tells us that calcium carbonate is a compound. The white solid A obtained from calcium carbonate can be further broken down into a solid and a gas in a definite ratio by mass, 71.5:28.5. But neither of these can be further decomposed, so they must be elements. The gas is identical to be oxygen obtained from the electrolysis of water; the solid is a metallic element called calcium. Similarly, the gas B, originally obtained from calcium carbonate, can be decomposed into two elements, carbon and oxygen, in a fixed mass ratio, 27.3:72.7. This sequence illustrates that a compound can be broken apart into simpler substances at a fixed mass ratio; those simpler substances may be either elements or simpler compounds.

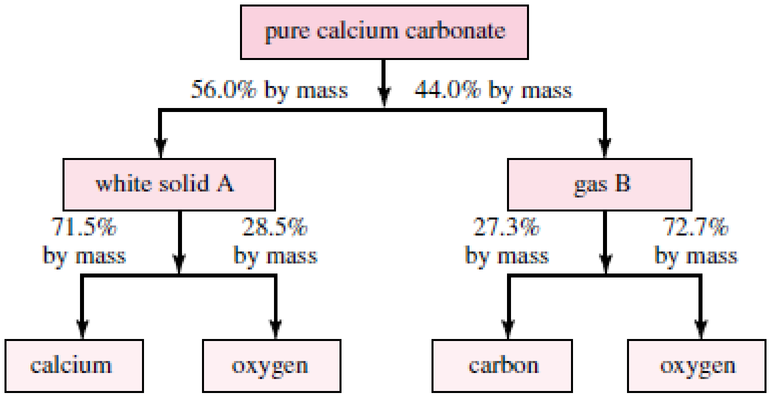
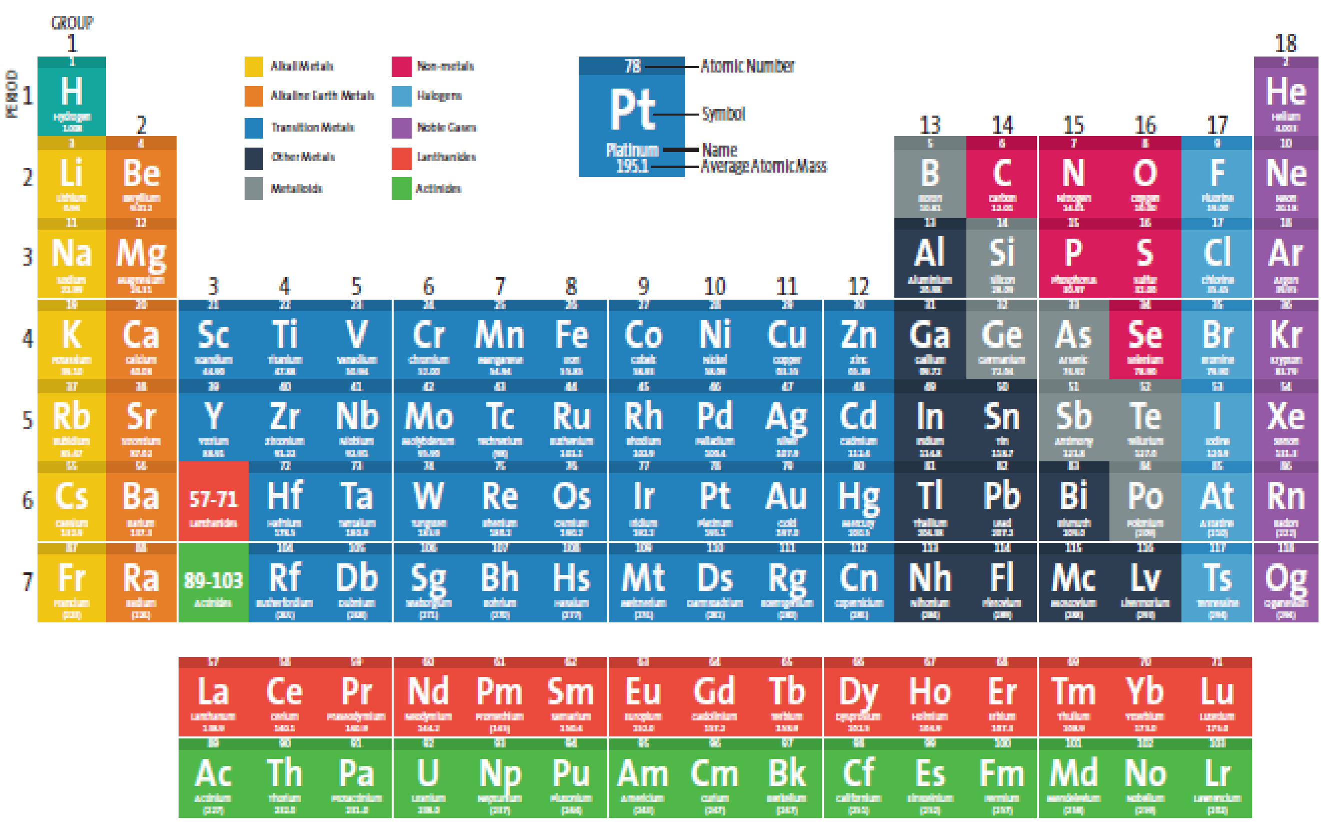
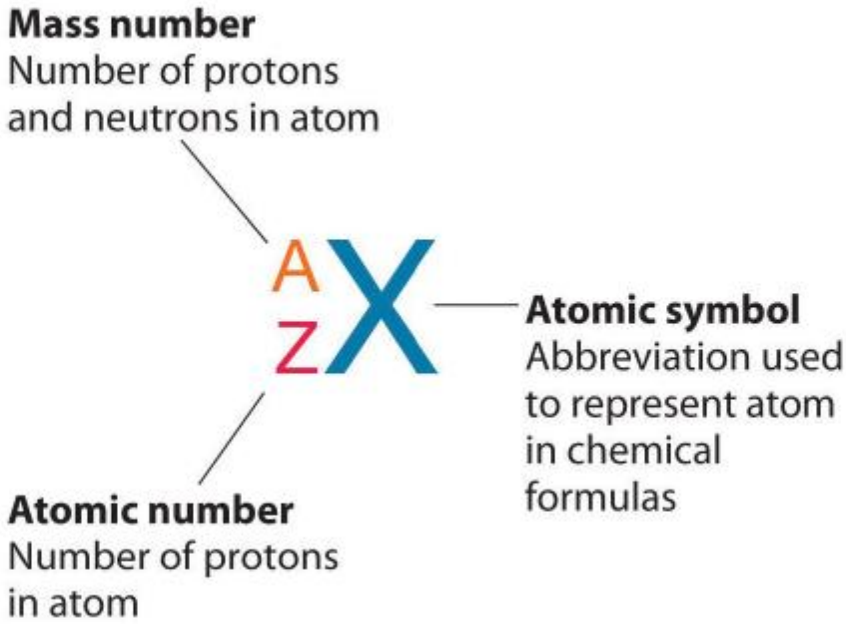


Diagram of the decomposition of calcium carbonate to give a white solid A (56.0 % by mass) and a gas B (44.0% by mass). This decomposition into simpler substances at a fixed ratio proves that calcium carbonate is a compound. The white solid A further decomposes to give the elements calcium (71.5 % by mass) and oxygen (28.5% by mass). This proves that the white solid A is a compound; it is known as calcium oxide. The gas B also can be broken down to give the elements carbon (27.3 % by mass) and oxygen (72.7 % by mass). This establishes that gas B is a compound; it is known as carbon dioxide.



THE PERIODIC TABLE WITH 118 ELEMENTS

We use a set of **symbols** to represent the elements. These symbols can be written more quickly than names, and they occupy less space. The symbols for the first 109 elements consist of either a capital letter *or* a capital letter and a lowercase letter, such as C (carbon) or Ca (calcium). Chemists use alphabetical symbols to represent the names of the elements. The first letter of the symbol for an element is ***always*** capitalized, but the second or third letter is *never* capitalized but given in small letter(s). For example, Co is the symbol for the element cobalt, whereas CO is the formula for carbon monoxide, which is made up of the elements carbon and oxygen. The general representation of atomic symbol, mass number and atomic number are shown in the next Figure.



The symbols for some elements are derived from their German or Latin names as shown below:

Symbols and Latin/German Roots for Elements

1. **Antimony-** Sb: Stibium (Latin)
2. **Copper-** Cu: Cuprum (Latin)
3. **Gold-** Au: Aurum (Latin)
4. **Iron -** Fe: Ferum (Latin)
5. **Lead -** Pb: Plumbum (Latin)
6. **Mercury -** Hg: Hydrargyrum (Latin)
7. **Potassium -** K: Kalium (Latin)
8. **Silver -** Ag: Argentum (Latin)
9. **Sodium -** Na: Natrium (Latin)
10. **Tin -** Sn: Stannum (Latin)
11. **Tungsten -** W: Wolfram (German)

MOLECULES

A **molecule** is the smallest particle of an element or compound that can have a stable independent existence. In nearly all molecules, two or more atoms are bonded together in very small, discrete units (particles) that are electrically neutral.

Molecules of an element constitute same type of atoms. Eg-O_2 , P_4 , S_8.
Molecules of compounds are formed by different type of atoms combined in different proportions: Eg-CO_2, CH_4 ,$C_6H_{12}O_6$ and so on.

Atomicity is the total number of atoms present in a molecule. Based on atomicity, molecules can be classified as:

1. Monoatomic (composed of one atom). Examples include He (helium), Ne (neon), Ar (argon), and Kr (krypton). All noble gases are monoatomic.
2. Diatomic (composed of two atoms). Examples include H_2 (hydrogen), N_2 (nitrogen), O_2 (oxygen), F_2 (fluorine), and Cl_2 (chlorine). Halogens are usually diatomic.
3. Triatomic (composed of three atoms). Examples include O_3 (ozone).
4. Polyatomic (composed of two or more atoms). Examples include S_8.

For example, each molecule of oxygen (O_2) is composed of two oxygen atoms. Individual oxygen atoms are not stable at room temperature and atmospheric pressure. Single atoms of oxygen mixed under these conditions quickly combine to form pairs. The oxygen with which we are all familiar is made up of two atoms of oxygen; it is a *diatomic* molecule, O_2. Therefore, the atomicity of oxygen is 2. Hydrogen, nitrogen, fluorine, chlorine, bromine, and iodine are other examples of diatomic molecules.

Some other elements exist as more complex molecules. One form of phosphorus molecules consists of four atoms, and sulfur exists as eight-atom molecules at ordinary temperatures and pressures. Molecules that contain two or more atoms are called *polyatomic* molecules. In modern terminology, O_2 is named dioxygen, H_2 is dihydrogen, P_4 is tetraphosphorus, and so on. Even though such terminology is officially preferred, it has not yet gained wide acceptance. Most chemists still refer to O_2 as oxygen, H_2 as hydrogen, P_4 as phosphorus, and so on.

Atoms are the components of molecules, and molecules are the stable forms of many elements and compounds. We are able to study samples of compounds and elements that consist of large numbers of atoms and molecules. With the scanning tunnelling microscope it is now possible to "see" atoms.

Name of the element	Atomicity	Molecules formula
Helium	Monoatomic	He
Neon	Monoatomic	Ne
Argon	Monoatomic	Ar
Sodium	Monoatomic	Na
Iron	Monoatomic	Fe
Aluminium	Monoatomic	Al
Hydrogen	Di-atomic	H_2
Oxygen	Di-atomic	O_2
Chlorine	Di-atomic	Cl_2
Nitrogen	Di-atomic	N_2
Phosphorus	Polyatomic (Tetra)	P_4
Sulphur	Polyatomic (Octa)	S_8

CHEMICAL REACTIONS: This is a process that leads to the chemical transformation of one set of chemical substances to another or a process in which reactants react chemically and convert into products by chemical transformation, or a process generally characterized by a chemical change in which the reactants are different from the products. In a chemical reaction, the substances that react are called **reactants** while the substances formed are known as **products**. Reactions often consist of a sequence of individual sub-steps, the so-called elementary reactions, and the information on the precise course of action is part of the reaction mechanism. Chemical reactions are described with chemical equations, which symbolically present the starting materials, end products, and sometimes intermediate products and reaction conditions.

CHARACTERISTICS OF A CHEMICAL REACTION

1. Evolution of gas
2. Change in colour
3. Change in temperature
4. Change in energy
5. Formation of precipitate
6. Change in state.

TYPES OF CHEMICAL REACTIONS

Main types of reactions: There are thousands of types of chemical reactions but the main types of reactions can be classified as 4, 5 or 6 types. The four main types are:

1. Synthesis reactions (or direct combinations reaction or *just* combination reactions)
2. Decomposition reactions (or analysis reactions)
3. Single displacement reactions (or displacement reactions or replacement reactions or substitution reactions)
4. Double displacement reactions (or metathesis reactions or double replacement reactions)

The 5 or 6 main types are the four above and then either acid-base and/or redox reaction.

1. **COMBINATION REACTIONS**

Reactions in which two or more substances combine to form a compound are called **Combination reactions.**

They may involve (1) the combination of two elements to form a compound, (2) the combination of an element and a compound to form a single new compound, or (3) the combination of two compounds to form a single new compound. Some of these reactions are as follows:

Element + Element → Compound

For this type of combination reaction, each element goes from an uncombined state, where its oxidation state is zero, to a combined state in a compound, where its oxidation state is not zero. Thus reactions of this type are oxidation–reduction reactions. Some examples are:

$$2Na(s) + Cl_2(g) \longrightarrow 2NaCl(s)$$

$$Mg(s) + F_2(g) \longrightarrow MgF_2(s)$$

$$P_4(s) + 6Cl_2(g) \longrightarrow 4\overset{+3}{P}Cl_3(\ell) \quad \text{(with limited } Cl_2\text{)} \quad \text{(mp } -112°C\text{)}$$

$$P_4(s) + 10Cl_2(g) \longrightarrow 4\overset{+5}{P}Cl_5(s) \quad \text{(with excess } Cl_2\text{)} \quad \text{(decomposes at } 167°C\text{)}$$

Compound + Element → Compound

$$\overset{+3}{P}Cl_3(\ell) + Cl_2(g) \longrightarrow \overset{+5}{P}Cl_5(s)$$

$$\overset{+4}{S}F_4(g) + F_2(g) \longrightarrow \overset{+6}{S}F_6(g) \quad \text{sulfur hexafluoride (mp } -50.5°C\text{)}$$

Compound + Compound → Compound

$$CaO(s) + CO_2(g) \longrightarrow CaCO_3(s)$$

Pyrosulfuric acid is produced by dissolving sulfur trioxide in concentrated sulfuric acid:

$$SO_3(g) + H_2SO_4(\ell) \longrightarrow H_2S_2O_7(\ell)$$

Pyrosulfuric acid, $H_2S_2O_7$, is then diluted with water to make H_2SO_4:

$$H_2S_2O_7(\ell) + H_2O(\ell) \longrightarrow 2H_2SO_4(\ell)$$

Oxides of the Group IA and IIA metals react with water to form metal hydroxides, e.g.:

$$CaO(s) + H_2O(\ell) \longrightarrow Ca(OH)_2(aq)$$

2. DECOMPOSITION REACTIONS

Decomposition reactions are those in which a compound decomposes to produce (1) two elements, (2) one or more elements *and* one or more compounds, or (3) two or more compounds. Examples of each type follow.

Compound → Element +Element

The electrolysis of water produces two elements by the decomposition of a compound. A compound that ionizes, such as H_2SO_4, is added to increase the conductivity of water and the rate of the reaction, but it does not participate in the reaction:

$$2H_2O(\ell) \xrightarrow{\text{electrolysis}} 2H_2(g) + O_2(g)$$

Small amounts of oxygen can be prepared by the thermal decomposition of certain oxygen-containing compounds. Some metal oxides, such as mercury (II) oxide, HgO, decompose on heating to produce oxygen:

$$\underset{\text{mercury(II) oxide}}{2HgO(s)} \xrightarrow{\text{heat}} 2Hg(\ell) + O_2(g)$$

Compound →Compound + Element

The alkali metal chlorates, such as $KClO_3$, decompose when heated to produce the corresponding chlorides and liberate oxygen. Potassium chlorate is a common laboratory source of small amounts of oxygen:

$$\underset{\text{potassium chlorate}}{2KClO_3(s)} \xrightarrow[MnO_2]{\text{heat}} \underset{\text{potassium chloride}}{2KCl(s)} + 3O_2(g)$$

Nitrate salts of alkali metals or alkaline earth metals decompose to form metal nitrites and oxygen gas.

$$2NaNO_3(s) \longrightarrow 2NaNO_2(s) + O_2(g)$$

Hydrogen peroxide decomposes to form water and oxygen.

$$H_2O_2(\ell) \longrightarrow 2H_2O(\ell) + O_2(g)$$

Compound → Compound +Compound

$$CaCO_3(s) \xrightarrow{\text{heat}} CaO(s) + CO_2(g)$$

$$Mg(OH)_2(s) \xrightarrow{\text{heat}} MgO(s) + H_2O(g)$$

$$(NH_4)_2SO_4(s) \xrightarrow{\text{heat}} 2NH_3(g) + H_2SO_4(\ell)$$

$$(NH_4)_2Cr_2O_7(s) \xrightarrow{\text{heat}} Cr_2O_3(s) + 4H_2O(g) + N_2(g)$$

3. DISPLACEMENT REACTIONS

Reactions in which one element displaces another from a compound are called **displacement reactions.**

These reactions are always redox reactions. The more readily a metal undergoes oxidation, the more active we say it is. Active metals displace less active metals or hydrogen from their compounds in aqueous solution to form the oxidized form of the more active metal and the reduced (free metal) form of the other metal or hydrogen.

In the Table below (activity series of some elements), the most active metals are listed at the top of the first column. These metals tend to react to form their oxidized forms (cations). Elements at the bottom of the activity series tend to remain in their reduced form. They are easily converted from their oxidized forms to their reduced forms.

Element				Common Reduced Form	Common Oxidized Forms
Li	Displace hydrogen from nonoxidizing acids	Displace hydrogen from steam	Displace hydrogen from cold water	Li	Li^{+}
K				K	K^{+}
Ca				Ca	Ca^{2+}
Na				Na	Na^{+}
Mg				Mg	Mg^{2+}
Al				Al	Al^{3+}
Mn				Mn	Mn^{2+}
Zn				Zn	Zn^{2+}
Cr				Cr	Cr^{3+}, Cr^{6+}
Fe				Fe	Fe^{2+}, Fe^{3+}
Cd				Cd	Cd^{2+}
Co				Co	Co^{2+}
Ni				Ni	Ni^{2+}
Sn				Sn	Sn^{2+}, Sn^{4+}
Pb				Pb	Pb^{2+}, Pb^{4+}
H (a nonmetal)				H_2	H^{+}
Sb (a metalloid)				Sb	Sb^{3+}
Cu				Cu	Cu^{+}, Cu^{2+}
Hg				Hg	Hg_2^{2+}, Hg^{2+}
Ag				Ag	Ag^{+}
Pt				Pt	Pt^{2+}, Pt^{4+}
Au				Au	Au^{+}, Au^{3+}

$$1\begin{bmatrix}\text{More Active Metal +} \\ \text{Salt of Less Active Metal}\end{bmatrix} \longrightarrow \begin{bmatrix}\text{Less Active Metal +} \\ \text{Salt of More Active Metal}\end{bmatrix}$$

$$Zn(s) + CuSO_4(aq) \longrightarrow Cu(s) + ZnSO_4(aq)$$
$$Zn(s) + [Cu^{2+}(aq) + SO_4{}^{2-}(aq)] \longrightarrow Cu(s) + [Zn^{2+}(aq) + SO_4{}^{2-}(aq)]$$
$$Zn(s) + Cu^{2+}(aq) \longrightarrow Cu(s) + Zn^{2+}(aq)$$

In this *displacement reaction*, the more active metal, zinc, displaces the ions of the less active metal, copper, from aqueous solution.

2 [Active Metal + Nonoxidizing Acid] ⟶ [Hydrogen + Salt of Acid]

$$Zn(s) + \underset{\text{strong acid}}{H_2SO_4(aq)} \longrightarrow \underset{\text{soluble salt}}{ZnSO_4(aq)} + H_2(g)$$

3 $\left[\begin{array}{l}\textbf{Active Nonmetal +}\\ \textbf{Salt of Less Active Nonmetal}\end{array}\right] \longrightarrow \left[\begin{array}{l}\textbf{Less Active Nonmetal +}\\ \textbf{Salt of More Active Nonmetal}\end{array}\right]$

The more active halogen, Cl_2, displaces the less active halogen, I_2, from its compounds.

$$Cl_2(g) + 2NaI(aq) \longrightarrow I_2(s) + 2NaCl(aq)$$
$$Cl_2(g) + 2[Na^+(aq) + I^-(aq)] \longrightarrow I_2(s) + 2[Na^+(aq) + Cl^-(aq)]$$
$$Cl_2(g) + 2I^-(aq) \longrightarrow I_2(s) + 2Cl^-(aq)$$

4. **METATHESIS REACTIONS (double displacement or double replacement reactions**)

In many reactions between two compounds in aqueous solution, the positive and negative ions appear to “change partners” to form two new compounds, with no change in oxidation numbers. Such reactions are called **metathesis reactions.** Here, two compounds exchange bonds or ions in order to form different compounds. In this reaction, ions get exchanged between two reactants which form a new compound.

We can represent such reactions by the following general equation, where A and B represent positive ions (cations) and X and Y represent negative ions (anions):

$$AX + BY \longrightarrow AY + BX$$

For example, when we mix silver nitrate and sodium chloride solutions, solid silver chloride is formed and sodium nitrate remains dissolved in water:

$$AgNO_3(aq) + NaCl(aq) \longrightarrow AgCl(s) + NaNO_3(aq)$$

Metathesis reactions or **double displacement** reactions result in the removal of ions from solution; this removal of ions can be thought of as the *driving force* for the reaction—the reason it occurs. The removal of ions can occur in three ways, which can be used to classify three types of metathesis reactions:

1. Formation of predominantly nonionized molecules (weak or nonelectrolytes) in solution; the most common of such nonelectrolyte product is water
2. Formation of an insoluble solid, called a *precipitate* (which separates from the solution)
3. Formation of a gas (which is evolved from the solution)

1. Acid–Base (Neutralization) Reactions: Formation of a Nonelectrolyte
Acid–base reactions are among the most important kinds of chemical reactions. Many acid–base reactions occur in nature in both plants and animals. Many acids and bases are essential compounds in an industrialized society. For example, approximately 350 pounds of sulphuric acid, H_2SO_4, and approximately 135 pounds of ammonia, NH_3, is required to support the lifestyle of an average American for one year.

The reaction of an acid with a metal hydroxide base produces a salt and water. Such reactions are called **neutralization reactions** because the typical properties of acids and bases are neutralized.

$$HCl(aq) + NaOH(aq) \longrightarrow H_2O(\ell) + NaCl(aq)$$
$$[H^+(aq) + Cl^-(aq)] + [Na^+(aq) + OH^-(aq)] \longrightarrow H_2O(\ell) + [Na^+(aq) + Cl^-(aq)]$$
$$H^+(aq) + OH^-(aq) \longrightarrow H_2O(\ell)$$

The net ionic equation for *all* reactions of strong acids with strong bases that form soluble salts and water is

$$H^+(aq) + OH^-(aq) \longrightarrow H_2O(\ell)$$

2 Precipitation Reactions: In **precipitation reactions** an insoluble solid, a **precipitate,** forms and then settles out of solution. The driving force for these reactions is the strong attraction between cations and anions. This results in the removal of ions from solution by the formation of a precipitate. Our teeth and bones were formed by very slow precipitation reactions in which mostly calcium phosphate $Ca_3(PO_4)_2$ was deposited in the correct geometric arrangements.
An example of a precipitation reaction is the formation of bright yellow insoluble lead (II) chromate when we mix solutions of the soluble ionic compounds lead(II) nitrate with potassium chromate.

The balanced formula unit, total ionic, and net ionic equations for this reaction follow.

$$Pb(NO_3)_2(aq) + K_2CrO_4(aq) \longrightarrow PbCrO_4(s) + 2KNO_3(aq)$$
$$[Pb^{2+}(aq) + 2\ NO_3^-(aq)] + [2K^+(aq) + CrO_4^{2-}(aq)] \longrightarrow PbCrO_4(s) + 2[K^+(aq) + NO_3^-(aq)]$$
$$Pb^{2+}(aq) + CrO_4^{2-}(aq) \longrightarrow PbCrO_4(s)$$

3 Gas-Formation Reactions
The formation of an insoluble or slightly soluble gas provides a driving force for a third type of metathesis reaction that we call a **gas-formation reaction.** The only common gases that are very soluble in water are HCl(g) and NH_3(g). All other gases are sufficiently insoluble to force a reaction to proceed if they are formed as a reaction product.

When an acid—for example, hydrochloric acid—is added to solid calcium carbonate, a reaction occurs in which carbonic acid, a weak acid, is produced.

$$2HCl(aq) + CaCO_3(s) \longrightarrow H_2CO_3(aq) + CaCl_2(aq)$$
$$2[H^+(aq) + Cl^-(aq)] + CaCO_3(s) \longrightarrow H_2CO_3(aq) + [Ca^{2+}(aq) + 2Cl^-(aq)]$$
$$2H^+(aq) + CaCO_3(s) \longrightarrow H_2CO_3(aq) + Ca^{2+}(aq)$$

The heat generated in the reaction causes thermal decomposition of carbonic acid to gaseous carbon dioxide and water:

$$H_2CO_3(aq) \longrightarrow CO_2(g) + H_2O(\ell)$$

Most of the CO_2 bubbles off, and the reaction goes to completion (with respect to the limiting reactant). The net effect is the conversion of ionic species into nonionized molecules of a gas (CO_2) and water.

$$2HCl(aq) + CaCO_3(s) \longrightarrow CO_2(g) + H_2O(\ell) + CaCl_2(aq)$$

5. OXIDATION–REDUCTION REACTIONS (REDOX REACTIONS)

The term "oxidation" originally referred to the combination of a substance with oxygen. This results in an increase in the oxidation number of an element in that substance. According to the original definition, the following reactions involve oxidation of the substance shown on the far left of each equation. Oxidation numbers are shown for *one* atom of the indicated kind.

The formation of rust, Fe_2O_3, iron (III) oxide:

$$4Fe(s) + 3O_2(g) \longrightarrow 2Fe_2O_3(s)$$

Oxidation state of Fe $\quad 0 \longrightarrow +3$

Combustion reactions: oxidation state of C

$$C(s) + O_2(g) \longrightarrow CO_2(g) \qquad 0 \longrightarrow +4$$
$$2CO(g) + O_2(g) \longrightarrow 2CO_2(g) \qquad +2 \longrightarrow +4$$
$$C_3H_8(g) + 5O_2(g) \longrightarrow 3CO_2(g) + 4H_2O(g) \qquad -8/3 \longrightarrow +4$$

Originally, *reduction* described the removal of oxygen from a compound. Oxide ores are reduced to metals (a very real reduction in mass). For example, tungsten for use in light bulb filaments can be prepared by reduction of tungsten(VI) oxide with hydrogen at 1200°C:

$$WO_3(s) + 3H_2(g) \longrightarrow W(s) + 3H_2O(g) \qquad +6 \longrightarrow 0$$

Tungsten is reduced, and its oxidation state decreases from +6 to zero. Hydrogen is oxidized from zero to the +1 oxidation state. The terms "oxidation" and "reduction" are now applied much more broadly.

Oxidation is an increase in oxidation number and corresponds to the loss, or apparent loss, of electrons.
Reduction is a decrease in oxidation number and corresponds to a gain, or apparent gain, of electrons.

Electrons are neither created nor destroyed in chemical reactions. So oxidation and reduction always occur simultaneously, and to the same extent, in ordinary chemical reactions.

In the four equations cited previously as *examples of oxidation,* the oxidation numbers of iron and carbon atoms increase as they are oxidized. In each case oxygen is reduced as its oxidation number decreases from zero to -2.

Because oxidation and reduction occur simultaneously in all of these reactions, they are referred to as oxidation–reduction reactions. For brevity, we usually call them **redox** reactions. Redox reactions occur in nearly every area of chemistry and biochemistry. We need to be able to identify oxidizing agents and reducing agents and to balance oxidation–reduction equations. These skills are necessary for the study of electrochemistry.

Oxidizing agents are species that (1) oxidize other substances, (2) contain atoms that are reduced, and (3) gain (or appear to gain) electrons.

Reducing agents are species that (1) reduce other substances, (2) contain atoms that are oxidized, and (3) lose (or appear to lose) electrons.

The following equations represent examples of redox reactions. Oxidation numbers are shown above the formulas, and oxidizing and reducing agents are indicated:

$$\underset{\text{red. agt.}}{2\overset{0}{Fe}(s)} + \underset{\text{ox. agt.}}{3\overset{0}{Cl_2}(g)} \longrightarrow 2\overset{+3}{Fe}\overset{-1}{Cl_3}(s)$$

$$\underset{\text{red. agt.}}{2\overset{+3}{Fe}\overset{-1}{Br_3}(aq)} + \underset{\text{ox. agt.}}{3\overset{0}{Cl_2}(g)} \longrightarrow 2\overset{+3}{Fe}\overset{-1}{Cl_3}(aq) + 3\overset{0}{Br_2}(\ell)$$

Equations for redox reactions can also be written as total ionic and net ionic equations. For example, the second equation may also be written as:

$$2[Fe^{3+}(aq) + 3Br^-(aq)] + 3Cl_2(g) \longrightarrow 2[Fe^{3+}(aq) + 3Cl^-(aq)] + 3Br_2(\ell)$$

We distinguish between oxidation numbers and actual charges on ions by denoting oxidation numbers as +*n* or -*n in red circles just above the symbols of the elements,* and actual charges as *n*+ or *n*- above and to the right of formulas of ions. The spectator ions, Fe^{3+}, do not participate in electron transfer. Their cancellation allows us to focus on the oxidizing agent, $Cl_2(g)$, and the reducing agent, Br^- (aq).

$$2Br^-(aq) + Cl_2(g) \longrightarrow 2Cl^-(aq) + Br_2(\ell)$$

A **disproportionation reaction** is a redox reaction in which the same element is oxidized and reduced. An example is:

$$\overset{0}{Cl_2} + H_2O \longrightarrow H\overset{-1}{Cl} + H\overset{+1}{Cl}O$$

ATOMIC STRUCTURE, SUBATOMIC PARTICLES , FUN*DAMENTAL PARTICLES*

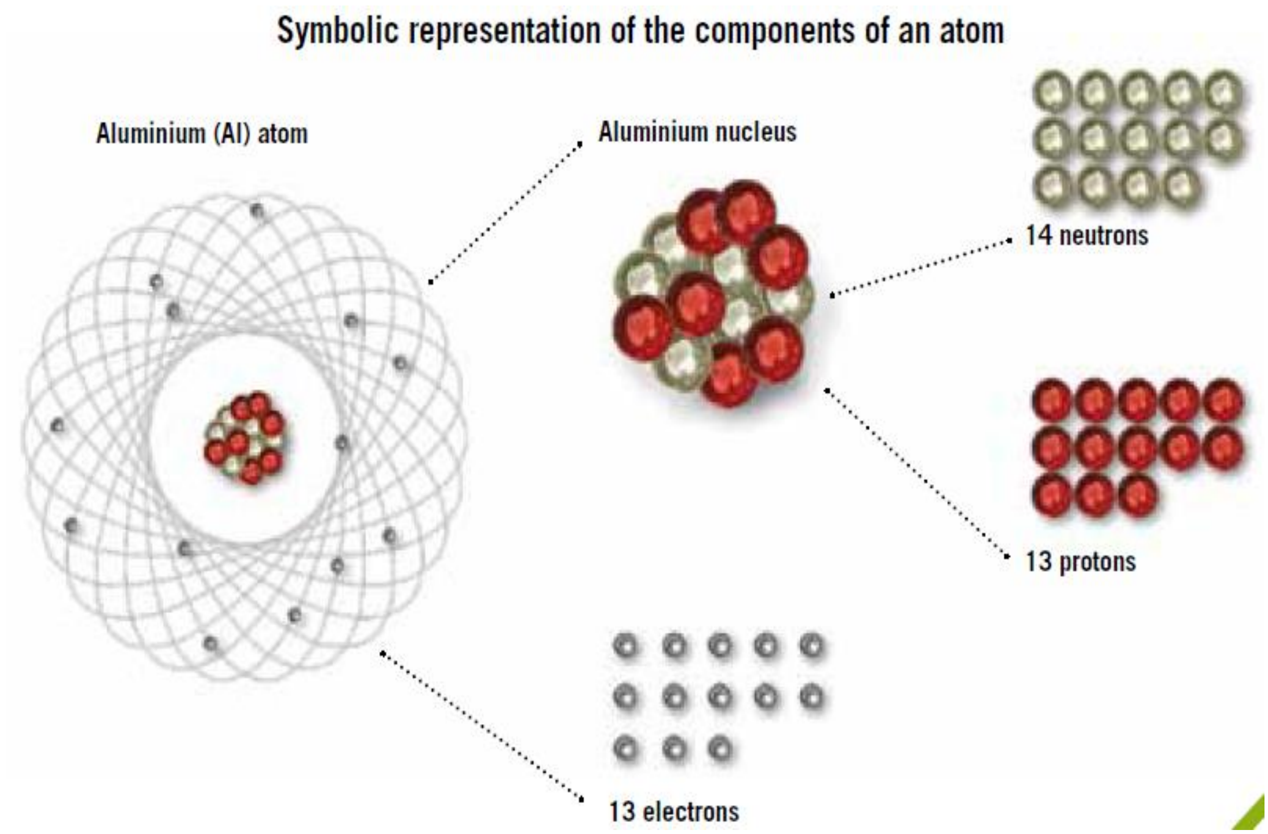
In our study of atomic structure, we look first at the **fundamental particles.** These are the basic building blocks of all atoms. Atoms, and hence *all* matter, consist principally of three fundamental particles: *electrons*, *protons*, and *neutrons*. Knowledge of the nature and functions of these particles is essential to understanding chemical interactions. The relative masses and charges of the three fundamental particles are shown in the Table below.

FUNDAMENTAL PARTICLES OF MATTER

Particle	**Mass**	**charge (relative scale)**
Electron (*e*-)	0.00054858 amu	- 1
Proton (*p* or *p*+)	1.0073 amu	+ 1
Neutron (*n* or n^0)	1.0087 amu	none

Many other subatomic particles, such as quarks, positrons, neutrinos, pions, and muons, have also been discovered.

The mass of an electron is very small compared with the mass of either a proton or a neutron. The charge on a proton is equal in magnitude, but opposite in sign, to the charge on an electron.



$^{27}_{13}Al$

Let us examine these particles in more detail.

THE DISCOVERY OF ELECTRONS

Some of the earliest evidence about atomic structure was supplied in the early 1800s by the English chemist Humphrey Davy (1778–1829). He found that when he passed electric current through some substances, the substances decomposed. He therefore suggested that the elements of a chemical compound are held together by electrical forces. In 1832–1833, Michael Faraday (1791–1867), Davy's student, determined the quantitative relationship between the amount of electricity used in electrolysis and the amount of chemical reaction that occurs. Studies of Faraday's work by George Stoney (1826–1911) led him to suggest in 1874 that units of electric charge are associated with atoms. In 1891, he suggested that they be named *electrons.*

The most convincing evidence for the existence of electrons came from experiments using *cathode-ray tubes*. Two electrodes are sealed in a glass tube containing gas at a very low pressure. When a high voltage is applied, current flows and rays are given off by the cathode (negative electrode). These rays travel in straight lines toward the anode (positive electrode) and cause the walls opposite the cathode to glow. An object placed in the path of the cathode rays casts a shadow on a zinc sulfide screen placed near the anode. The shadow shows that the rays travel from the cathode toward the anode.

The rays must therefore be negatively charged. Furthermore, they are deflected by electric and magnetic fields in the directions expected for negatively charged particles.

In 1897 J. J. Thomson (1856–1940) studied these negatively charged particles more carefully. He called them **electrons,** the name Stoney had suggested in 1891. By studying the degree of deflections of cathode rays in different electric and magnetic fields, Thomson determined the ratio of the charge (*e*) of the electron to its mass (*m*). The modern value for this ratio is

e/m = 1.75882 x 108 coulomb (C)/gram

This ratio is the same regardless of the type of gas in the tube, the composition of the electrodes, or the nature of the electric power source. The clear implication of Thomson's work was that electrons are fundamental particles present in all atoms. We now know that this is true and that all atoms contain integral numbers of electrons.

Once the charge-to-mass ratio for the electron had been determined, additional experiments were necessary to determine the value of either its mass or its charge, so that the other could be calculated.

Robert Millikan (1868–1953) solved this dilemma with the famous "oil-drop experiment," in which he determined the charge of the electron. All of the charges measured by Millikan turned out to be integral multiples of the same number. He assumed that this smallest charge was the charge on one electron. This value is 1.60218×10^{19} coulomb (modern value). The charge-to-mass ratio, $e/m = 1.75882 \times 10^8$ C/g, can be used in inverse form to calculate the mass of the electron:

$$m = \frac{1 \text{ g}}{1.75882 \times 10^8 \text{ C}} \times 1.60218 \times 10^{-19} \text{ C}$$
$$= 9.10940 \times 10^{-28} \text{ g}$$

This is only about 1/1836 the mass of a hydrogen atom, the lightest of all atoms. Millikan's simple oil-drop experiment stands as one of the cleverest, yet most fundamental, of all classic scientific experiments. It was the first experiment to suggest that atoms contain integral numbers of electrons; we now know this to be true.

CANAL RAYS AND PROTONS

In 1886, Eugen Goldstein (1850–1930) first observed that a cathode-ray tube also generates a stream of positively charged particles that moves towards the cathode. These were called **canal rays** because they were observed occasionally to pass through a channel, or "canal," drilled in the negative electrode. These *positive* rays, or *positive ions,*are created when the gaseous atoms in the tube lose electrons. Positive is ions are formed by the process.

$$\text{Atom} \longrightarrow \text{cation}^{+} + e^{-} \qquad \text{or} \qquad X \longrightarrow X^{+} + e^{-} \qquad \text{(energy absorbed)}$$

Different elements give positive ions with different *e/m* ratios. The regularity of the *e/m* values for different ions led to the idea that there is a unit of positive charge and that it resides in the **proton.** The proton is a fundamental particle with a charge equal in magnitude but opposite in sign to the charge on the electron. Its mass is almost 1836 times that of the electron. **Canal ray** is a stream of positively charged particles (cations) that moves towards the negative electrode in a cathode-ray tube; observed to pass through canals in the negative electrode.

Ernest Rutherford discovered the nucleus of an atom.

Robert Millikan determined the charge of the electron by "oil-drop experiment".

James Chadwick discovered neutron.

The basis of Rutherford's model of the atom is that atoms consist of very small and highly dense positively charged **nuclei surrounded by clouds of electrons.**

Only a few years after Rutherford's scattering experiments, H. G. J. Moseley (1887–1915) studied X-rays given off by various elements. Moseley showed that the X-ray wavelengths could be better correlated with the atomic number. **On the basis of his mathematical analysis of these X-ray data, he concluded that each element differs from the preceding element by having one more positive charge in its nucleus. For the first time it was possible to arrange all known elements in order of increasing nuclear charge.** We now know that every nucleus contains an integral number of protons exactly equal to the number of electrons in a neutral atom of the element.

MASS NUMBER, ISOBARS, ISOTONES AND ISOTOPES

The **mass number** of an atom is the sum of the number of protons and the number of neutrons in its nucleus; that is:

Mass number = number of protons + number of neutrons
= atomic number + neutron number

Isobars are the atoms with same mass number but different atomic numbers. Or, Isobars are atoms with the same number of nucleons from different chemical elements. Examples: ${}^{14}_{6}C$ and ${}^{14}_{7}N$. They share similar physical properties but differ in chemical properties.

Isotones are atoms that have the same neutron number but different proton number. For example, ${}^{36}{}_{16}S$, ${}^{37}{}_{17}Cl$, ${}^{38}{}_{18}Ar$, ${}^{39}{}_{19}K$, and ${}^{40}{}_{20}Ca$ are all isotones of 20 since they all contain 20 neutrons.

Most elements consist of atoms of different masses, called **isotopes.** The isotopes of a given element contain the same number of protons (and also the same number of electrons) because they are atoms of the same element. They differ in mass because they contain different numbers of neutrons in their nuclei.

Isotopes are atoms of the same element with different masses; they are atoms containing the same number of protons but different numbers of neutrons.

For example, there are three distinct kinds of hydrogen atoms, commonly called hydrogen (or **protium**), deuterium, tritium. (This is the only element for which we give each isotope a different name.) Each contains one proton in the atomic nucleus. The predominant form of hydrogen contains no neutrons, but each deuterium atom contains one neutron and each tritium atom contains two neutrons in its nucleus. All three forms of hydrogen display very similar chemical properties.

The mass number for normal hydrogen (protium) atoms is 1; for deuterium, 2; and for tritium, 3. The composition of a nucleus is indicated by its **nuclide symbol.** This consists of the symbol for the element (*E*), with the atomic number (*Z*) written as a subscript at the lower left and the mass number (*A*) as a superscript at the upper left, ${}^{A}_{Z}E$. By this system, the three isotopes of hydrogen are designated as ${}^{1}_{1}H, {}^{2}_{1}H\ and\ {}^{3}_{1}H$.

The Three Isotopes of Hydrogen

Name	Symbol	Nuclide Symbol	Mass (amu)	Atomic Abundance in Nature	No. of Protons	No. of Neutrons	No. of Electrons (in neutral) atoms)
hydrogen	H	${}^{1}_{1}H$	1.007825	99.985%	1	0	1
deuterium	D	${}^{2}_{1}H$	2.01400	0.015%	1	1	1
tritium*	T	${}^{3}_{1}H$	3.01605	0.000%	1	2	1

EXAMPLE *Determination of Atomic Makeup*
Determine the number of protons, neutrons, and electrons in each of the following species. Are the members within each pair isotopes?
(a) ${}^{35}_{17}Cl$ and ${}^{37}_{17}Cl$ (b) ${}^{63}_{29}Cu$ and ${}^{65}_{29}Cu$

Plan
Knowing that the number at the bottom left of the nuclide symbol is the atomic number or number of protons; we can verify the identity of the element in addition to knowing the number or protons per nuclide. From the mass number at the top left, we know the number of protons plus neutrons. The number of protons (atomic number) minus the number of electrons must equal the charge, if any, shown at the top right. From these data one can determine if two nuclides have the same number of protons and are therefore the same element. If they are the same element, they will be isotopes only if their mass numbers differ.

Solution

(a) For $^{35}_{17}Cl$: Atomic number = 17. There are therefore 17 protons per nucleus.
Mass number = 35. There are therefore 35 protons plus neutrons or, because we know that there are 17 protons, there are 18 neutrons.
Because no charge is indicated, there must be equal numbers of protons and electrons, or 17 electrons.

For $^{37}_{17}Cl$: There are 17 protons, 20 neutrons, and 17 electrons per atom.

These are isotopes of the same element. Both have 17 protons, but they differ in their numbers of neutrons: one has 18 neutrons and the other has 20.

(b) For $^{63}_{29}Cu$: Atomic number = 29. There are 29 protons per nucleus.
Mass number = 63. There are 29 protons plus 34 neutrons.
Because no charge is indicated, there must be equal numbers of protons and electrons, or 29 electrons.

For $^{65}_{29}Cu$: There are 29 protons, 36 neutrons, and 29 electrons per atom.

These are isotopes. Both have 29 protons, but they differ in their numbers of neutrons: one isotope has 34 neutrons and the other has 36.

Example: Determine the number of protons, neutrons, and electrons in each of the following species.
(i) $^{35}_{17}Cl^{-}$ and $^{37}_{17}Cl$
(ii) $^{63}_{29}Cu$ and $^{65}_{29}Cu^{2+}$.

Plan: The number of protons which is the atomic number remains the same in a neutral or charged atom since it determines the chemical reactivity of the element unlike the electrons that gain or lose. Neutron which is electrically neutral also remains the same, and therefore not affected by gain or loss of electrons.

Species	Protons	Neutrons	Electrons
(i) $^{35}_{17}Cl^{-}$	17 (atomic number)	18 (i.e 35 -17)	18 (i.e 17 +1, electron gain)
$^{37}_{17}Cl$	17 (atomic number)	20 (i.e 37 -17)	17 (i.e 17 same as proton as atom is neutral)
(ii) $^{63}_{29}Cu$	29 (atomic number)	34 (i.e 63 - 29)	29 (i.e 29 same as proton as atom is neutral)
$^{65}_{29}Cu^{2+}$	29 (atomic number)	36 (i.e 65 - 29)	27 (i.e 29 – 2, electron loss)

Some Naturally Occurring Isotopic Abundances

Element	Atomic Weight (amu)	Isotope	% Natural Abundance	Mass (amu)
boron	10.811	$^{10}_{5}B$	19.91	10.01294
		$^{11}_{5}B$	80.09	11.00931
oxygen	15.9994	$^{16}_{8}O$	99.762	15.99492
		$^{17}_{8}O$	0.038	16.99913
		$^{18}_{8}O$	0.200	17.99916
chlorine	35.4527	$^{35}_{17}Cl$	75.770	34.96885
		$^{37}_{17}Cl$	24.230	36.96590
uranium	238.0289	$^{234}_{92}U$	0.0055	234.0409
		$^{235}_{92}U$	0.720	235.0439
		$^{238}_{92}U$	99.2745	238.0508

The 20 elements that have only one naturally occurring isotope are $^{9}_{4}Be$, $^{19}_{9}F$, $^{23}_{11}Na$, $^{27}_{13}Al$, $^{31}_{15}P$, $^{45}_{21}Sc$, $^{55}_{25}Mn$, $^{59}_{27}Co$, $^{75}_{33}As$, $^{89}_{39}Y$, $^{93}_{41}Nb$, $^{103}_{45}Rh$, $^{127}_{53}I$, $^{133}_{55}Cs$, $^{141}_{59}Pr$, $^{159}_{65}Tb$, $^{165}_{67}Ho$, $^{169}_{69}Tm$, $^{197}_{79}Au$, *and* $^{209}_{83}Bi$. *There are however other, artificially produced isotopes of these elements.*

Note the following:

1. The *atomic number, Z,* is an integer equal to the number of protons in the nucleus of an atom of the element. It is also equal to the number of electrons in a neutral atom. It is the same for all atoms of an element.
2. The *mass number, A,* is an integer equal to the *sum* of the number of protons and the number of neutrons in the nucleus of an atom of a *particular isotope* of an element. It is different for different isotopes of the same element.
3. Many elements occur in nature as mixtures of isotopes. The *atomic weight* of such an element is the weighted average of the masses of its isotopes. Atomic weights are fractional numbers, not integers.
4. The atomic weight that we determine experimentally (for an element that consists of more than one isotope) is such a weighted average. The following example shows how an atomic weight can be calculated from measured isotopic abundances.

EXAMPLE *Calculation of Atomic Weight*

1. Three isotopes of magnesium occur in nature. Their abundances and masses, determined by mass spectrometry, are listed in the following table. Use this information to calculate the atomic weight of magnesium.

Isotope	% Abundance	Mass (amu)
$^{24}_{12}Mg$	78.99	23.98504
$^{25}_{12}Mg$	10.00	24.98584
$^{26}_{12}Mg$	11.01	25.98259

Plan

We multiply the fraction of each isotope by its mass and add these numbers to obtain the atomic weight of magnesium.

Solution

$$\begin{aligned}\text{Atomic weight} &= 0.7899(23.98504\text{ amu}) + 0.1000(24.98584\text{ amu}) + 0.1101(25.98259\text{ amu}) \\ &= 18.946\text{ amu} \quad + 2.4986\text{ amu} \quad + 2.8607\text{ amu} \\ &= \boxed{24.30\text{ amu}} \quad \text{(to four significant figures)}\end{aligned}$$

The two heavier isotopes make small contributions to the atomic weight of magnesium because most magnesium atoms are the lightest isotope.

2. Bromine is composed of $^{79}{}_{35}Br$, 78.9183 amu, and $^{81}{}_{35}Br$, 80.9163 amu. The percent composition of a sample is 50.69 % Br-79 and 49.31 % Br-81. Based on this sample, calculate the atomic weight of bromine.
   Weight of Br = $\frac{(50.69 \times 78.9183) + (49.31\ X\ 80.9163)}{100} = 79.9 \approx 80\ amu$

   OR $\frac{50.69 \times 78.9183}{100} + \frac{49.31\ X\ 80.9163}{100}$
   $= 79.9 \approx 80\ amu.$

3. An element Y has isotopic masses of 6 and 7. If the relative abundance is 1 to 12.5 respectively, what is the Relative Atomic Mass (RAM) of Y?

SOLUTION: Total Relative abundance = 1+ 12.5 = 13.5

$$RAM\ of\ Y = \frac{1\times6}{13.5} + \frac{12.5\times7}{13.5} = 6.93$$

Calculation of Isotopic Abundance

The atomic weight of gallium is 69.72 amu. The masses of the naturally occurring isotopes are 68.9257 amu for $^{69}_{31}Ga$ and 70.9249 amu for $^{71}_{31}Ga$ respectively. Calculate the percent abundance of each isotope.

Plan

We represent the fraction of each isotope algebraically. Atomic weight is the weighted average of the masses of the constituent isotopes. So the fraction of each isotope is multiplied by its mass, and the sum of the results is equal to the atomic weight.

Solution

Let x = fraction of ${}^{69}_{31}Ga$. Then $(1 - x)$ = fraction of ${}^{71}_{31}Ga$.

$$x(68.9257 \text{ amu}) + (1 - x)(70.9249 \text{ amu}) = 69.72 \text{ amu}$$
$$68.9257x + 70.9249 - 70.9249x = 69.72$$
$$-1.9992x = -1.20$$
$$x = 0.600$$
$$x = 0.600 = \text{fraction of } {}^{69}_{31}Ga \quad \therefore \quad \boxed{60.0\% \ {}^{69}_{31}Ga}$$
$$(1 - x) = 0.400 = \text{fraction of } {}^{71}_{31}Ga \quad \therefore \quad \boxed{40.0\% \ {}^{71}_{31}Ga}$$

THE ELECTRONIC STRUCTURES OF ATOMS

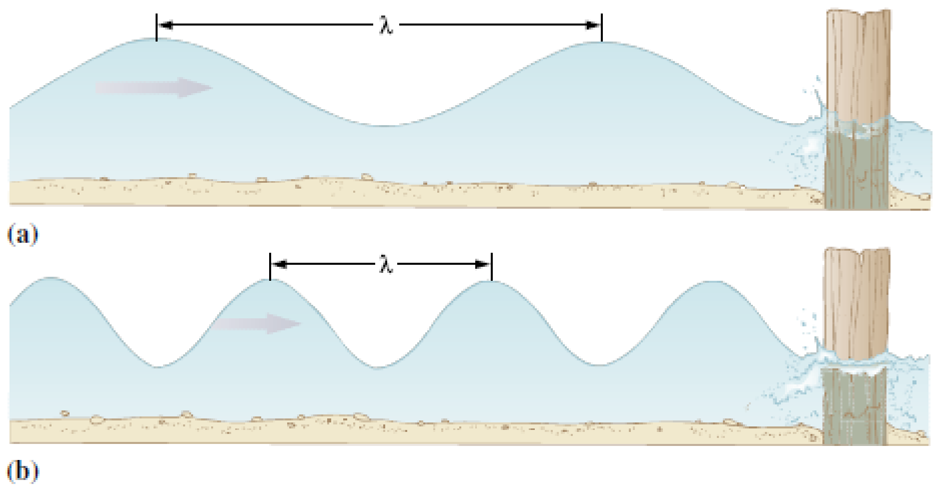
The Rutherford model of the atom is consistent with the evidence presented so far, but it has some serious limitations. It does not answer such important questions as: *Why* do different elements have such different chemical and physical properties? *Why* does chemical bonding occur at all? *Why* does each element form compounds with characteristic formulas? *How* can atoms of different elements give off or absorb light only of characteristic colours (as was known long before 1900)? To improve our understanding, we must first learn more about the arrangements of electrons in atoms. The theory of these arrangements is based largely on the study of the light given off and absorbed by atoms. Then we will develop a detailed picture of the *electron configurations* of different elements. Knowledge of these arrangements will help us to understand the periodic table and chemical bonding.

ELECTROMAGNETIC RADIATION

Our ideas about the arrangements of electrons in atoms have evolved slowly. Much of the information has been derived from **atomic emission spectra.** These are the lines, or bands, produced on photographic film by radiation that has passed through a refracting glass prism after being emitted from electrically or thermally excited atoms. To help us understand the nature of atomic spectra, we first describe electromagnetic radiation.

All types of electromagnetic radiation, or radiant energy, can be described in the terminology of waves. To help characterize any wave, we specify its *wavelength* (or its *frequency*).

Let us use a familiar kind of wave, that on the surface of water (Figure below), to illustrate these terms. The significant feature of wave motion is its repetitive nature.



The **wavelength, λ**, is the distance between any two adjacent identical points of the wave, for instance, two adjacent crests. The **frequency** is the number of wave crests passing a given point per unit time; it is represented by the symbol ν (Greek letter "nu") and is usually expressed in cycles per second or, more commonly, simply as 1/s or s^{-1} with "cycles" understood. For a wave that is "traveling" at some speed, the wavelength and the frequency are related to each other by:

$$\lambda\nu = \text{speed of propagation of the wave} \quad \text{or} \quad \lambda\nu = c$$

Thus, wavelength and frequency are inversely proportional to each other; for the same wave speed, shorter wavelengths correspond to higher frequencies. For water waves, it is the surface of the water that changes repetitively; for a vibrating violin string, it is the displacement of any point on the string. Electromagnetic radiation is a form of energy that consists of electric and magnetic fields that vary repetitively. The electromagnetic radiation most obvious to us is visible light. It has wavelengths ranging from about 4.0×10^{-7} m (violet) to about 7.5×10^{-7} m (red). Expressed in frequencies, this range is about 7.5×10^{14} Hz (violet) to about 4.0 × 1014 Hz (red).

Isaac Newton (1642–1727) first recorded the separation of sunlight into its component colors by allowing it to pass through a prism. Because sunlight (white light) contains all wavelengths of visible light, it gives the *continuous spectrum* observed in a rainbow. Visible light represents only a tiny segment of the electromagnetic radiation spectrum. In addition to all wavelengths of visible light, sunlight also contains shorter wavelength (ultraviolet) radiation as well as longer wavelength (infrared) radiation. Neither of these can be detected by the human eye. Both may be detected and recorded photographically or by detectors designed for that purpose. Many other familiar kinds of radiation are simply electromagnetic radiation of longer or shorter wavelengths.
In a vacuum, the speed of electromagnetic radiation, *c*, is the same for all wavelengths, 2.99792458 $\times 10^8$ m/s. The relationship between the wavelength and frequency of electromagnetic radiation, with *c* rounded to three significant figures, is

$$\lambda\nu = c = 3.00 \times 10^8 \text{ m/s}$$

EXAMPLE *Wavelength of Light*
Light near the middle of the ultraviolet region of the electromagnetic radiation spectrum has a frequency of 2.73×10^{16} s^{-1}. Yellow light near the middle of the visible region of the spectrum has a frequency of 5.26×10^{14} s^{-1}. Calculate the wavelength that corresponds to each of these two frequencies of light.

Plan

Wavelength and frequency are inversely proportional to each other, $\lambda\nu = c$. We solve this relationship for λ and calculate the wavelengths.

Solution

$$\text{(ultraviolet light)}\ \lambda = \frac{c}{\nu} = \frac{3.00 \times 10^{8}\ \text{m}\cdot\text{s}^{-1}}{2.73 \times 10^{16}\ \text{s}^{-1}} = \boxed{1.10 \times 10^{-8}\ \text{m}\ (1.10 \times 10^{2}\ \text{Å})}$$

$$\text{(yellow light)}\ \lambda = \frac{c}{\nu} = \frac{3.00 \times 10^{8}\ \text{m}\cdot\text{s}^{-1}}{5.26 \times 10^{14}\ \text{s}^{-1}} = \boxed{5.70 \times 10^{-7}\ \text{m}\ (5.70 \times 10^{3}\ \text{Å})}$$

We have described light in terms of wave behaviour. Under certain conditions, it is also possible to describe light as composed of *particles*, or **photons.** According to the ideas presented by Max Planck (1858–1947) in 1900, each photon of light has a particular amount (a **quantum**) of energy. The amount of energy possessed by a photon depends on the frequency of the light. The energy of a photon of light is given by Planck's equation:

$$E = h\nu \qquad \text{or} \qquad E = \frac{hc}{\lambda}$$

where h is Planck's constant, $6.6260755 \times 10^{-34}$ Js, and ν is the frequency of the light. Thus, energy is directly proportional to frequency.

Planck's equation is used in the next Example to show that a photon of ultraviolet light has more energy than a photon of yellow light.

EXAMPLE *Energy of Light*

Calculate the energy, in joules, of an individual photon of each of the wavelengths of ultraviolet light of frequency $2.73 \times 10^{16}\ s^{-1}$ and of yellow light of frequency $5.26 \times 10^{14}\ s^{-1}$. Compare these photons by calculating the ratio of their energies.

Plan

We use each frequency to calculate the photon energy from the relationship $E = h\nu$. Then we calculate the required ratio.

Solution

$$\text{(ultraviolet light)}\ E = h\nu = (6.626 \times 10^{-34}\ \text{J}\cdot\text{s})(2.73 \times 10^{16}\ \text{s}^{-1}) = \boxed{1.81 \times 10^{-17}\ \text{J}}$$

$$\text{(yellow light)}\ E = h\nu = (6.626 \times 10^{-34}\ \text{J}\cdot\text{s})(5.26 \times 10^{14}\ \text{s}^{-1}) = \boxed{3.49 \times 10^{-19}\ \text{J}}$$

Now, we compare the energies of these two photons.

$$\frac{E_{\text{uv}}}{E_{\text{yellow}}} = \frac{1.81 \times 10^{-17}\ \text{J}}{3.49 \times 10^{-19}\ \text{J}} = \boxed{51.9}$$

A photon of light near the middle of the ultraviolet region is more than 51 times more energetic than a photon of light near the middle of the visible region.

Example: Calculate the frequency of radiation and energy of photon with a wavelength of 8973 Å. ($c = 3 \times 10^{8}\ ms^{-1}$, $h = 6.626 \times 10^{-34}\ Kgm^2/s$, $1\ Å = 10^{-10}$ m).

$v = c/\lambda$

8973 Å = 8973×10^{-10} m
OR 8.973×10^{-7} m

$V = \frac{3 \times 10^{8}}{8.973 \times 10^{-7}}$
$= 3.34 \times 10^{14} s^{-1}$

$E = hv$
$E = 6.626 \times 10^{-34} \times 3.34 \times 10^{14}$

$= 2.21 \times 10^{-19}$

ATOMIC SPECTRA AND THE BOHR ATOM
Incandescent ("red hot" or "white hot") solids, liquids, and high-pressure gases give continuous spectra. When an electric current is passed through a gas in a vacuum tube at very low pressures, however, the light that the gas emits can be dispersed by a prism into distinct lines. Such an **emission spectrum** is described as a *bright line spectrum.*
BOHR'S MODEL OF THE ATOM (Assumptions of Bohr, also known as quantum theory)

1. An electron in an atom can only exist in certain states characterized by definite energy levels.
2. The energy of an electron can only change by some definite whole number multiple of a unit called the quantum, e.g. 1, 2, 3, …,etc. Quanta cannot have non-integral values such as 1.5 or 2.2. This means that the energy of an electron is quantized.
3. As the energy of an electron is quantized, the radius of the orbit should also be quantized. Different orbits represent different energy levels. The energy level with the lowest energy is the one nearest to the nucleus.

An energy level having higher energy content is further away from the nucleus. When an electron moves from one orbit to another, it must emit or absorb a definite amount of energy to bring it to that orbit. When the electron jumps from a higher energy state E_2 to a lower energy state of energy E_1, energy is emitted. The energy emitted is related to the frequency of light recorded in the spectrum by the following equation.
$\Delta E = E_2 - E_1 = hv$
Where ΔE = difference in energy between two energy levels
h= Planck's constant having a value of 6.626×10^{-34} J and
v = frequency of light emitted
Similarly, energy has to be absorbed before the electron can jump from a lower energy state, E2 to a higher energy state, E_3. The amount of energy absorbed is also quantized to the frequency absorbed in the spectrum by same relationship.
$\Delta E = E_3 - E_2 = hv'$
When sufficient energy is supplied to an atom, it is possible to promote or excite an electron from a lower energy level to a higher one. This process is called excitation. Since an electron is unstable at a higher energy level, it will fall back to a lower energy level. Excess energy is emitted as

radiation. As the energy difference between higher and lower energy levels can only have certain fixed values, this explains why discrete lines of fixed frequencies are found in the spectrum.

LIMITATIONS OF BOHR'S MODEL OF THE ATOM

1. It could not explain the more complicated spectral lines observed in spectra other than hydrogen.
2. There is still no experimental evidence that proves the basis of electrons moving around the nucleus of an atom in fixed orbits.

EXAMPLE *Energy of Light*

A green line of wavelength 4.86 x 10^{-7} m is observed in the emission spectrum of hydrogen. Calculate the energy of one photon of this green light.

Plan

We know the wavelength of the light, and we calculate its frequency so that we can then calculate the energy of each photon.

Solution

$$E = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34}\ \text{J}\cdot\text{s})(3.00 \times 10^{8}\ \text{m/s})}{(4.86 \times 10^{-7}\ \text{m})} = \boxed{4.09 \times 10^{-19}\ \text{J/photon}}$$

To gain a better appreciation of the amount of energy involved, let's calculate the total energy, in kilojoules, emitted by one mole of atoms. (Each atom emits one photon.)

$$\frac{?\ \text{kJ}}{\text{mol}} = 4.09 \times 10^{-19}\ \frac{\text{J}}{\text{atom}} \times \frac{1\ \text{kJ}}{1 \times 10^{3}\ \text{J}} \times \frac{6.02 \times 10^{23}\ \text{atoms}}{\text{mol}} = \boxed{2.46 \times 10^{2}\ \text{kJ/mol}}$$

This calculation shows that when each atom in one mole of hydrogen atoms emits light of wavelength 4.86 × 10^{-7} m, the mole of atoms loses 246 kJ of energy as green light. (This would be enough energy to operate a 100-watt light bulb for more than 40 minutes.)

When an electric current is passed through hydrogen gas at very low pressures, several series of lines in the spectrum of hydrogen are produced. These lines were studied intensely by many scientists. In the late nineteenth century, Johann Balmer (1825–1898) and Johannes Rydberg (1854–1919) showed that the wavelengths of the various lines in the hydrogen spectrum can be related by a mathematical equation:

$$\frac{1}{\lambda} = R\left(\frac{1}{n_1^{2}} - \frac{1}{n_2^{2}}\right)$$

Here R is 1.097 × 107 m^{-1} and is known as the Rydberg constant. The n's are positive integers, and n_1 is smaller than n_2. The Balmer-Rydberg equation was derived from numerous observations, not theory. It is thus an empirical equation.

THE WAVE NATURE OF THE ELECTRON

Einstein's idea that light can exhibit both wave properties and particle properties suggested to Louis de Broglie (1892–1987) that very small particles, such as electrons, might also display wave properties under the proper circumstances. In his doctoral thesis in 1925, de Broglie predicted that a particle with a mass *m* and velocity *v* should have the wavelength associated with it. The numerical value of this de Broglie wavelength is given by:

$\lambda = h/mv$ (where h = Planck's constant)

Two years after de Broglie's prediction, C. Davisson (1882–1958) and L. H. Germer (1896–1971) at the Bell Telephone Laboratories demonstrated diffraction of electrons by a crystal of nickel. This behaviour is an important characteristic of waves. It shows conclusively that electrons do have wave properties. Davisson and Germer found that the wavelength associated with electrons of known energy is exactly that predicted by de Broglie. Similar diffraction experiments have been successfully performed with other particles, such as neutrons.

EXAMPLE *de Broglie Equation*

(a) Calculate the wavelength in meters of an electron traveling at 1.24×10^7 m/s. The mass of an electron is 9.11×10^{-28} g. (b) Calculate the wavelength of a baseball of mass 5.25 oz traveling at 92.5 mph. Recall that 1 J = 1 kgm^2/s^2.

Plan

For each calculation, we use the de Broglie equation

$$\lambda = \frac{h}{mv}$$

where

$$h \text{ (Planck's constant)} = 6.626 \times 10^{-34}\ \text{J} \cdot \text{s} \times \frac{1 \dfrac{\text{kg} \cdot \text{m}^2}{\text{s}^2}}{1\ \text{J}}$$

$$= 6.626 \times 10^{-34}\ \frac{\text{kg} \cdot \text{m}^2}{\text{s}}$$

For consistency of units, mass must be expressed in kilograms. In part (b), we must also convert the speed to meters per second.

Solution

(a) $$m = 9.11 \times 10^{-28}\ \text{g} \times \frac{1\ \text{kg}}{1000\ \text{g}} = 9.11 \times 10^{-31}\ \text{kg}$$

Substituting into the de Broglie equation,

$$\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34}\ \dfrac{\text{kg} \cdot \text{m}^2}{\text{s}}}{(9.11 \times 10^{-31}\ \text{kg})\left(1.24 \times 10^7\ \dfrac{\text{m}}{\text{s}}\right)} = \boxed{5.87 \times 10^{-11}\ \text{m}}$$

Though this seems like a very short wavelength, it is similar to the spacing between atoms in many crystals. A stream of such electrons hitting a crystal gives measurable diffraction patterns.

(b)

$$m = 5.25 \text{ oz} \times \frac{1 \text{ lb}}{16 \text{ oz}} \times \frac{1 \text{ kg}}{2.205 \text{ lb}} = 0.149 \text{ kg}$$

$$v = \frac{92.5 \text{ miles}}{\text{h}} \times \frac{1 \text{ h}}{3600 \text{ s}} \times \frac{1.609 \text{ km}}{1 \text{ mile}} \times \frac{1000 \text{ m}}{1 \text{ km}} = 41.3 \frac{\text{m}}{\text{s}}$$

Now, we substitute into the de Broglie equation.

$$\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34} \frac{\text{kg}\cdot\text{m}^2}{\text{s}}}{(0.149 \text{ kg})\left(41.3 \frac{\text{m}}{\text{s}}\right)} = \boxed{1.08 \times 10^{-34} \text{ m}}$$

This wavelength is far too short to give any measurable effects. Recall that atomic diameters are in the order of 10^{-10} m, which is 24 powers of 10 greater than the baseball "wavelength."

THE QUANTUM MECHANICAL PICTURE OF THE ATOM

Through the work of de Broglie, Davisson and Germer, and others, we now know that electrons in atoms can be treated as waves more effectively than as small compact particles traveling in circular or elliptical orbits. Large objects such as golf balls and moving automobiles obey the laws of classical mechanics (Isaac Newton's laws), but very small particles such as electrons, atoms, and molecules do not. A different kind of mechanics, called **quantum mechanics,** which is based on the *wave* properties of matter, describes the behaviour of very small particles much better. Quantization of energy is a consequence of these properties.

One of the underlying principles of quantum mechanics is that we cannot determine precisely the paths that electrons follow as they move about atomic nuclei.

The **Heisenberg Uncertainty Principle,** stated in 1927 by Werner Heisenberg (1901–1976), is a theoretical assertion that is consistent with all experimental observations. The **Heisenberg Uncertainty Principle states that** it is impossible to determine accurately both the momentum and the position of an electron (or any other very small particle) simultaneously. Momentum is mass times velocity, *mv*. Because electrons are so small and move so rapidly, their motion is usually detected by electromagnetic radiation. Photons that interact with electrons have about the same energies as the electrons. Consequently, the interaction of a photon with an electron severely disturbs the motion of the electron. It is not possible to determine simultaneously both the position and the velocity of an electron, so we resort to a statistical approach and speak of the probability of finding an electron within specified regions in space.

With these ideas in mind, we list some basic ideas of quantum mechanics.

1. Atoms and molecules can exist only in certain energy states. In each energy state, the atom or molecule has a definite energy. When an atom or molecule changes its energy state, it must emit or absorb just enough energy to bring it to the new energy state (the quantum condition).

Atoms and molecules possess various forms of energy. Let us focus our attention on their *electronic energies.*

2. When atoms or molecules emit or absorb radiation (light), they change their energies. The energy change in the atom or molecule is related to the frequency or wavelength of the light emitted or absorbed by the equations:

$$\Delta E = h\nu \qquad \text{or} \qquad \Delta E = hc/\lambda$$

This gives a relationship between the energy change, ΔE, and the wavelength, $\boldsymbol{\lambda}$, of the radiation emitted or absorbed. *The energy lost (or gained) by an atom as it goes from higher to lower (or lower to higher) energy states is equal to the energy of the photon emitted (or absorbed) during the transition.*

3. The allowed energy states of atoms and molecules can be described by sets of numbers called *quantum numbers.*

The mathematical approach of quantum mechanics involves treating the electron in an atom as a *standing wave.* A standing wave is a wave that does not travel and therefore has at least one point at which it has zero amplitude, called a node.

QUANTUM NUMBERS

The solutions of the Schrödinger and Dirac equations for hydrogen atoms give wave functions,ψ, that describe the various states available to hydrogen's single electron. Each of these possible states is described by four quantum numbers. We can use these quantum numbers to designate the electronic arrangements in all atoms, their so-called **electron configurations (electronic configurations).** These quantum numbers play important roles in describing the energy levels of electrons and the shapes of the orbitals that describe distributions of electrons in space. The interpretation will become clearer when we discuss atomic orbitals in the following section.

An **atomic orbital** is a region of space in which the probability of finding an electron is high.

We define each quantum number and describe the range of values it may take.
The four quantum numbers assigned to each electron in an atom are as follows:

1. The **principal quantum number,** *n*, describes the *main energy level,* or shell, an electron occupies. It may be any positive integer: $n = 1, 2, 3, 4, \ldots$
2. The **angular momentum (or azimuthal or subsidiary) quantum number,** *l,* designates the *shape of the region* in space that an electron occupies. Within a shell (defined by the value of *n,* the principal quantum number) different sublevels or subshells are possible, each with a characteristic shape. The angular momentum quantum number designates a *sublevel,* or specific *shape* of atomic orbital that an electron may occupy. This number,l, may take integral values from 0 up to and including (*n* - 1):

   $$\ell = 0, 1, 2, \ldots, (n-1)$$

   Thus, the maximum value of l is (*n* - 1). We give a letter notation to each value of l. Each letter corresponds to a different sublevel (subshell).

   $$\begin{array}{l} \ell = 0,\ 1,\ 2,\ 3, \ldots, (n-1) \\ \quad\ \ s\ \ \ p\ \ \ d\ \ \ f \end{array}$$

In the first shell, the maximum value of l is zero, which tells us that there is only an *s* subshell and no *p* subshell. In the second shell, the permissible values of l are 0 and 1, which tells us that there are only *s* and *p* subshells. (NB: s = sharp, p =principal, d = diffuse and f = fundamental).

3. The **magnetic quantum number,** m_l, designates the specific orbital within a subshell. Orbitals within a given subshell differ in their orientations in space, but not in their energies. Within each subshell, m_l may take any integral values from
   -l through zero up to and including +l:

   $$m_\ell = (-\ell), \ldots, 0, \ldots, (+\ell)$$

   The maximum value of m_l depends on the value of l. For example when l=1 which designates the *p* subshell, there are three permissible values of m_l = -1, 0, and +1. Thus, three distinct regions of space, called atomic orbitals, are associated with a *p* subshell. We refer to these orbitals as the *px*, *py*, and *pz* orbitals. The d-subshells has five orbitals viz dxy, dxz, dyz, dx^2-y^2 and $d_z{}^2$.

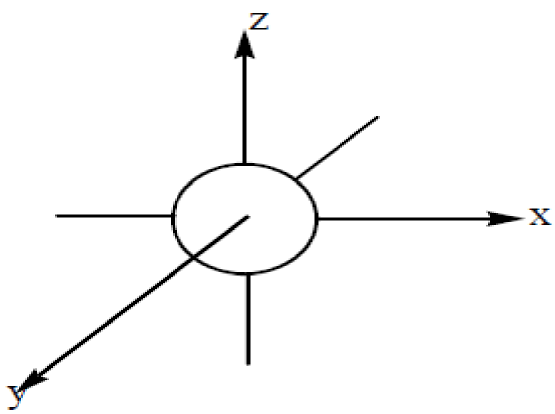
4. The **spin quantum number, m_s** : The **spin quantum number, m_s**, refers to the spin of an electron and the orientation of the magnetic field produced by this spin. An electron spins either in the clockwise or counterclockwise direction and this results in two spin states. For every set of *n*, *l* and *m* values, *s* can take the value of +1/2 or -1/2. This interprets that electrons in an orbital have opposite spins indicated by ↑ and ↓. A combination of the four quantum numbers indicate that for example n=3,l=2, that there are five orbitals corresponding to the m_l ie -2,-1,0,+1,+2.for each of the five orbitals, there are two spins +1/2 and -1/2

TABLE ***Permissible Values of the Quantum Numbers Through*** **n = 4**

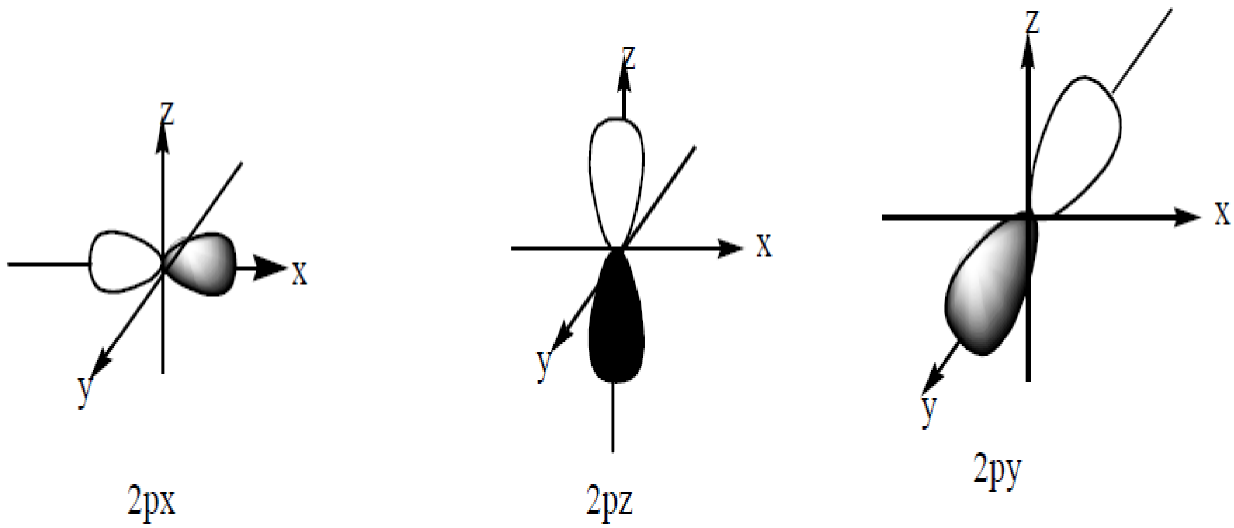
n	ℓ	m_ℓ	m_s	Electron Capacity of Subshell = $4\ell + 2$	Electron Capacity of Shell = $2n^2$
1	0 (1*s*)	0	$+\frac{1}{2}, -\frac{1}{2}$	2	2
2	0 (2*s*)	0	$+\frac{1}{2}, -\frac{1}{2}$	2	8
	1 (2*p*)	−1, 0, +1	$\pm\frac{1}{2}$ for each value of m_ℓ	6	
3	0 (3*s*)	0	$+\frac{1}{2}, -\frac{1}{2}$	2	18
	1 (3*p*)	−1, 0, +1	$\pm\frac{1}{2}$ for each value of m_ℓ	6	
	2 (3*d*)	−2, −1, 0, +1, +2	$\pm\frac{1}{2}$ for each value of m_ℓ	10	
4	0 (4*s*)	0	$+\frac{1}{2}, -\frac{1}{2}$	2	32
	1 (4*p*)	−1, 0, +1	$\pm\frac{1}{2}$ for each value of m_ℓ	6	
	2 (4*d*)	−2, −1, 0, +1, +2	$\pm\frac{1}{2}$ for each value of m_ℓ	10	
	3 (4*f*)	−3, −2, −1, 0, +1, +2, +3	$\pm\frac{1}{2}$ for each value of m_ℓ	14	

ATOMIC ORBITALS
The shapes of the orbitals.

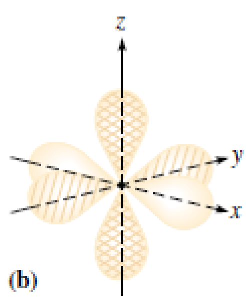
The s-subshell has only one orbital, the p-subshell has three orbitals , the d-subshell has five orbitals and the f-subshell has seven orbitals. It is important to mention that the s, p,d and f terms are spectroscopic terms and they mean sharp, principal, diffuse and fundamental respectively. Each electron is said to occupy an atomic orbital defined by a set of quantum numbers *n*, l, m and *s*. In any atom, each orbital can hold a maximum of two electrons. Within each atom, these atomic orbitals, taken together, can be represented as a diffuse cloud of electrons .The main shell of each atomic orbital in an atom is indicated by the principal quantum number *n.* The value *n* = 1 describes the first, or innermost shell. These shells have been referred to as electron energy levels. Successive shells are at increasingly greater distances from the nucleus. For example, the *n* =2 shell is farther from the nucleus than the *n* =1 shell. When l=0, the orbital is called the s-orbital and it is spherical in shape, when l=1, it is p-orbital and it is dumbbell in shape ; when l=2, it is d-orbital and the shape is double dumbbell but when l=3 it is f-orbital but the shape is complex.



s-orbital



the three p-orbitals



A model of three p orbitals (p_x, p_y and p_z) of a single set of orbitals

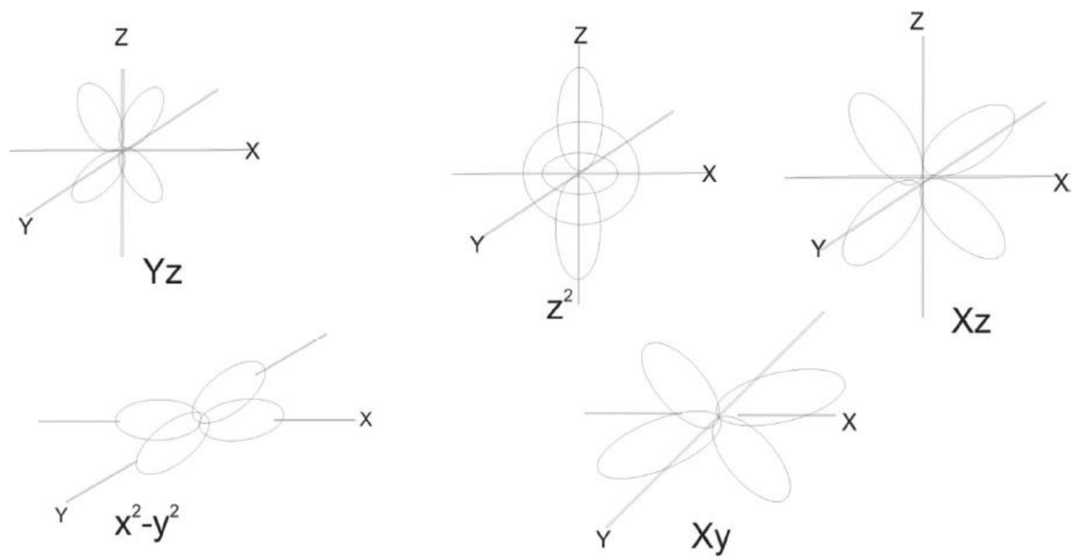


Fig 1a: Spatial arrangement of the five d-orbitals.

Shell n	Number of Subshells per Shell n	Number of Atomic Orbitals n^2	Maximum Number of Electrons $2n^2$
1	1	1 (1*s*)	2
2	2	4 (2*s*, $2p_x$, $2p_y$, $2p_z$)	8
3	3	9 (3*s*, three 3*p*'s, five 3*d*'s)	18
4	4	16	32
5	5	25	50

EXAMPLES

1. Write an acceptable set of four quantum numbers for each electron in a nitrogen atom.
   Recall that:

(a) Principal quantum number, *n*, describes the *main energy level*, or shell, an electron occupies. It may be any positive integer: $n = 1, 2, 3, 4, \ldots$ **It therefore means that electrons number 1 and 2 occupy the first shell n =1 (n = 1, 1= $1s^2$). Electrons number 3 and 4 occupy the second shell, subshell n =2 (n = 2, 2= $2s^2$). Electrons number 5, 6 and 7 occupy the second shell, the third subshell n =2 (i.e 2p) (n = 2, 2, 2= $2p^3$).**

(b) The **angular momentum (or azimuthal or subsidiary) quantum number,** *l*, designates the *shape of the region* in space that an electron occupies. This number,l, may take integral values from 0 up to and including (*n* - 1):

$$\ell = 0, 1, 2, \ldots, (n - 1)$$

Hence, s, p, d and f subshells correspond to 0, 1, 2 and 4 respectively. **Therefore s = 0 = (0, 0 for 1s and 2s) and p = 1 = (1, 1, 1) for the three p orbitals (*px*, *py*, and *pz*) in a nitrogen atom.**

(c) The **magnetic quantum number,** m_l, designates the specific orbital within a subshell. m_l may take any integral values from -l through zero up to and including +l:

$$m_\ell = (-\ell), \ldots, 0, \ldots, (+\ell)$$

The maximum value of m_l depends on the value of l. For example when l=1 which designates the *p* subshell, there are three permissible values of m_l = -1, 0, and +1. Thus, three distinct regions of space, called atomic orbitals, are associated with a *p* subshell. We refer to these orbitals as the *px*, *py*, and *pz* orbitals. **Therefore nitrogen has m_l = 0's for the 4 electrons in 1s and 2s and m_l = -1, 0, and +1 for the *p* subshell with orbitals as *px*, *py*, and *pz*.**

(d) The **spin quantum number, m_s**, refers to the spin of an electron and the orientation of the magnetic field produced by this spin. A combination of the four quantum numbers indicate that for example n=3,l=2, that there are five orbitals corresponding to the m_s ie -2,-1,0,+1,+2.for each of the five orbitals, there are two spins +1/2 and -1/2. **Thus for nitrogen atom with the following electronic configuration of, N = $1s^22s^22p^3$, 2 elecrons in 1s will have +1/2, -1/2 and 2 elecrons in 2s will have +1/2, -1/2. An electron in p orbitals will have, *px* = +1/2 or -1/2, *py* =+1/2 or -1/2 and *pz* = +1/2 or -1/2.**

Solution

Electron	n	ℓ	m_ℓ	m_s	e^- Configuration
1,2	1	0	0	$+\frac{1}{2}$	$1s^2$
	1	0	0	$-\frac{1}{2}$	
3,4	2	0	0	$+\frac{1}{2}$	$2s^2$
	2	0	0	$-\frac{1}{2}$	
5, 6, 7	2	1	-1	$+\frac{1}{2}$ or $-\frac{1}{2}$	$2p_x{}^1$
	2	1	0	$+\frac{1}{2}$ or $-\frac{1}{2}$	$2p_y{}^1$ or $2p^3$
	2	1	$+1$	$+\frac{1}{2}$ or $-\frac{1}{2}$	$2p_z{}^1$

2.

An oxygen atom has a total of eight electrons. Write the four quantum numbers for each of the eight electrons in the ground state.

O	↑↓	↑↓	↑↓ ↑ ↑
	$1s^2$	$2s^2$	$2p^4$

The results are summarized in the following table:

Electron	n	ℓ	m_ℓ	m_s	Orbital
1	1	0	0	$+\frac{1}{2}$	$1s$
2	1	0	0	$-\frac{1}{2}$	
3	2	0	0	$+\frac{1}{2}$	$2s$
4	2	0	0	$-\frac{1}{2}$	
5	2	1	-1	$+\frac{1}{2}$	$2p_x$, $2p_y$, $2p_z$
6	2	1	0	$+\frac{1}{2}$	
7	2	1	1	$+\frac{1}{2}$	
8	2	1	1	$-\frac{1}{2}$	

Of course, the placement of the eighth electron in the orbital labeled $m_\ell = 1$ is completely arbitrary. It would be equally correct to assign it to $m_\ell = 0$ or $m_\ell = -1$.

Example: In a tabular form, give the names, magnetic quantum numbers, and number of orbitals for each sub-shell with the following quantum numbers: (i) n = 3, *l* = 2; (ii) n = 2, *l* = 0.

Solution

Plan: Sub-shell = n – 1, magnetic quantum numbers ranges from -1 through 0 to +1 depending on the number of orbitals, and Number of orbitals = (2*l* -1).

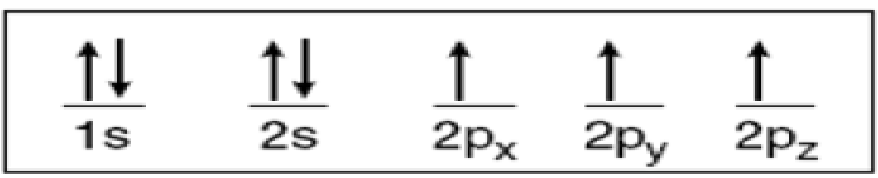
E.g, In (i) n = 3, *l* = 2, Sub-shell = 3 – 1 = 2, corresponding to d-orbital (recall; 0 = s, p = 1, d = 2 and f = 3), but n = 3, therefore sub-shell name 3d. magnetic quantum numbers ranges from -1 through 0 to +1 depending on the number of orbitals, and Number of orbitals = (2*l* -1).

Hence, 2*l* - 1= (2 × 3) - 1 = 5 (5 orbitals). Therefore, magnetic quantum numbers = (-2,-1, 0, +1,+2).

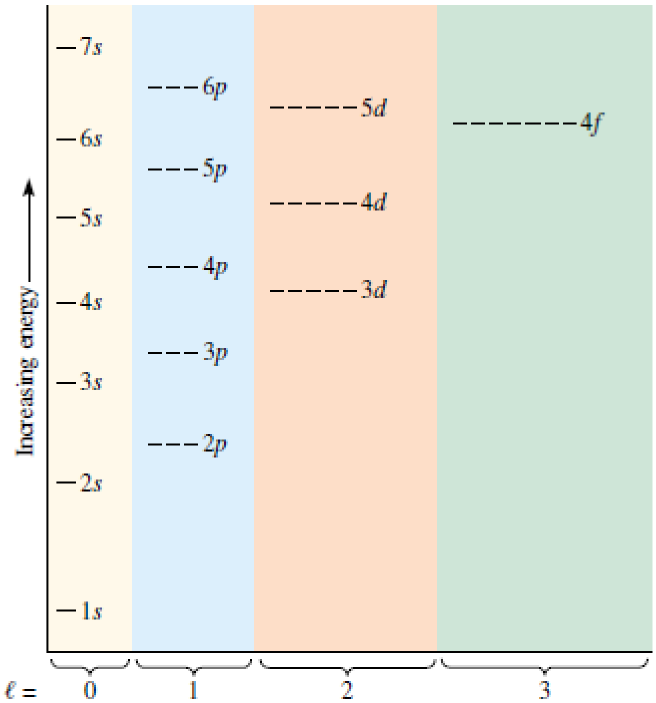
	N	*L*	sub-shell names	magnetic quantum numbers	number of orbitals
(i)	3	2	3d	-2,-1,0, +1,+2	5
(ii)	2	0	2s	0	1

ELECTRON CONFIGURATIONS (OR ELECTRONIC CONFIGURATIONS)
This is the description of electron occupancy of the various orbitals of a given atom. There are many available orbitals for an electron to occupy but the challenge remains which particular orbital would the electron occupy? **There are basic rules for filling of electron into the orbitals of an atom and they include:**

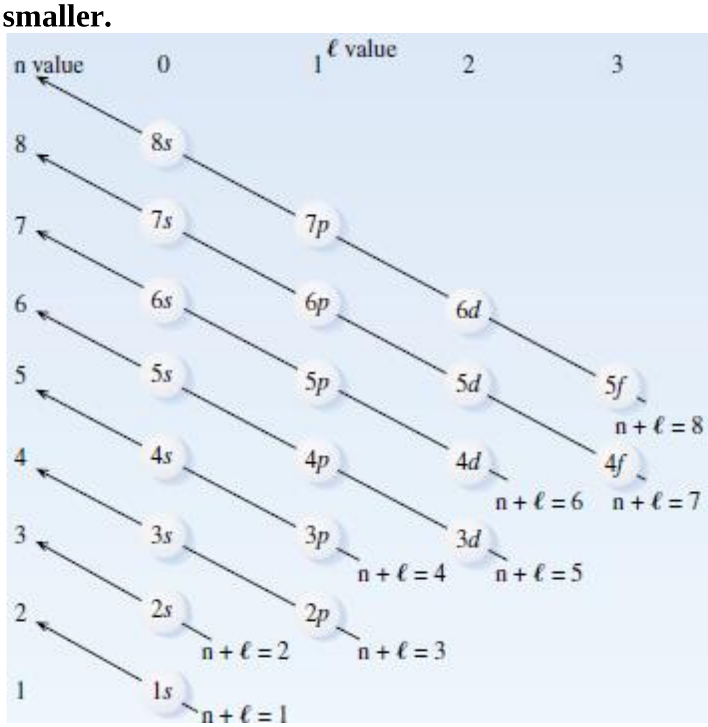
1. Hund's rule of maximum multiplicity: this states that electrons are filled singly first before pairing when they enter subshell with more than one orbital. When pairing the electrons are in opposite spin because the repulsive effect resulting from the two electrons having the same negative charge and in close proximity is much greater. E.g. A diagram of nitrogen is shown below (total of 7 electrons).



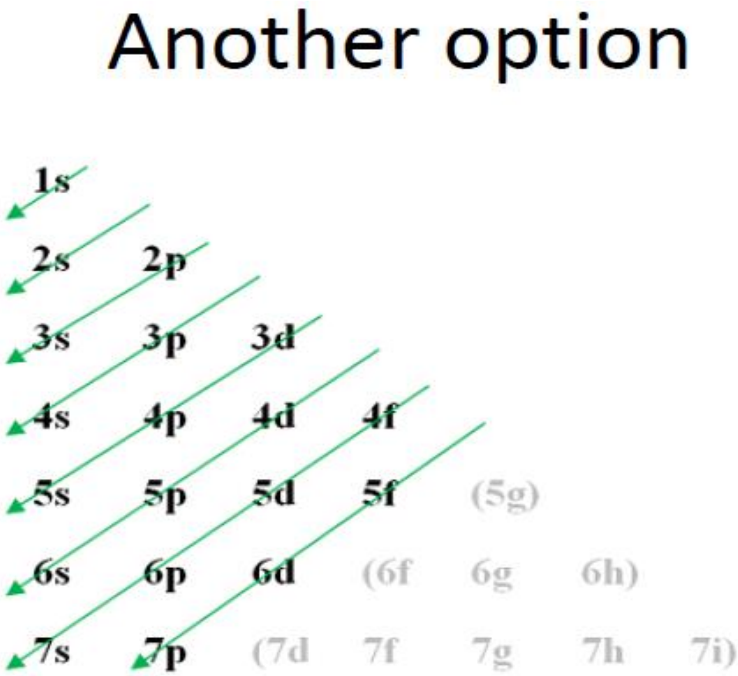
2. Aufbau's principle: this states that electrons occupy the lowest energy orbitals first to obtain the most stable configuration. the orbital with the lowest energy is the one with the least (n+l) value. For example 1S , 2S, 2P have the following (n+l) values 1+0 = 1, 2+0 =2, 2+1=3 respectively. We should recall that s=0, p=1,d=2,f=3 etc. according to this principle the order of filling electrons is 1s,2s,2p,3s,3p,4s,3d,4p,5s,4d,5p,6s,4f,5d,6p,7s, and 6d. When we have the n+l values of the subshell to be the same, then the one with lower n-value has lower energy and comes first.



The usual order of filling (Aufbau order) of the orbitals of an atom. The relative energies are different for different elements, but the following main features should be noted: (1) The largest energy gap is between the 1*s* and 2*s* orbitals. (2) The energies of orbitals are generally closer together at higher energies. (3) The gap between *np* and (*n* + 1)*s* (e.g., between 2*p* and 3*s* or between 3*p* and 4*s*) is fairly large. (4) The gap between (*n* -1)*d* and *ns* (e.g., between 3*d* and 4*s*) is quite small. (5) The gap between (*n* -2)*f* and *ns* (e.g., between 4*f* and 6*s*) is even smaller.



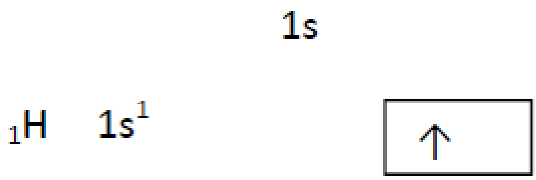
An aid to remembering the usual order of filling of atomic orbitals. Write each shell (value of *n*) on one horizontal line, starting with *n* = 1 at the bottom. Write all like subshells (same *l* values) in the same vertical column. Subshells are filled in order of increasing (*n* + *l*). When subshells have the same (*n* + *l*), the subshell with the lower *n* fills first. To use the diagram, we follow the diagonal arrows in order, reading bottom to top.



3. Pauli Exclusion Principle: this principle states that it is not possible for two electrons in an atom to have the same set of four quantum numbers. In other words, only two electrons can occupy the same orbital and the two electrons must have opposite spins. Therefore, any atomic orbital can be populated by only zero, one, or two electrons. Any two electrons in the same orbital could be described by the same n,*l*,and m_l but the s cannot be the same; one is either +1/2 and the other -1/2. Wolfgang Pauli formulated this principle in 1925.

An orbital is described by a particular allowed set of values for *n, l,* and m_s. Two electrons can occupy the same orbital only if they have opposite spins, m_s. Two of such electrons in the same orbital are *paired.* For simplicity, we shall indicate atomic orbitals as — and show an unpaired electron as ↑ (an electron that occupies an orbital singly) and spin-paired electrons as ⇅. Here, one of the electrons spins clockwise, and the other anticlockwise. Because electrons may spin in two directions, the spin quantum number has two possible values, +1/2 (spin up) and -1/2(spin down). When two electrons have opposite spins, the attraction due to their opposite magnetic fields helps to overcome the repulsion of their like charges. This permits two electrons to occupy the same region (orbital).

To illustrate the rules further we can use the box diagram.



A Hydrogen atom has 1s subshell with 1 'up' electron, $1s^1$. A neutral **helium atom** has 2 bound electrons and they occupy the outermost shell with opposite spins. The 2 electrons are in 1s subshell; where n = 1, l = 0 and m_l = 0 are the same but their spin quantum number, m_s, will be different. One will be m_s = +1/2 (spin up) and the other will be m_s = -1/2 (spin down) as represented on the box diagram above.

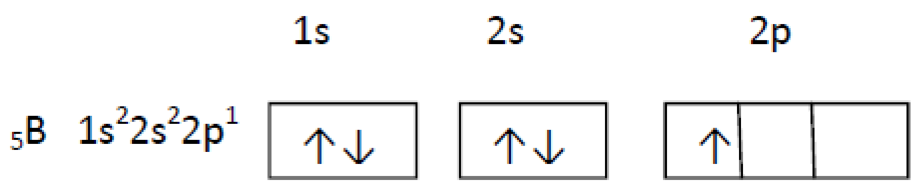


Illustration using orbital notation and simplified notation

Row 1 of the periodic Table

	Orbital Notation 1*s*	Simplified Notation
$_1H$	↑	$1s^1$
$_2He$	↑↓	$1s^2$

Row 2. Elements of atomic numbers 3 through 10 occupy the second period, or horizontal row, in the periodic table. In neon atoms the second main shell is filled completely. Neon, a noble gas, is extremely stable. No reactions of it are known.

	Orbital Notation			Simplified Notation		
	1*s*	2*s*	2*p*			
$_{3}Li$	↑↓	↑		$1s^22s^1$	or	[He] $2s^1$
$_{4}Be$	↑↓	↑↓		$1s^22s^2$		[He] $2s^2$
$_{5}B$	↑↓	↑↓	↑ — —	$1s^22s^22p^1$		[He] $2s^22p^1$
$_{6}C$	↑↓	↑↓	↑ ↑ —	$1s^22s^22p^2$		[He] $2s^22p^2$
$_{7}N$	↑↓	↑↓	↑ ↑ ↑	$1s^22s^22p^3$		[He] $2s^22p^3$
$_{8}O$	↑↓	↑↓	↑↓ ↑ ↑	$1s^22s^22p^4$		[He] $2s^22p^4$
$_{9}F$	↑↓	↑↓	↑↓ ↑↓ ↑	$1s^22s^22p^5$		[He] $2s^22p^5$
$_{10}Ne$	↑↓	↑↓	↑↓ ↑↓ ↑↓	$1s^22s^22p^6$		[He] $2s^22p^6$

Ground state electronic configurations of the elements up to Z = 103.

Atomic number	Element	Ground state electronic configuration	Atomic number	Element	Ground state electronic configurati
1	H	$1s^1$	53	I	$[Kr]5s^2 4d^{10} 5p^5$
2	He	$1s^2 = [He]$	54	Xe	$[Kr]5s^2 4d^{10} 5p^6 = [Xe]$
3	Li	$[He]2s^1$	55	Cs	$[Xe]6s^1$
4	Be	$[He]2s^2$	56	Ba	$[Xe]6s^2$
5	B	$[He]2s^2 2p^1$	57	La	$[Xe]6s^2 5d^1$
6	C	$[He]2s^2 2p^2$	58	Ce	$[Xe]4f^1 6s^2 5d^1$
7	N	$[He]2s^2 2p^3$	59	Pr	$[Xe]4f^3 6s^2$
8	O	$[He]2s^2 2p^4$	60	Nd	$[Xe]4f^4 6s^2$
9	F	$[He]2s^2 2p^5$	61	Pm	$[Xe]4f^5 6s^2$
10	Ne	$[He]2s^2 2p^6 = [Ne]$	62	Sm	$[Xe]4f^6 6s^2$
11	Na	$[Ne]3s^1$	63	Eu	$[Xe]4f^7 6s^2$
12	Mg	$[Ne]3s^2$	64	Gd	$[Xe]4f^7 6s^2 5d^1$
13	Al	$[Ne]3s^2 3p^1$	65	Tb	$[Xe]4f^9 6s^2$
14	Si	$[Ne]3s^2 3p^2$	66	Dy	$[Xe]4f^{10} 6s^2$
15	P	$[Ne]3s^2 3p^3$	67	Ho	$[Xe]4f^{11} 6s^2$
16	S	$[Ne]3s^2 3p^4$	68	Er	$[Xe]4f^{12} 6s^2$
17	Cl	$[Ne]3s^2 3p^5$	69	Tm	$[Xe]4f^{13} 6s^2$
18	Ar	$[Ne]3s^2 3p^6 = [Ar]$	70	Yb	$[Xe]4f^{14} 6s^2$
19	K	$[Ar]4s^1$	71	Lu	$[Xe]4f^{14} 6s^2 5d^1$
20	Ca	$[Ar]4s^2$	72	Hf	$[Xe]4f^{14} 6s^2 5d^2$
21	Sc	$[Ar]4s^2 3d^1$	73	Ta	$[Xe]4f^{14} 6s^2 5d^3$
22	Ti	$[Ar]4s^2 3d^2$	74	W	$[Xe]4f^{14} 6s^2 5d^4$
23	V	$[Ar]4s^2 3d^3$	75	Re	$[Xe]4f^{14} 6s^2 5d^5$
24	Cr	$[Ar]4s^1 3d^5$	76	Os	$[Xe]4f^{14} 6s^2 5d^6$
25	Mn	$[Ar]4s^2 3d^5$	77	Ir	$[Xe]4f^{14} 6s^2 5d^7$
26	Fe	$[Ar]4s^2 3d^6$	78	Pt	$[Xe]4f^{14} 6s^1 5d^9$
27	Co	$[Ar]4s^2 3d^7$	79	Au	$[Xe]4f^{14} 6s^1 5d^{10}$
28	Ni	$[Ar]4s^2 3d^8$	80	Hg	$[Xe]4f^{14} 6s^2 5d^{10}$
29	Cu	$[Ar]4s^1 3d^{10}$	81	Tl	$[Xe]4f^{14} 6s^2 5d^{10} 6p^1$
30	Zn	$[Ar]4s^2 3d^{10}$	82	Pb	$[Xe]4f^{14} 6s^2 5d^{10} 6p^2$
31	Ga	$[Ar]4s^2 3d^{10} 4p^1$	83	Bi	$[Xe]4f^{14} 6s^2 5d^{10} 6p^3$
32	Ge	$[Ar]4s^2 3d^{10} 4p^2$	84	Po	$[Xe]4f^{14} 6s^2 5d^{10} 6p^4$
33	As	$[Ar]4s^2 3d^{10} 4p^3$	85	At	$[Xe]4f^{14} 6s^2 5d^{10} 6p^5$
34	Se	$[Ar]4s^2 3d^{10} 4p^4$	86	Rn	$[Xe]4f^{14} 6s^2 5d^{10} 6p^6 = [$
35	Br	$[Ar]4s^2 3d^{10} 4p^5$	87	Fr	$[Rn]7s^1$
36	Kr	$[Ar]4s^2 3d^{10} 4p^6 = [Kr]$	88	Ra	$[Rn]7s^2$
37	Rb	$[Kr]5s^1$	89	Ac	$[Rn]6d^1 7s^2$
38	Sr	$[Kr]5s^2$	90	Th	$[Rn]6d^2 7s^2$
39	Y	$[Kr]5s^2 4d^1$	91	Pa	$[Rn]5f^2 7s^2 6d^1$
40	Zr	$[Kr]5s^2 4d^2$	92	U	$[Rn]5f^3 7s^2 6d^1$
41	Nb	$[Kr]5s^1 4d^4$	93	Np	$[Rn]5f^4 7s^2 6d^1$
42	Mo	$[Kr]5s^1 4d^5$	94	Pu	$[Rn]5f^6 7s^2$
43	Tc	$[Kr]5s^2 4d^5$	95	Am	$[Rn]5f^7 7s^2$
44	Ru	$[Kr]5s^1 4d^7$	96	Cm	$[Rn]5f^7 7s^2 6d^1$
45	Rh	$[Kr]5s^1 4d^8$	97	Bk	$[Rn]5f^9 7s^2$
46	Pd	$[Kr]5s^0 4d^{10}$	98	Cf	$[Rn]5f^{10} 7s^2$
47	Ag	$[Kr]5s^1 4d^{10}$	99	Es	$[Rn]5f^{11} 7s^2$
48	Cd	$[Kr]5s^2 4d^{10}$	100	Fm	$[Rn]5f^{12} 7s^2$
49	In	$[Kr]5s^2 4d^{10} 5p^1$	101	Md	$[Rn]5f^{13} 7s^2$
50	Sn	$[Kr]5s^2 4d^{10} 5p^2$	102	No	$[Rn]5f^{14} 7s^2$
51	Sb	$[Kr]5s^2 4d^{10} 5p^3$	103	Lr	$[Rn]5f^{14} 7s^2 6d^1$
52	Te	$[Kr]5s^2 4d^{10} 5p^4$			

EXAMPLE *Electron Configurations and Quantum Numbers*
Write an acceptable set of four quantum numbers for each electron in a chlorine atom.

Plan

Chlorine is element number 17. Its first seven electrons have the same quantum numbers as those of nitrogen in one the earlier Example. Electrons 8, 9, and 10 complete the filling of the 2*p* subshell $(n = 2, \ell = 1)$ and therefore also the second energy level. Electrons 11 through 17 fill the 3*s* subshell (*n* =3, *l* =0) and partially fill the 3*p* subshell (*n* = 3, *l* =1).

Electron	n	ℓ	m_ℓ	m_s	e^- Configuration
1, 2	1	0	0	$\pm\frac{1}{2}$	$1s^2$
3, 4	2	0	0	$\pm\frac{1}{2}$	$2s^2$
5–10	2	1	−1	$\pm\frac{1}{2}$	$2p^6$
	2	1	0	$\pm\frac{1}{2}$	
	2	1	+1	$\pm\frac{1}{2}$	
11, 12	3	0	0	$\pm\frac{1}{2}$	$3s^2$
13–17	3	1	−1	$\pm\frac{1}{2}$	$3p^5$
	3	1	0	$\pm\frac{1}{2}$	
	3	1	+1	$+\frac{1}{2}$ or $-\frac{1}{2}$*	

**The 3p orbital with only a single electron can be any one of the set, not necessarily the one with $m_\ell = +1$.*

THE PERIODIC TABLE AND ELECTRONIC CONFIGURATIONS (OR ELECTRON CONFIGURATIONS)

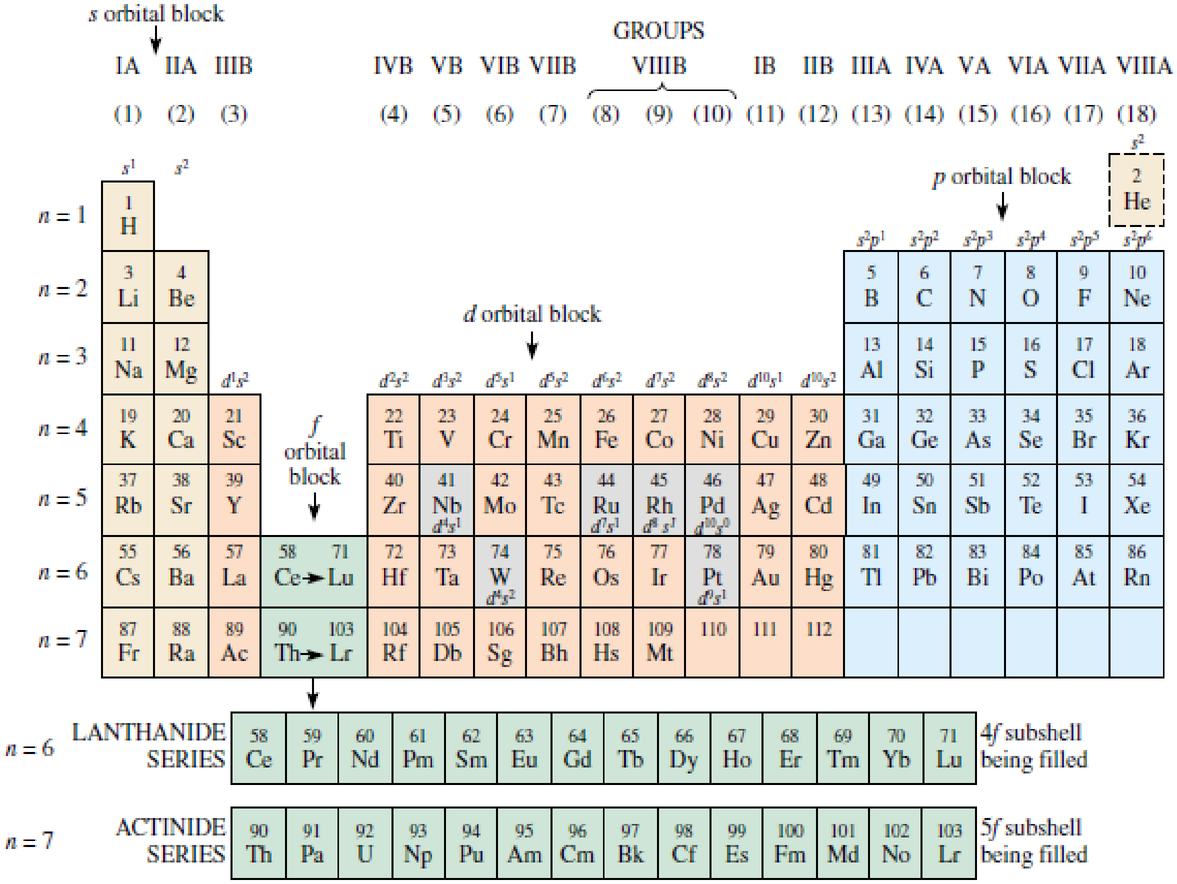
In this section, we view the periodic table from a modern, much more useful perspective—as a systematic representation of the electron configurations of the elements. In the periodic table, elements are arranged in blocks based on the kinds of atomic orbitals that are being filled (Tables A and Bbelow). The periodic tables in this text are divided into “A” and “B” groups. The A groups contain elements in which *s* and *p* orbitals are being filled. Elements within any particular A group have similar electron configurations and chemical properties. The B groups are those in which there are one or two electrons in the *s* orbital of the outermost occupied shell, and the *d* orbitals, one shell smaller, are being filled. Lithium, sodium, and potassium, elements of the leftmost column of the periodic table (Group IA), have a single electron in their outermost *s* orbital (ns^1). Beryllium and magnesium, of Group IIA, have two electrons in their outermost shell, ns^2, while boron and aluminum (Group IIIA) have three electrons in their outermost shell, ns^2np^1. Similar observations can be made for other A group elements. The electron configurations of the A group elements and the noble gases can be predicted reliably. However, there are some more pronounced irregularities in the B groups below the fourth period. In the heavier B group elements, the higher energy subshells in different principal shells have energies that are very nearly equal (Figure below). It is easy for an electron to jump from one orbital to another of nearly the same energy, even in a different set. This is because the orbital energies are *perturbed* (change slightly) as the nuclear charge changes, and an extra electron is added in going from one element to the next.

Group	IA	IIA	IIIB	IVB	VB	VIB	VIIB	VIIIB	VIIIB	VIIIB	IB	IIB	IIIA	IVA	VA	VIA	VIIA	VIIIA
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Period 1	1 H																1 H	2 He
	1s																1s	
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
	2s												2p					
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
	3s												3p					
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
	4s		3d										4p					
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
	5s		4d										5p					
6	55 Cs	56 Ba	57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
	6s		5d										6p					
7	87 Fr	88 Ra	89 Ac	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110	111	112						
	7s		6d															

6	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
	4f													
7	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr
	5f													

Table A: **A periodic table colored to show the kinds of atomic orbitals (subshells) being filled and the symbols of blocks of elements. The electronic structures of the A group elements are quite regular and can be predicted from their positions in the periodic table, but many exceptions occur in the *d* and *f* blocks.**

*The s, p, d, and f Blocks of the Periodic Table B**



**n* is the principal quantum number. The d^1s^2, d^2s^2, . . . *designations represent known configurations. They refer to* $(n-1)d$ *and ns orbitals. Several exceptions to the configurations indicated above each group are shown in gray.*

EXAMPLE *Electron Configurations*

Determine the electron configurations of (a) magnesium, Mg; (b) germanium, Ge; and (c) molybdenum, Mo.

Solution

(a) Magnesium, Mg, is in Group IIA, which has the general configuration s^2; it is in Period 3 (third row). The last filled noble gas configuration is that of neon, or [Ne]. The electron configuration of Mg is [Ne] $3s^2$.

(b) Germanium, Ge, is in Group IVA, for which shows the general configuration s^2p^2. It is in Period 4 (the fourth row), so we interpret this as $4s^24p^2$. The last filled noble gas configuration is that of argon, Ar, accounting for 18 electrons. In addition, Ge lies beyond the *d* orbital block, so we know that the 3*d* orbitals are completely filled. The electron configuration of Ge is:
[Ar] $4s^23d^{10}4p^2$ or [Ar] $3d^{10}4s^24p^2$.

(c) Molybdenum, Mo, is in Group VIB, with the general configuration d^5s^1; it is in Period 5, which begins with 5*s* and is beyond the noble gas krypton. The electron configuration of Mo is
[Kr] $5s^14d^5$ or [Kr] $4d^55s^1$.

EXAMPLE ***Unpaired Electrons***
Determine the number of unpaired electrons in an atom of tellurium, Te.
Plan
Te is in Group VIA in the periodic table, which tells us that its configuration is s^2p^4. All other shells are completely filled, so they contain only paired electrons. We need only to find out how many unpaired electrons are represented by s^2p^4.

Solution
The notation s^2p^4 is a short representation for $s\ \underline{\uparrow\downarrow}\ p\ \underline{\uparrow\downarrow}\ \underline{\uparrow}\ \underline{\uparrow}$. This shows that an atom of Te contains two unpaired electrons.

EXERCISES

Particles and the Nuclear Atom

1. List the three fundamental particles of matter, and indicate the mass and charge associated with each.
2. (a) How do we know that canal rays have charges opposite in sign to cathode rays? What are canal rays? (b) Why are cathode rays from all samples of gases identical, whereas canal rays are not?

Atom Composition, Isotopes, and Atomic Weights

1. How do the isotopes of a given element differ?
2. Define and illustrate the following terms clearly and concisely: (a) atomic number, (b) isotope, (c) mass number, (d) nuclear charge.
3. Complete Charts A and B for neutral atoms.

Chart A

Kind of Atom	Atomic Number	Mass Number	Isotope	Number of Protons	Number of Electrons	Number of Neutrons
______	______	______	$^{21}_{10}Ne$	______	______	______
chlorine	______	35	______	______	______	______
______	28	58	______	______	______	______
______	______	40	______	18	______	______

Chart B

Kind of Atom	Atomic Number	Mass Number	Isotope	Number of Protons	Number of Electrons	Number of Neutrons
selenium	______	______	______	______	______	40
______	______	______	$^{11}_{5}B$	______	______	______
______	______	______	______	______	35	46
______	______	104	______	______	45	______

4. Determine the number of protons, neutrons, and electrons in each of the following species: (a) $^{40}_{20}Ca$; (b) $^{45}_{21}Sc$;

(c) $^{91}_{40}Zr$; (d) $^{39}_{19}K^+$; (e) $^{65}_{30}Zn^{2+}$; (f) $^{108}_{47}Ag^+$.

5. What is the symbol of the species composed of each of the following sets of subatomic particles? (a) 24*p*, 27*n*, 24*e*; (b) 20*p*, 20*n*, 18*e*; (c) 34*p*, 44*n*, 34*e*; (d) 53*p*, 74*n*, 54*e*.

6. What is the symbol of the species composed of each of the following sets of subatomic particles? (a) 94*p*, 150*n*, 94*e*; (b) 79*p*, 118*n*, 76*e*; (c) 34*p*, 45*n*, 36*e*; (d) 52*p*, 76*n*, 54*e*.

7. The atomic weight of lithium is 6.941 amu. The two naturally occurring isotopes of lithium have the following masses: 6Li, 6.01512 amu; ^{7}Li, 7.01600 amu. Calculate the percent of ^{6}Li in naturally occurring lithium.

8. The atomic weight of rubidium is 85.4678 amu. The two naturally occurring isotopes of rubidium have the following masses: ^{85}Rb, 84.9118 amu; ^{87}Rb, 86.9092 amu. Calculate the percent of ^{85}Rb in naturally occurring rubidium.

9. What is the atomic weight of a hypothetical element that consists of the following isotopes in the indicated relative abundances?

Isotope	Isotopic Mass (amu)	% Natural Abundance
1	94.9	12.4
2	95.9	73.6
3	97.9	14.0

10. Naturally occurring iron consists of four isotopes with the abundances indicated here. From the masses and relative abundances of these isotopes, calculate the atomic weight of naturally occurring iron.

Isotope	Isotopic Mass (amu)	% Natural Abundance
^{54}Fe	53.9396	5.82
^{56}Fe	55.9349	91.66
^{57}Fe	56.9354	2.19
^{58}Fe	57.9333	0.33

11. Calculate the atomic weight of silicon using the following data for the percent natural abundance and mass of each isotope: (a) 92.23% ^{28}Si (27.9769 amu); (b) 4.67% ^{29}Si (28.9765 amu); (c) 3.10% ^{30}Si (29.9738 amu).

12. Calculate the atomic weight of chromium using the following data for the percent natural abundance and mass of each isotope: (a) 4.35% ^{50}Cr (49.9461 amu); (b) 83.79% ^{52}Cr (51.9405 amu); (c) 9.50% ^{53}Cr (52.9406 amu); (d) 2.36% ^{54}Cr (53.9389 amu)

Electromagnetic Radiation

1. Calculate the wavelengths, in meters, of radiation of the following frequencies: (a) 5.60 ×1015 s^{-1}; (b) 2.11 × 1014 s^{-1}; (c) 3.89 ×1012 s^{-1}.

2. Calculate the frequency of radiation of each of the following wavelengths: (a) 8973 Å; (b) 492 nm; (c) 4.92 cm; (d) 4.55 × 10^{-9} cm.

3. What is the energy of a photon of each of the radiations in Exercise 37? Express your answer in joules per photon.

4. Excited lithium ions emit radiation at a wavelength of 670.8 nm in the visible range of the spectrum. (This characteristic color is often used as a qualitative analysis test for the presence of Li^+.) Calculate (a) the frequency and (b) the energy of a photon of this radiation.

5. During photosynthesis, chlorophyll-α absorbs light of wavelength 440 nm and emits light of wavelength 670 nm. What is the energy available for photosynthesis from the absorption–emission of a mole of photons?

6. (a) What is the de Broglie wavelength of a proton moving at a speed of 2.50×10^7 m/s? The proton mass is 1.67×10^{-24} g. (b) What is the de Broglie wavelength of a stone with a mass of 30.0 g moving at 2.00×10^3 m/h (≈100 mph)?

7. What is the wavelength corresponding to a neutron of mass 1.67×10^{-27} kg moving at 2360 m/s?

Quantum Numbers and Atomic Orbitals

1. (a) What is a quantum number? What is an atomic orbital? (b) How many quantum numbers are required to specify a single atomic orbital? What are they?

2. An electron is in one of the 3*p* orbitals. What are the possible values of the quantum numbers n, l, m_l, and m_s for the electron?

3. What is the maximum number of electrons in an atom that can have the following quantum numbers? (a) $n = 3$;

(b) $n = 3$ and $\ell = 1$; (c) $n = 3$, $\ell = 1$, and $m_\ell = -1$;

((d) $n = 3$, $\ell = 1$, $m_\ell = -1$, and $m_s = -\frac{1}{2}$.

4. What are the values of n and l for the following subshells?

(a) 1*s*; (b) 4*s*; (c) 3*p*; (d) 3*d*; (e) 4*f*.

5. How many individual orbitals are there in the third shell? Write out n, l, and m_l quantum numbers for each one, and label each set by the *s, p, d, f* designations.

6. (a) Write the possible values of l when n=5. (b) Write the allowed number of orbitals (1) with the quantum numbers $n = 4$, $l = 3$; (2) with the quantum number $n = 4$; (3) with the quantum numbers $n = 4$, $l = 2$, $m_l = 0$; (4) with the quantum numbers $n = 6$, $l = 5$.

7. Write the subshell notations that correspond to (a) $n = 3$, $l = 0$; (b) $n = 3$, $l = 1$; (c) $n = 7$, $l = 0$; (d) $n = 3$, $l = 2$.

8. What values can m_l take for (a) a 3*d* orbital, (b) a 1*s* orbital, and (c) a 3*p* orbital?

Electron Configurations and the Periodic Table

1. Draw representations of ground state electron configurations using the orbital notation $(\uparrow\downarrow)$ for the following elements. (a) N; (b) Fe; (c) Cl; (d) Rh.

2. Determine the number of electrons in the outer occupied shell of each of the following elements, and indicate the principal quantum number of that shell. (a) Na; (b) Al; (c) Ca; (d) Sr; (e) Ba; (f) Br.

3. Identify the element, or elements possible, given only the number of electrons in the outermost shell and the principal quantum number of that shell. (a) 1 electron, first shell; (b) 2 electrons, second shell; (c) 3 electrons, third shell; (d) 2 electrons, seventh shell; (e) 4 electrons, fourth shell; (f) 8 electrons, sixth shell.

4. State the Pauli Exclusion Principle. Would any of the following electron configurations violate this rule: (a) $1s^2$; (b) $1s^2 2p^7$; (c) $1s^3$? Explain.

5. In nature, potassium and sodium are often found together.

6. Which elements are represented by the following electron configurations?

(a) $1s^22s^22p^63s^23p^63d^{10}4s^24p^5$
(b) [Kr] $4d^{10}4f^{14}5s^25p^65d^{10}5f^{14}6s^26p^66d^17s^2$
(c) [Kr] $4d^{10}4f^{14}5s^25p^65d^{10}6s^2\ 6p^6$
(d) [Kr] $4d^55s^2$
(e) $1s^22s^22p^63s^23p^63d^34s^2$

7. Find the total number of *s*, *p*, and *d* electrons in each of the following: (a) Si; (b) Ar; (c) Ni; (d) Zn; (e) V.

8. How many unpaired electrons are in atoms of Na, Ne, B, Be, As, and Ti?

CHEMICAL PERIODICITY

The periodic law: It states that the properties of the elements are periodic functions of their atomic numbers.

In the long form of the periodic table, elements are arranged in blocks based on the kinds of atomic orbitals being filled.

The elements can be classified according to their electronic configurations, which is a very useful system as follows:

1. **Noble Gases:** The Group VIIIA elements—the noble gases—are called inert gases because no chemical reactions were known for them. We now know that the heavier members do form compounds, mostly with fluorine and oxygen. Except for helium, each of these elements has eight electrons in its outermost occupied shell. Their outer shell may be represented as having the electron configuration . . . *ns*2*np*6.
2. **Representative Elements:** The A group elements in the periodic table are called representative elements. Their "last" electron is assigned to an outer shell *s* or *p* orbital. These elements show distinct and fairly regular variations in their properties with changes in atomic number.
3. **The *d*-Transition Elements:** Elements in the B groups in the periodic table are known as the *d*-transition elements or, more simply, as transition elements or transition metals. The elements of four transition series are all metals and are characterized by electrons being assigned to *d* orbitals. Stated differently, the *d*-transition elements contain electrons in both the *ns* and (*n* - 1)*d* orbitals, but not in the *np* orbitals. The first transition series, Sc through Zn, has electrons in the 4*s* and 3*d* orbitals, but not in the 4*p* orbitals. They are referred to as

First transition series:	$_{21}$Sc through $_{30}$Zn
Second transition series:	$_{39}$Y through $_{48}$Cd
Third transition series:	$_{57}$La and $_{72}$Hf through $_{80}$Hg
Fourth transition series:	$_{89}$Ac and $_{104}$Rf through element 112

4. ***f*-Transition Elements:** Sometimes known as *inner transition elements,* these are elements in which electrons are being added to *f* orbitals. In these elements, the second from the outermost occupied shell is building from 18 to 32 electrons. All are metals. The *f*-transition elements are located between Groups IIIB and IVB in the periodic table.

They are:
First *f*-transition series (lanthanides): ^{58}Ce through ^{71}Lu
Second *f*-transition series (actinides): ^{90}Th through ^{103}Lr

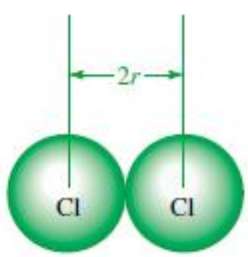
The A and B designations for groups of elements in the periodic table are somewhat arbitrary, and they are reversed in some periodic tables. In another designation, the groups are numbered 1 through 18. Elements with the same group numbers, but with different letters, have relatively few similar properties. The origin of the A and B designations is the fact that some compounds of elements with the same group numbers have similar formulas but quite different properties, for example, NaCl (IA) and AgCl (IB), $MgCl_2$ (IIA) and $ZnCl_2$ (IIB). As we shall see, variations in the properties of the B groups across a row are not nearly as regular and dramatic as the variations observed across a row of A group elements.

PERIODIC PROPERTIES OF THE ELEMENTS

Now we investigate the nature of periodicity. Knowledge of periodicity is valuable in understanding bonding in simple compounds. Many physical properties, such as melting points, boiling points, and atomic volumes, show periodic variations. For now, we describe the variations that are most useful in predicting chemical behavior. The variations in these properties depend on electron configurations, especially the configurations in the outermost occupied shell, and on how far away that shell is from the nucleus.

ATOMIC RADII

The size of an atom is determined by its immediate environment, especially its interaction with surrounding atoms. By analogy, suppose we arrange some golf balls in an orderly array in a box. If we know how the balls are positioned, the number of balls, and the dimensions of the box, we can calculate the diameter of an individual ball. Application of this reasoning to solids and their densities leads us to values for the atomic sizes of many elements. In other cases, we derive atomic radii from the observed distances between atoms that are combined with one another. For example, the distance between atomic centers (nuclei) in the Cl_2 molecule is measured to be 2.00 Å. This suggests that the radius of *each* Cl atom is half the interatomic distance, or 1.00 Å.



The radius of an atom, r, is taken as half of the distance between nuclei in *homonuclear* molecules such as Cl_2.

We collect the data obtained from many such measurements to indicate the *relative* sizes of individual atoms.

The top of Figure below displays the relative sizes of atoms of the representative elements and the noble gases. It shows the periodicity in atomic radii.

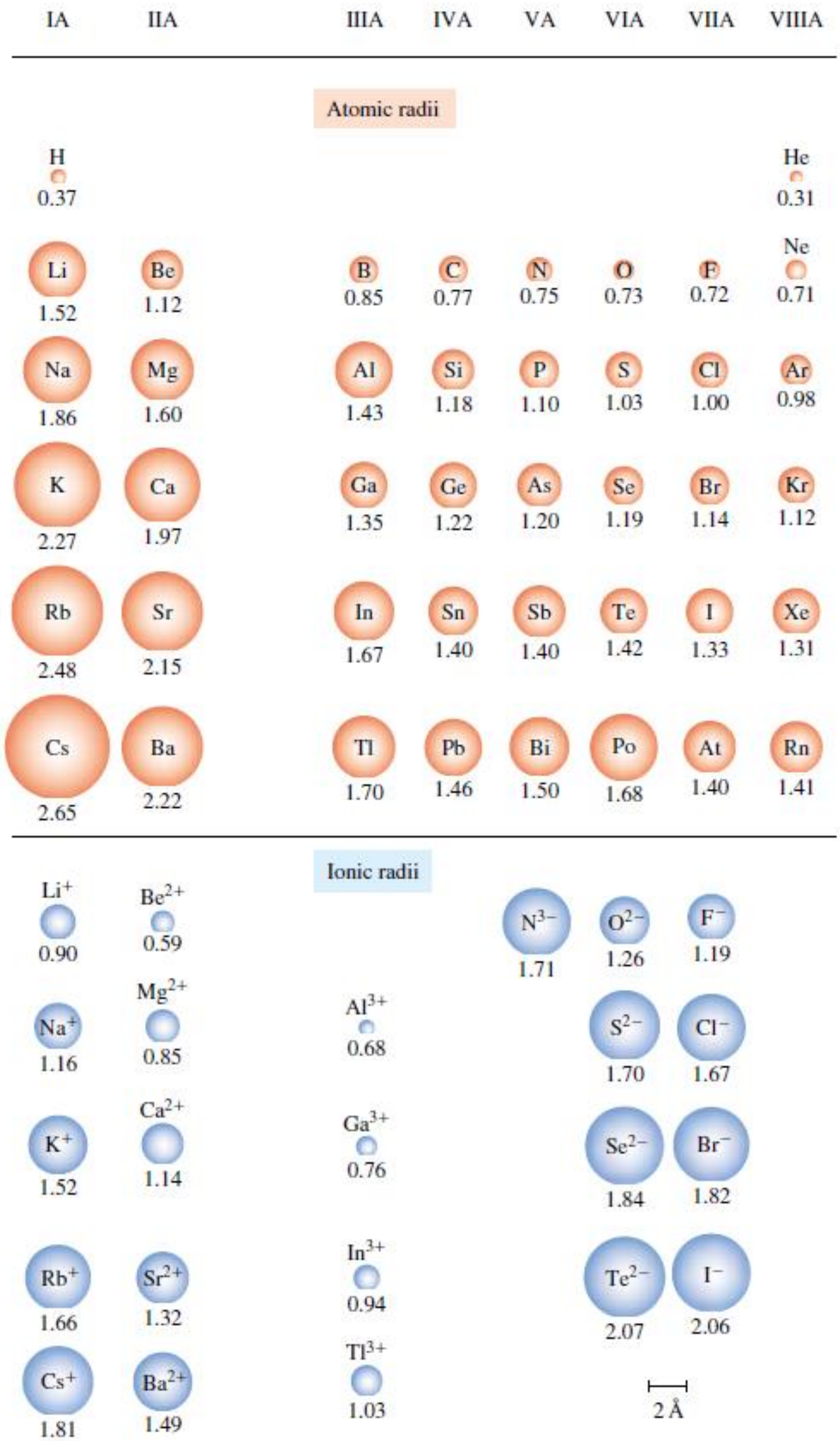


Figure 6-1 (*Top*) Atomic radii of the A group (representative) elements and the noble gases, in angstroms, Å (Section 6-2). Atomic radii *increase going down a group* because electrons are being added to shells farther from the nucleus. Atomic radii *decrease from left to right within a given period* owing to increasing effective nuclear charge. Hydrogen atoms

The **effective nuclear charge,** Z_{eff}, experienced by an electron in an outer shell is less than the actual nuclear charge, *Z.* This is because the *attraction* of outer-shell electrons by the nucleus is partly counterbalanced by the repulsion of these outer-shell electrons by electrons in inner shells. We say that the electrons in inner shells *screen,* or *shield,* electrons in outer shells from the full effect of the nuclear charge. This concept of a **screening,** or **shielding, effect** helps us understand many periodic trends in atomic properties. Consider an atom of lithium; it has two electrons in a filled shell, $1s^2$, and one electron in the 2*s* orbital, $2s^1$. The electron in the 2*s* orbital is fairly effectively screened from the nucleus by the two electrons in the filled 1*s* orbital, so the 2*s* electron does not "feel" the full +3 charge of the nucleus. The effective nuclear charge, Z_{eff}, experienced by the electron in the 2*s* orbital, however, is not 1 (3 minus 2) either. The electron in the outer shell of lithium has some probability of being found close to the nucleus.

We say that, to some extent, it *penetrates* the region of the 1*s* electrons; that is, the 1*s* electrons do not completely shield the outer-shell electrons from the nucleus. The electron in the 2*s* shell "feels" an effective nuclear charge a little larger than +1. Sodium, element number 11, has ten electrons in inner shells, $1s_2 2s_2 2p^6$, and one electron in an outer shell, $3s^1$. The ten inner-shell electrons of the sodium atom screen (shield) the outer-shell electron somewhat from the nucleus, counteracting some of the +11 nuclear charge. But the 3*s* electron of sodium penetrates the inner shells to a significant extent, so the effective nuclear charge felt by the outermost (3*s*) electron is actually greater than it is for lithium (2*s*). The somewhat increased attraction for the outermost electron in sodium is outweighed, however, by the fact that the "outer" electron in a sodium atom is in the third shell, whereas in lithium it is in the second shell. Recall from Chapter 5 that the third shell ($n = 3$) is farther from the nucleus than the second shell ($n = 2$). Thus, we see why sodium atoms are larger than lithium atoms. Similar reasoning explains why potassium atoms are larger than sodium atoms and why the sizes of the elements in each column of the periodic table are related in a similar way.

Atomic radii *increase going down a group* because electrons are being added to shells farther from the nucleus. Atomic radii *decrease from left to right within a given period* owing to increasing effective nuclear charge. Hydrogen atoms are the smallest and cesium atoms are the largest naturally occurring atoms.

Within a family (vertical group on the periodic table) of representative elements, atomic radii increase from top to bottom as electrons are added to shells farther from the nucleus.

As we move *across* the periodic table, atoms become smaller due to increasing effective nuclear charges. Consider the elements Boron, B ($Z = 5$, $1s^2 2s^2 2p^1$) to Fluorine, F ($Z = 9$, $1s2^2 s2^2 p^5$). In B there are two electrons in a noble gas configuration, $1s^2$, and three electrons in the second shell, $2s^2 2p^1$. The two electrons in the noble gas configuration fairly effectively screen out the effect of two protons in the nucleus. So the electrons in the second shell of B "feel" a greater effective nuclear charge than do those of Be. By similar arguments, we see that in carbon ($Z =6$, $1s^2 2s^2 2p^2$) the electrons in the second shell "feel" an effective nuclear charge greater than those of B. So we expect C atoms to be smaller than B atoms, and they are. In nitrogen ($Z= 7$, $1s^2 2s^2 2p^3$), the electrons in the second shell "feel" an even greater effective nuclear charge, and so N atoms are smaller than C atoms.

As we move from left to right *across a period* in the periodic table, atomic radii of representative elements *decrease* as a proton is added to the nucleus and an electron is added to a particular shell. For the transition elements, the variations are not so regular because electrons are being added to an inner shell. All transition elements have smaller radii than the preceding Group IA and IIA elements in the same period.

EXAMPLE *Trends in Atomic Radii*
Arrange the following elements in order of increasing atomic radii. Justify your order.
Cs, F, K, Cl

Plan
Both K and Cs are Group IA metals, whereas F and Cl are halogens (VIIA nonmetals). Atomic radii increase as we descend a group, so K <Cs and F < Cl. Atomic radii decrease from left to right.

Solution
The order of increasing atomic radii is F < Cl <K < Cs.

IONIZATION ENERGY
The **first ionization energy (IE1),** also called *first ionization potential*, is the minimum amount of energy required to remove the most loosely bound electron from an isolated gaseous atom to form an ion with a +1 charge.

First Ionization Energies (kJ/mol of atoms) of Some Elements

H 1312																	He 2372
Li 520	Be 899											B 801	C 1086	N 1402	O 1314	F 1681	Ne 2081
Na 496	Mg 738											Al 578	Si 786	P 1012	S 1000	Cl 1251	Ar 1521
K 419	Ca 599	Sc 631	Ti 658	V 650	Cr 652	Mn 717	Fe 759	Co 758	Ni 757	Cu 745	Zn 906	Ga 579	Ge 762	As 947	Se 941	Br 1140	Kr 1351
Rb 403	Sr 550	Y 617	Zr 661	Nb 664	Mo 685	Tc 702	Ru 711	Rh 720	Pd 804	Ag 731	Cd 868	In 558	Sn 709	Sb 834	Te 869	I 1008	Xe 1170
Cs 377	Ba 503	La 538	Hf 681	Ta 761	W 770	Re 760	Os 840	Ir 880	Pt 870	Au 890	Hg 1007	Tl 589	Pb 715	Bi 703	Po 812	At 890	Rn 1037

For calcium, for example, the first ionization energy, IE1, is 590 kJ/mol:

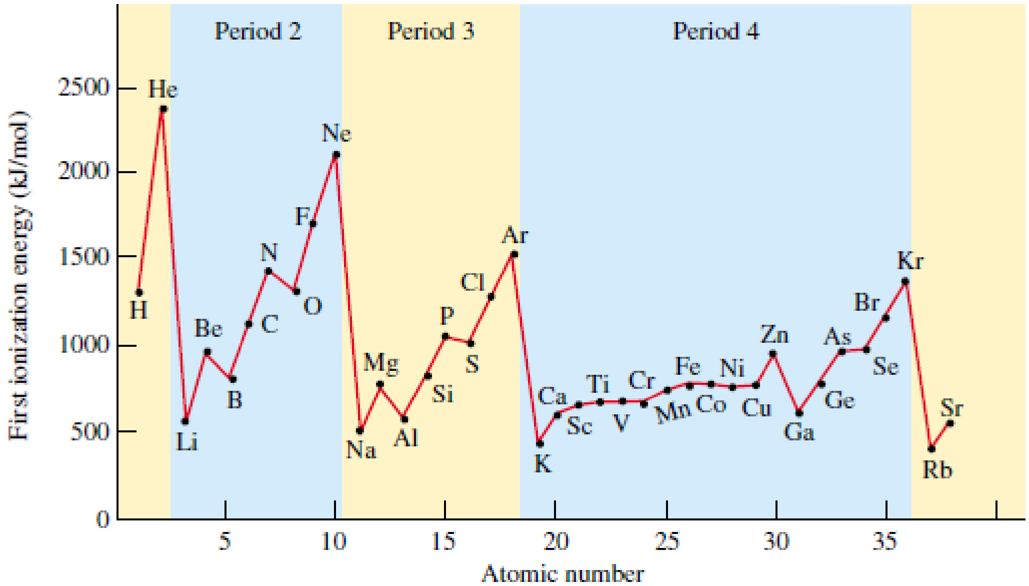
$$Ca(g) + 590\ kJ \longrightarrow Ca^{+}(g) + e^{-}$$

The **second ionization energy (IE2)** is the amount of energy required to remove the second electron. For calcium, it may be represented as

$$Ca^{+}(g) + 1145\ kJ \longrightarrow Ca^{2+}(g) + e^{-}$$

For a given element, *IE_2 is always greater than IE1* because it is always more difficult to remove an electron from a positively charged ion than from the corresponding neutral atom. The Table above gives first ionization energies.

Ionization energies measure how tightly electrons are bound to atoms. Ionization always requires energy to remove an electron from the attractive force of the nucleus. Low ionization energies indicate ease of removal of electrons, and hence ease of positive ion (cation) formation. Figure below shows a plot of first ionization energy versus atomic number for several elements.



Elements with low ionization energies (IE) lose electrons easily to form cations. We see that in each period of Figure above, the noble gases have the highest first ionization energies. This should not be surprising, because the noble gases are known to be very unreactive elements. It requires more energy to remove an electron from a helium atom (slightly less than 4.0 $\times 10^{-18}$ J/atom, or 2372 kJ/mol) than to remove one from a neutral atom of any other element.

$$He(g) + 2372\ kJ \longrightarrow He^{+}(g) + e^{-}$$

The Group IA metals (Li, Na, K, Rb, Cs) have very low first ionization energies. Each of these elements has only one electron in its outermost shell (. . . *ns*1), and they are the largest atoms in their periods. The first electron added to a shell is easily removed to form a noble gas configuration. As we move down the group, the first ionization energies become smaller. The force of attraction of the positively charged nucleus for electrons decreases as the square of the distance between them increases. So as atomic radii increase in a given group, first ionization energies decrease because the outermost electrons are farther from the nucleus.

Effective nuclear charge, Z_{eff}, increases going from left to right across a period. The increase in effective nuclear charge causes the outermost electrons to be held more tightly, making them harder to remove. The first ionization energies therefore generally *increase* from left to right across the periodic table. The reason for the trend in first ionization energies is the same as that used to explain trends in atomic radii. The first ionization energies of the Group IIA elements (Be, Mg, Ca, Sr, Ba) are significantly higher than those of the Group IA elements in the same periods. This is because the Group IIA elements have higher Z_{eff} values and smaller atomic radii. Thus, their outermost electrons are held more tightly than those of the neighboring IA metals. It is harder to remove an electron from a pair in the filled outermost *s* orbitals of the Group IIA elements than to remove the single electron from the half-filled outermost *s* orbitals of the Group IA elements.

The first ionization energies for the Group IIIA elements (B, Al, Ga, In, Tl) are exceptions to the general horizontal trends. They are *lower* than those of the IIA elements in the same periods because the IIIA elements have only a single electron in their outermost *p* orbitals. Less energy is required to remove the first *p* electron than the second *s* electron from the outermost shell, because the *p* orbital is at a higher energy (less stable) than an *s* orbital within the same shell (*n* value).

Going from Groups IIIA to VA, electrons are going singly into separate *np* orbitals, where they do not shield one another significantly. The general left-to-right increase in IE1 for each period is interrupted by a dip between Groups VA (N, P, As, Sb, Bi) and VIA elements (O, S, Se, Te, Po). Presumably, this behavior is because the fourth *np* electron in the Group VIA elements is paired with another electron in the same orbital, so it experiences greater repulsion than it would in an orbital by itself. This increased repulsion apparently outweighs the increase in Z_{eff}, so the fourth *np* electron in an outer shell (Group VIA elements) is somewhat easier to remove (lower ionization energy) than is the third *np* electron in an outer shell (Group VA elements). After the dip between Groups VA and VIA, the importance of the increasing Z_{eff} outweighs the repulsion of electrons needing to be paired, and the general left-to-right increases in first ionization energies resume.

Knowledge of the relative values of ionization energies assists us in predicting whether an element is likely to form ionic or molecular (covalent) compounds. Elements with low ionization energies form ionic compounds by losing electrons to form **cations** (positively charged ions). Elements with intermediate ionization energies generally form molecular compounds by sharing electrons with other elements. Elements with very high ionization energies, such as Groups VIA and VIIA, often gain electrons to form **anions** (negatively charged ions).

One factor that favors an atom of a *representative* element forming a monatomic ion in a compound is the formation of a stable noble gas electron configuration. Energy considerations are consistent with this observation. For example, as one mole of Li from Group IA forms one mole of Li^+ ions, it absorbs 520 kJ per mole of Li atoms. The IE2 value is 14 times greater, 7298 kJ/mol, and is prohibitively large for the formation of Li^{2+} ions under ordinary conditions. For Li^{2+} ions to form, an electron would have to be removed from the filled first shell. We recognize that this is unlikely. The other alkali metals behave in the same way, for similar reasons.

The first two ionization energies of Be (Group IIA) are 899 and 1757 kJ/mol, but IE3 is more than eight times larger, 14,849 kJ/mol. So Be forms Be^{2+} ions, but not Be^{3+} ions.

The other alkaline earth metals—Mg, Ca, Sr, Ba, and Ra—behave in a similar way. Only the lower members of Group IIIA, beginning with Al, form +3 ions. Bi and some *d*- and *f*-transition metals do so, too.

EXAMPLE *Trends in First IEs*
Arrange the following elements in order of increasing first ionization energy. Justify your order.
Na, Mg, Al, Si
Plan
Table above shows that first ionization energies generally increase from left to right in the same period, but there are exceptions at Groups IIIA and VIA. Al is a IIIA element with only one electron in its outer *p* orbitals, $1s^2 2s^2 2p^6 3s^2 3p^1$.
Solution
There is a slight dip at Group IIIA in the plot of first IE versus atomic number. The order of increasing first ionization energy is Na < Al <Mg< Si.

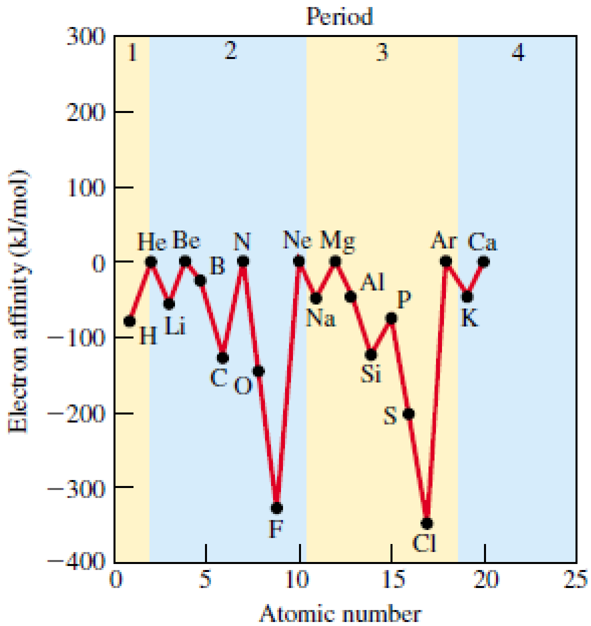
ELECTRON AFFINITY

The **electron affinity (EA)** of an element may be defined as the amount of energy *absorbed* when an electron is added to an isolated gaseous atom to form an ion with a -1 charge. The convention is to assign a positive value when energy is absorbed and a negative value when energy is released. Most elements have no affinity for an additional electron and thus have an electron affinity (EA) equal to zero. We can represent the electron affinities of helium and chlorine as

$$He(g) + e^- \not\longrightarrow He^-(g) \qquad EA = \quad 0\ kJ/mol$$

$$Cl(g) + e^- \longrightarrow Cl^-(g) + 349\ kJ \qquad EA = -349\ kJ/mol$$

The first equation tells us that helium will not add an electron. The second equation tells us that when one mole of gaseous chlorine atoms gain one electron each to form gaseous chloride ions, 349 kJ of energy is *released (exothermic).* Figure below shows a plot of electron affinity versus atomic number for several elements.



Electron affinity involves the *addition* of an electron to a neutral gaseous atom. The process by which a neutral atom X gains an electron (EA),

$$X(g) + e^- \longrightarrow X^-(g) \qquad (EA)$$

is *not* the reverse of the ionization process,

$$X^+(g) + e^- \longrightarrow X(g) \qquad (\text{reverse of } IE_1)$$

The first process begins with a neutral atom, whereas the second begins with a positive ion. Thus, IE1 and EA are *not* simply equal in value with the signs reversed. We see from Figure above that electron affinities generally become more negative from left to right across a row in the periodic table (excluding the noble gases). This means that most representative elements in

Groups IA to VIIA show a greater attraction for an extra electron from left to right. Halogen atoms, which have the outer electron configuration *ns*2*np*5, have the most negative electron affinities. They form stable anions with noble gas configurations, . . . ns^2np^6, by gaining one electron.

Electron Affinity Values (kJ/mol) of Some Elements*

1	H −73								He 0
2	Li −60	Be (−0)		B −29	C −122	N 0	O −141	F −328	Ne 0
3	Na −53	Mg (−0)	Cu −118	Al −43	Si −134	P −72	S −200	Cl −349	Ar 0
4	K −48	Ca (−0)	Ag −125	Ga −29	Ge −119	As −78	Se −195	Br −324	Kr 0
5	Rb −47	Sr (−0)	Au −282	In −29	Sn −107	Sb −101	Te −190	I −295	Xe 0
6	Cs −45	Ba (−0)		Tl −19	Pb −35	Bi −91			

**Estimated values are in parentheses.*

Elements with very negative electron affinities gain electrons easily to form negative ions (anions). “Electron affinity” is a precise and quantitative term, like “ionization energy,” but it is difficult to measure. The Table above shows electron affinities for several elements.

For many reasons, the variations in electron affinities are not regular across a period. The general trend is: the electron affinities of the elements become more negative from left to right in each period. Noteworthy exceptions are the elements of Groups IIA and VA, which have less negative values than the trends suggest. It is very difficult to add an electron to a IIA metal atom because its outer *s* subshell is filled. The values for the VA elements are slightly less negative than expected because they apply to the addition of an electron to a half-filled set of *np* orbitals ($ns^2np^3 \rightarrow ns^2np^4$), which requires pairing. The resulting repulsion overcomes the increased attractive force of the nucleus.

Energy is always required to bring a negative charge (electron) closer to another negative charge (anion). So the addition of a second electron to a -1 anion to form an ion with a -2 charge is always endothermic. Thus, electron affinities of *anions* are always positive.

EXAMPLE *Trends in EAs*

Arrange the following elements in order of increasing values of electron affinity, that is, from most negative to least negative: K, Br, Cs, Cl.

Plan

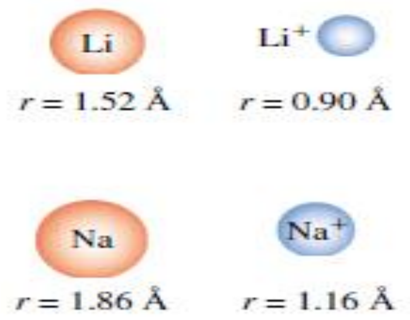
Table above shows that electron affinity values generally become more negative from left to right across a period, with major exceptions at Groups IIA (Be) and VA (N). They generally become more negative from bottom to top.

Solution

The order of increasing values of electron affinity is (most negative EA) Cl <Br < K <Cs (least negative EA)

IONIC RADII

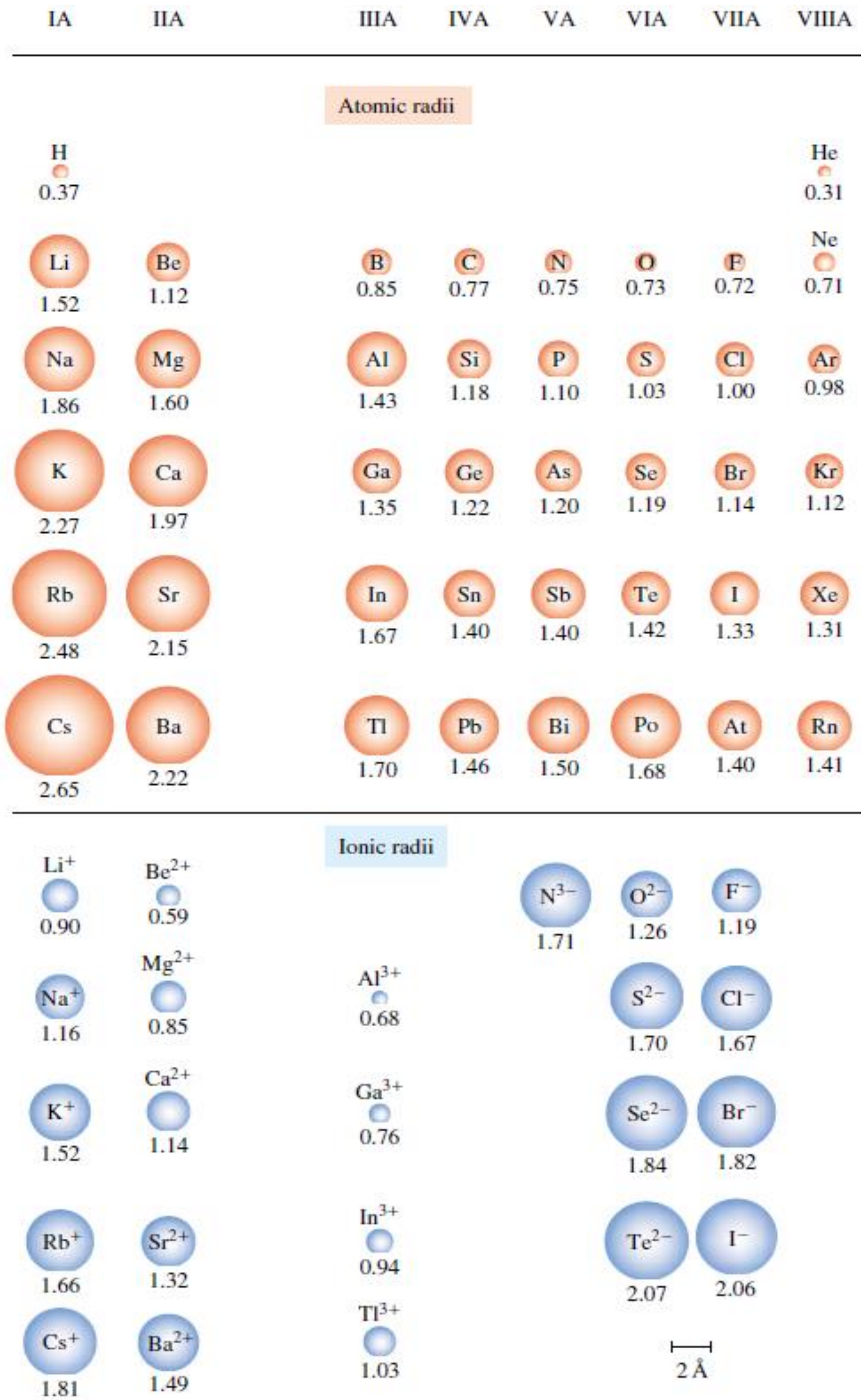
Many elements on the left side of the periodic table react with other elements by *losing* electrons to form positively charged ions. Each of the Group IA elements (Li, Na, K, Rb, Cs) has only one electron in its outermost shell (electron configuration . . . *ns*1).

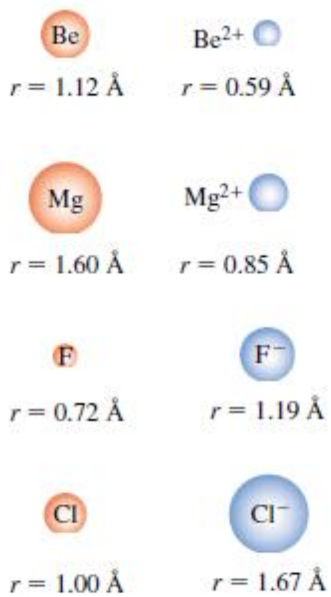


The nuclear charge remains constant when the ion is formed.

These elements react with other elements by losing one electron to attain noble gas configurations.

They form the ions Li^+, Na^+, K^+, Rb^+, and Cs^+. A neutral lithium atom, Li, contains three protons in its nucleus and three electrons, with its outermost electron in the 2*s* orbital. A lithium ion, Li^+, however, contains three protons in its nucleus but only two electrons, both in the 1*s* orbital. So a Li^+ ion is much smaller than a neutral Li atom. Likewise, a sodium ion, Na^+, is considerably smaller than a sodium atom, Na. The relative sizes of atoms and common ions of some representative elements are shown in Figure below.


IA IIA IIIA IVA VA VIA VIIA VIIIA
Atomic radii
H 0.37
He 0.31
Li 1.52
Be 1.12
B 0.85
C 0.77
N 0.75
O 0.73
F 0.72
Ne 0.71
Na 1.86
Mg 1.60
Al 1.43
Si 1.18
P 1.10
S 1.03
Cl 1.00
Ar 0.98
K 2.27
Ca 1.97
Ga 1.35
Ge 1.22
As 1.20
Se 1.19
Br 1.14
Kr 1.12
Rb 2.48
Sr 2.15
In 1.67
Sn 1.40
Sb 1.40
Te 1.42
I 1.33
Xe 1.31
Cs 2.65
Ba 2.22
Tl 1.70
Pb 1.46
Bi 1.50
Po 1.68
At 1.40
Rn 1.41
Ionic radii
Li^{+} 0.90
Be^{2+} 0.59
N^{3-} 1.71
O^{2-} 1.26
F^{-} 1.19
Na^{+} 1.16
Mg^{2+} 0.85
Al^{3+} 0.68
S^{2-} 1.70
Cl^{-} 1.67
K^{+} 1.52
Ca^{2+} 1.14
Ga^{3+} 0.76
Se^{2-} 1.84
Br^{-} 1.82
Rb^{+} 1.66
Sr^{2+} 1.32
In^{3+} 0.94
Te^{2-} 2.07
I^{-} 2.06
Cs^{+} 1.81
Ba^{2+} 1.49
Tl^{3+} 1.03
2 Å



Isoelectronic species have the same number of electrons. We see that the ions formed by the Group IIA elements (Be^{2+}, Mg^{2+}, Ca^{2+}, Sr^{2+}, Ba^{2+}) are significantly smaller than the *isoelectronic* ions formed by the Group IA elements in the same period. The radius of the Li^+ ion is 0.90 Å, whereas the radius of the Be^{2+} ion is only 0.59 Å. This is what we should expect. A beryllium ion, Be^{2+}, is formed when a beryllium atom, Be, loses both of its 2*s* electrons while the 4+ nuclear charge remains constant. We expect the 4+ nuclear charge in Be^{2+} to attract the remaining two electrons quite strongly. Comparison of the ionic radii of the IIA elements with their atomic radii indicates the validity of our reasoning. Similar reasoning indicates that the ions of the Group IIIA metals (Al^{3+}, Ga^{3+}, In^{3+}, $Tl^{3+)}$ should be even smaller than the ions of Group IA and Group IIA elements in the same periods. Now consider the Group VIIA elements (F, Cl, Br, I). These have the outermost electron configuration . . . *ns* 2*np*^{5}. These elements can completely fill their outermost *p* orbitals by *gaining* one electron to attain noble gas configurations. Thus, when a fluorine atom (with seven electrons in its outer shell) gains one electron, it becomes a fluoride ion, F^-, with eight electrons in its outer shell. These eight electrons repel one another more strongly than the original seven, so the electron cloud expands. The F^- ion is much larger than the neutral F atom (see figure in the margin). Similar reasoning indicates that a chloride ion, Cl^-, should be larger than a neutral chlorine atom, Cl. Observed ionic radii verify this prediction.

Comparing the sizes of an oxygen atom (Group VIA) and an oxide ion, O^{2-}, again we find that the negatively charged ion is larger than the neutral atom. The oxide ion is also larger than the isoelectronic fluoride ion because the oxide ion contains ten electrons held by a nuclear charge of only -8, whereas the fluoride ion has ten electrons held by a nuclear charge of -9. Comparison of radii is not a simple issue when we try to compare atoms, positive and negative ions, and ions with varying charge. Sometimes we compare atoms to their ions, atoms or ions that are vertically or horizontally positioned on the periodic table, or isoelectronic species. The following guidelines are often considered in the order given.

1. Simple positively charged ions (cations) are always smaller than the neutral atoms from which they are formed.

2. Simple negatively charged ions (anions) are always larger than the neutral atoms from which they are formed.
3. The sizes of cations decrease from left to right across a period.
4. The sizes of anions decrease from left to right across a period.
5. Within an isoelectronic series, radii decrease with increasing atomic number because of increasing nuclear charge.
6. Both cation and anion sizes increase going down a group.

An isoelectronic series of ions

	N^{3-}	O^{2-}	F^{-}	Na^{+}	Mg^{2+}	Al^{3+}
Ionic radius (Å)	1.71	1.26	1.19	1.16	0.85	0.68
No. of electrons	10	10	10	10	10	10
Nuclear charge	+7	+8	+9	+11	+12	+13

EXAMPLE *Trends in Ionic Radii*
Arrange the following ions in order of increasing ionic radii: (a) Ca^{2+}, K^{+}, Al^{3+}; (b) Se^{2-}, Br^{-}, Te^{2}.
Plan
Some of the pairs of ions are isoelectronic, so we can compare their sizes on the basis of nuclear charges. Other comparisons can be made based on the outermost occupied shell (highest value of *n*).
Solution
(a) Ca^{2+} and K^{+} are isoelectronic (18 electrons each) with an outer-shell electron configuration of $3s^23p^6$. Because Ca^{2+} has a higher nuclear charge (+20) than K^{+} (+19), Ca^{2+}holds its 18 electrons more tightly, and Ca^{2+} is smaller than K^{+}. Al^{3+} has electrons only in the second main shell (outer-shell electron configuration of $2s^22p^6$), so it is smaller than either of the other two ions.

$Al^{3+} < Ca^{2+} < K^{+}$

(b) Br^{-} and Se^{2-} are isoelectronic (36 electrons each) with an outer-shell electron configuration of $4s^24p^6$. Because Br^{-} has a higher nuclear charge (+35) than Se^{2+} (+34), Br^{-} holds its 36 electrons more tightly, and Br^{-} is smaller than Se^{2-}. Te^{2-} has electrons in the fifth main shell (outer configuration of $5s^25p^6$), so it is larger than either of the other two ions.

$Br^{-} < Se^{2-} < Te^{2-}$

Example: Give the reasons why:

i. The radii of cations are not bigger than their parent atoms;
ii. The radii of anions are not smaller than their parent atoms.

(i) The ionic radius of a cation is smaller than the radius of the atom from which it is formed. It is because after losing the outermost shell electron(s) in forming the positive ion(or cation), the number of electronic shells decreases. Also, the number of protons is greater than the number of electrons, therefore the nuclear attractive force for the remaining electrons must be stronger than that in its neutral atom resulting in contraction of size.

(ii) The radii of anions are larger than their parent atoms: The ionic radius of an anion is larger than the radius of the atom from which it is formed. This is due to the repulsion between the newly added electron(s) and the electrons already present. Also, the number of protons is smaller than the number of protons in an anion, so the nuclear attractive force for the electron cloud becomes weakened resulting in the increase or expansion of size.

ELECTRONEGATIVITY

The **electronegativity (EN)** of an element is a measure of the relative tendency of an atom to attract electrons to itself *when it is chemically combined with another atom.* Elements with high electronegativities (nonmetals) often gain electrons to form anions. Elements with low electronegativities (metals) often lose electrons to form cations.

Electronegativity Values of the Elements

Metals / Nonmetals / Metalloids

	IA	IIA	IIIB	IVB	VB	VIB	VIIB	VIIIB	VIIIB	VIIIB	IB	IIB	IIIA	IVA	VA	VIA	VIIA	VIIIA
1	1 H 2.1																	2 He
2	3 Li 1.0	4 Be 1.5											5 B 2.0	6 C 2.5	7 N 3.0	8 O 3.5	9 F 4.0	10 Ne
3	11 Na 1.0	12 Mg 1.2											13 Al 1.5	14 Si 1.8	15 P 2.1	16 S 2.5	17 Cl 3.0	18 Ar
4	19 K 0.9	20 Ca 1.0	21 Sc 1.3	22 Ti 1.4	23 V 1.5	24 Cr 1.6	25 Mn 1.6	26 Fe 1.7	27 Co 1.7	28 Ni 1.8	29 Cu 1.8	30 Zn 1.6	31 Ga 1.7	32 Ge 1.9	33 As 2.1	34 Se 2.4	35 Br 2.8	36 Kr
5	37 Rb 0.9	38 Sr 1.0	39 Y 1.2	40 Zr 1.3	41 Nb 1.5	42 Mo 1.6	43 Tc 1.7	44 Ru 1.8	45 Rh 1.8	46 Pd 1.8	47 Ag 1.6	48 Cd 1.6	49 In 1.6	50 Sn 1.8	51 Sb 1.9	52 Te 2.1	53 I 2.5	54 Xe
6	55 Cs 0.8	56 Ba 1.0	57 La 1.1 *	72 Hf 1.3	73 Ta 1.4	74 W 1.5	75 Re 1.7	76 Os 1.9	77 Ir 1.9	78 Pt 1.8	79 Au 1.9	80 Hg 1.7	81 Tl 1.6	82 Pb 1.7	83 Bi 1.8	84 Po 1.9	85 At 2.1	86 Rn
7	87 Fr 0.8	88 Ra 1.0	89 Ac 1.1 †															

*	58 Ce 1.1	59 Pr 1.1	60 Nd 1.1	61 Pm 1.1	62 Sm 1.1	63 Eu 1.1	64 Gd 1.1	65 Tb 1.1	66 Dy 1.1	67 Ho 1.1	68 Er 1.1	69 Tm 1.1	70 Yb 1.0	71 Lu 1.2
†	90 Th 1.2	91 Pa 1.3	92 U 1.5	93 Np 1.3	94 Pu 1.3	95 Am 1.3	96 Cm 1.3	97 Bk 1.3	98 Cf 1.3	99 Es 1.3	100 Fm 1.3	101 Md 1.3	102 No 1.3	103 Lr 1.5

[1]*Electronegativity values are given at the bottoms of the boxes.*

Electronegativities of the elements are expressed on a somewhat arbitrary scale, called the Pauling scale.

The electronegativity of fluorine (4.0) is higher than that of any other element. This tells us that when fluorine is chemically bonded to other elements, it has a greater tendency to attract electron density to itself than does any other element.

Oxygen is the second most electronegative element. For the representative elements, electronegativities usually increase from left to right across periods and decrease from top to bottom within groups. Variations among the transition metals are not as regular. In general, both ionization energies and electronegativities are low for elements at the lower left of the periodic table and high for those at the upper right.

EXAMPLE *Trends in ENs*
Arrange the following elements in order of increasing electronegativity.
B, Na, F, O
Plan
The Table above shows that electronegativities increase from left to right across a period and decrease from top to bottom within a group.
Solution
The order of increasing electronegativity is $Na < B < O < F$.
Although the electronegativity scale is somewhat arbitrary, we can use it with reasonable confidence to make predictions about bonding. Two elements with quite different electronegativities (a metal and a nonmetal) tend to react with each other to form ionic compounds. The less electronegative element gives up its electron(s) to the more electronegative element. Two nonmetals with similar electronegativities tend to form covalent bonds with each other. That is, they share their electrons. In this sharing, the more electronegative element attains a greater share.

EXERCISES

Classification of the Elements

1. Define and illustrate the following terms clearly and concisely: (a) representative elements; (b) *d*-transition elements; (c) inner transition elements.
2. Explain why Period 1 contains two elements and Period 2 contains eight elements.
3. Explain why Period 4 contains 18 elements.
4. The third shell (n = 3) has *s*, *p*, and *d* subshells. Why does Period 3 contain only eight elements?
5. Account for the number of elements in Period 6.
6. What would be the atomic number of the as-yetundiscovered alkaline earth element of Period 8?
7. Identify the group, family, or other periodic table location of each element with the outer electron configuration
   (a) ns^2np^4
   (b) ns^2
   (c) $ns^2(n-1)d^{0-2}(n-2)f^{1-14}$
8. Write the outer electron configurations for the (a) alkaline earth metals; (b) first column of *d*-transition metals; and (c) halogens.
9. Which of the elements in the following periodic table is (are) (a) alkali metals; (b) an element with the outer configuration of d^7s^2; (c) lanthanides; (d) *p*-block representative

elements; (e) elements with partially filled *f*-subshells; (f) halogens; (g) *s*-block representative elemen (h) actinides; (i) *d*-transition elements; (j) noble gases?

10. Identify the elements and the part of the periodic table in which the elements with the following configurations are found.
    (a) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
    (b) [Kr] $4d^8 5s^2$
    (c) [Xe] $4f^{14} 5d^5 6s^1$
    (d) [Xe] $4f^{12} 6s^2$
    (e) [Kr] $4d^{10} 5s^2 5p^3$
    (f) [Kr] $4d^{10} 4f^{14} 5s^2 5p^6 5d^{10} 6s^2 6p^2$
11. Which of the following species are iso-electronic?
    O^{2-}, F^-, Ne, Na, Mg^{2+}, Al^{3+}
12. Which of the following species are iso-electronic?
    P^{3+}, S^{2-}, Cl^-, Ar, K^+, Ca^+

Atomic Radii

1. What is meant by nuclear shielding? What effect does it have on trends in atomic radii?
2. Why do atomic radii decrease from left to right within a period in the periodic table?
3. Consider the elements in Group VIA. Even though it has not been isolated or synthesized, what can be predicted about the ionic radius of element number 116?
4. Variations in the atomic radii of the transition elements are not so pronounced as those of the representative elements. Why?
5. Arrange each of the following sets of atoms in order of increasing atomic radii: (a) the alkaline earth elements; (b) the noble gases; (c) the representative elements in the third period; (d) N, Ba, B, Sr, and Sb.
6. Arrange each of the following sets of atoms in order of increasing atomic volume: (a) O, Mg, Al, Si; (b) O, S, Se,Te; (c) Ca, Sr, Ga, As.

Ionization Energy

1. Define (a) first ionization energy and (b) second ionization energy.
2. Why is the second ionization energy for a given element always greater than the first ionization energy?
3. What is the usual relationship between atomic radius and first ionization energy, other factors being equal?
4. What is the usual relationship between nuclear charge and first ionization energy, other factors being equal?
5. Going across a period on the periodic table, what is the relationship between shielding and first ionization energy?
6. Within a group on the periodic table, what is the relationship between shielding and first ionization energy?
7. Arrange the members of each of the following sets of elements in order of increasing first ionization energies: (a) the alkali metals; (b) the halogens; (c) the elements in the second period; (d) Br, F, B, Ga, Cs, and H.
8. The second ionization energy value is generally much larger than the first ionization energy. Explain why this is so.

9. What is the general relationship between the sizes of the atoms of Period 2 and their first ionization energies? Rationalize the relationship.
10. In a plot of first ionization energy versus atomic number for Periods 2 and 3, “dips” occur at the IIIA and VIA elements. Account for these dips.
11. ***30.** On the basis of electron configurations, would you expect a Na^{2+} ion to exist in compounds? Why or why not? How about Mg^{2+}?

Electron Affinity

1. Arrange the following elements in order of increasing negative values of electron affinity: P, S, Cl, and Br.
2. Arrange the members of each of the following sets of elements in order of increasingly negative electron affinities: (a) the Group IA metals; (b) the Group VIIA elements; (c) the elements in the second period; (d) Li, K, C, F, and Cl.
3. The electron affinities of the halogens are much more negative than those of the Group VIA elements. Why is this so?
4. The addition of a second electron to form an ion with a -2 charge is always endothermic. Why is this so?
5. Write the equation for the change described by each of the following, and write the electron configuration for each atom or ion shown: (a) the electron affinity of oxygen; (b) the electron affinity of chlorine; (c) the electron affinity of magnesium.

Ionic Radii

1. Compare the sizes of cations and the neutral atoms from which they are formed by citing three specific examples.
2. Arrange the members of each of the following sets of cations in order of increasing ionic radii: (a) K^+, Ca^{2+},
   Ga^{3+}; (b) Ca^{2+}, Be^{2+}, Ba^{2+}, Mg^{2+}; (c) Al^{3+}, Sr^{2+}, Rb^+,
   K^+; (d) K^+, Ca^{2+}, Rb^+.
3. Compare the sizes of anions and the neutral atoms from which they are formed by citing three specific examples.
4. Explain the trend in size of either the atom or ion as one moves down a group of periodic table.
5. Most transition metals can form more than one simple positive ion. For example, iron forms both Fe^{2+} and Fe^{3+} ions, and tin forms both Sn^{2+} and Sn^{4+} ions. Which is the smaller ion of each pair, and why?

Electronegativity

1. What is electronegativity?
2. Arrange the members of each of the following sets of elements
3. in order of increasing electronegativities: (a) Pb, C, Sn, Ge; (b) S, Na, Mg, Cl; (c) P, N, Sb, Bi; (d) Se, Ba, F, Si, Sc.
4. Which of the following statements is better? Why? (a) Magnesium has a weak attraction for electrons in a chemical bond because it has a low electronegativity. (b) The electronegativity of magnesium is low because magnesium has a weak attraction for electrons in a chemical bond.

CHEMICAL BONDING

Chemical bonding refers to the attractive forces that hold atoms together in compounds. There are two major classes of bonding. (1) **Ionic bonding** results from electrostatic interactions among ions, which often results from the net *transfer* of one or more electrons from one atom or group of atoms to another. (2) **Covalent bonding** results from *sharing* one or more electron pairs between two atoms. These two classes represent two extremes; all bonds between atoms of different elements have at least some degree of both ionic and covalent character. Compounds containing predominantly ionic bonding are called **ionic compounds.** Those that are held together mainly by covalent bonds are called **covalent compounds.** Some nonmetallic elements, such as H_2, Cl_2, N_2, and P_4, also involve covalent bonding. Some properties usually associated with many simple ionic and covalent compounds are summarized in the following list. The differences in these properties can be accounted for by the differences in bonding between the atoms or ions.

Ionic Compounds	Covalent Compounds
1. They are solids with high melting points (typically >400°C).	1. They are gases, liquids, or solids with low melting points (typically <300°C).
2. Many are soluble in polar solvents such as water.	2. Many are insoluble in polar solvents.
3. Most are insoluble in nonpolar solvents, such as hexane, C_6H_{14}, and carbon tetrachloride, CCl_4.	3. Most are soluble in nonpolar solvents, such as hexane, C_6H_{14}, and carbon tetrachloride, CCl_4.
4. Molten compounds conduct electricity well because they contain mobile charged particles (ions).	4. Liquid and molten compounds do not conduct electricity.
5. Aqueous solutions conduct electricity well because they contain mobile charged particles (ions).	5. Aqueous solutions are *usually* poor conductors of electricity because most do not contain charged particles.
6. They are often formed between two elements with quite different electronegativities, usually a metal and a nonmetal.	6. They are often formed between two elements with similar electronegativities, usually nonmetals.

LEWIS DOT FORMULAS OF ATOMS

The number and arrangements of electrons in the outermost shells of atoms determine the chemical and physical properties of the elements as well as the kinds of chemical bonds they form. We write **Lewis dot formulas** (or **Lewis dot representations,** or just **Lewis formulas**) as a convenient bookkeeping method for keeping track of these "chemically important electrons."

Lewis Dot Formulas for Representative Elements

Group	IA	IIA	IIIA	IVA	VA	VIA	VIIA	VIIIA
Number of electrons in valence shell	1	2	3	4	5	6	7	8 (*except He*)
Period 1	H ·							He :
Period 2	Li ·	Be :	B ·	C ·	· N ·	· O :	· F :	: Ne :
Period 3	Na ·	Mg :	Al ·	Si ·	· P ·	· S :	· Cl :	: Ar :
Period 4	K ·	Ca :	Ga ·	Ge ·	· As ·	· Se :	· Br :	: Kr :
Period 5	Rb ·	Sr :	In ·	Sn ·	· Sb ·	· Te :	· I :	: Xe :
Period 6	Cs ·	Ba :	Tl ·	Pb ·	· Bi ·	· Po :	· At :	: Rn :
Period 7	Fr ·	Ra :						

Chemical bonding usually involves only the outermost electrons of atoms, also called **valence electrons.** In Lewis dot representations, only the electrons in the outermost occupied *s* and *p* orbitals are shown as dots. Paired and unpaired electrons are also indicated. The Table above shows Lewis dot formulas for the representative elements. All elements in a given group have the same outer-shell electron configuration. It is somewhat arbitrary on which side of the atom symbol we write the electron dots. We do, however, represent an electron pair as a pair of dots and an unpaired electron as a single dot.

IONIC BONDING

FORMATION OF IONIC COMPOUNDS

An **ion** is an atom or a group of atoms that carries an electrical charge. An ion in which the atom or group of atoms has fewer electrons than protons is positively charged, and is called a **cation;** one that has more electrons than protons is negatively charged, and is called an **anion.** An ion that consists of only one atom is described as a **monatomic ion.** Examples include the chloride ion, Cl^-, and the magnesium ion, Mg^{2+}. An ion that contains more than one atom is called a **polyatomic ion.** Examples include the ammonium ion, NH_4^+; the hydroxide ion, OH^-; and the sulfate ion, SO_4^{2-}. The atoms of a polyatomic ion are held together by covalent bonds. In this section we shall discuss how ions can be formed from individual atoms; polyatomic ions will be discussed along with other covalently bonded species.

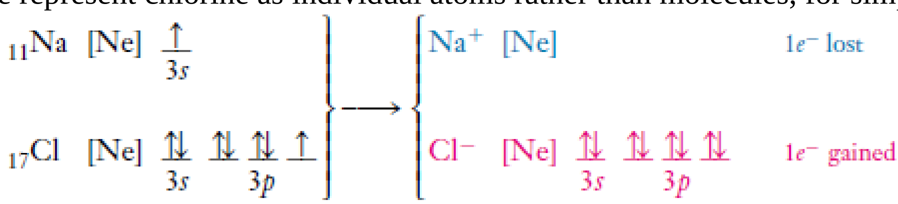
Ionic bonding (electrovalent bonding) is the attraction of oppositely charged ions (cations and anions) in large numbers to form a solid. Such a solid compound is called an *ionic solid.* As our previous discussions of ionization energy, electronegativity, and electron affinity would suggest, ionic bonding can occur easily when elements that have low ionization energies (metals) react with

elements having high electronegativities and very negative electron affinities (nonmetals). Many metals are easily *oxidized*—that is, they lose electrons to form cations; and many nonmetals are readily *reduced*—that is, they gain electrons to form anions. When the electronegativity difference, Δ(EN), between two elements is large, as between a metal and a nonmetal, the elements are likely to form a compound by ionic bonding (transfer of electrons).
Let us describe some combinations of metals with nonmetals to form ionic compounds.

Group IA Metals and Group VIIA Nonmetals
Consider the reaction of sodium (a Group IA metal) with chlorine (a Group VIIA nonmetal). Sodium is a soft silvery metal (mp 98 °C), and chlorine is a yellowish-green corrosive gas at room temperature. Both sodium and chlorine react with water, sodium vigorously. By contrast, sodium chloride is a white solid (mp 801 °C) that dissolves in water with no reaction and with the absorption of just a little heat. We can represent the reaction for its formation as

$$\underset{\text{sodium}}{2Na(s)} + \underset{\text{chlorine}}{Cl_2(g)} \longrightarrow \underset{\text{sodium chloride}}{2NaCl(s)}$$

We can understand this reaction better by showing electron configurations for all species.
We represent chlorine as individual atoms rather than molecules, for simplicity.



In this reaction, Na atoms lose one electron each to form Na^+ ions, which contain only ten electrons, the same number as the *preceding* noble gas, neon. We say that sodium ions have the neon electronic structure: Na^+ is *isoelectronic* with Ne. In contrast, Cl atoms gain one electron each to form Cl^- ions, which contain 18 electrons. This is the same number as the *following* noble gas, argon; Cl^- is *isoelectronic* with Ar. These processes can be represented compactly as

$$Na \longrightarrow Na^+ + e^- \qquad \text{and} \qquad Cl + e^- \longrightarrow Cl^-$$

Similar observations apply to most ionic compounds formed by reactions between *representative metals and representative nonmetals.*

We can use Lewis dot formulas to represent the reaction.

$$Na\cdot + :\ddot{\underset{..}{Cl}}\cdot \longrightarrow Na^+[:\ddot{\underset{..}{Cl}}:]^-$$

The formula for sodium chloride is NaCl because the compound contains Na^+ and Cl^- in a 1:1 ratio. This is the formula we predict based on the fact that each Na atom contains only one electron in its outermost occupied shell and each Cl atom needs only one electron to fill completely its outermost *p* orbitals.

The chemical formula NaCl does not explicitly indicate the ionic nature of the compound, only the ratio of ions. Furthermore, values of electronegativities are not always available. So we must learn to recognize, from positions of elements in the periodic table and known trends in electronegativity, when the difference in electronegativity is large enough to favor ionic bonding.

The farther apart across the periodic table two Group A elements are, the more ionic their bonding will be.

Formulas for some of these are

LiF	NaF	KF
LiCl	NaCl	KCl
LiBr	NaBr	KBr
LiI	NaI	KI

Group IA Metals and Group VIA Nonmetals

$$\underset{\text{lithium}}{4Li(s)} + \underset{\text{oxygen}}{O_2(g)} \longrightarrow \underset{\text{lithium oxide}}{2Li_2O(s)}$$

$$\left.\begin{array}{llllll} {}_3Li & \underset{1s}{\uparrow\downarrow} & \underset{2s}{\uparrow} & & & \\ {}_3Li & \underset{1s}{\uparrow\downarrow} & \underset{2s}{\uparrow} & & & \\ {}_8O & \underset{1s}{\uparrow\downarrow} & \underset{2s}{\uparrow\downarrow} & \uparrow\downarrow & \underset{2p}{\uparrow} & \uparrow \end{array}\right\} N \longrightarrow \left\{\begin{array}{llllll} Li^+ & \underset{1s}{\uparrow\downarrow} & \underset{2s}{\_} & & & 1\ e^-\ \text{lost} \\ Li^+ & \underset{1s}{\uparrow\downarrow} & \underset{2s}{\_} & & & 1\ e^-\ \text{lost} \\ O^{2-} & \underset{1s}{\uparrow\downarrow} & \underset{2s}{\uparrow\downarrow} & \uparrow\downarrow\ \underset{3p}{\uparrow\downarrow}\ \uparrow\downarrow & & 2\ e^-\ \text{gained} \end{array}\right.$$

In a compact representation,

$$2[Li \longrightarrow Li^+ + e^-] \qquad \text{and} \qquad O + 2e^- \longrightarrow O^{2-}$$

The Lewis dot formulas for the atoms and ions are

$$2Li\cdot + :\ddot{O}\cdot \longrightarrow 2Li^+[:\ddot{O}:]^{2-}$$

Lithium ions, Li^+, are isoelectronic with helium atoms ($2\ e^-$). Oxide ions, O^{2-}, are isoelectronic with neon atoms ($10\ e^-$).

Group IIA Metals and Group VIA Nonmetals

As our final illustration of ionic bonding, consider the reaction of calcium (Group IIA) with oxygen (Group VIA). This reaction forms calcium oxide, a white solid ionic compound with a very high melting point, 2580°C.

$$\underset{\text{calcium}}{2Ca(s)} + \underset{\text{oxygen}}{O_2(g)} \longrightarrow \underset{\text{calcium oxide}}{2CaO(s)}$$

Again, we show the electronic structure of the atoms and ions, representing the inner electrons by the symbol of the preceding noble gas.

$$\left.\begin{array}{lllll} {}_{20}Ca & [Ar] & \underset{4s}{\uparrow\downarrow} & & \\ {}_8O & [He] & \underset{2s}{\uparrow\downarrow} & \uparrow\downarrow\ \underset{2p}{\uparrow}\ \uparrow & \end{array}\right\} \longrightarrow \left\{\begin{array}{lllll} Ca^{2+} & [Ar] & \underset{4s}{\_} & & 2\ e^-\ \text{lost} \\ O^{2-} & [He] & \underset{2s}{\uparrow\downarrow} & \uparrow\downarrow\ \underset{2p}{\uparrow\downarrow}\ \uparrow\downarrow & 2\ e^-\ \text{gained} \end{array}\right.$$

The Lewis dot notation for the reacting atoms and the resulting ions is:

$$Ca: + :\ddot{O}\cdot \longrightarrow Ca^{2+}[:\ddot{O}:]^{2-}$$

Calcium ions, Ca^{2+}, are isoelectronic with argon ($18\ e^-$), the preceding noble gas. Oxide ions, O^{2-}, are isoelectronic with neon ($10\ e^-$), the following noble gas.

COVALENT BONDING

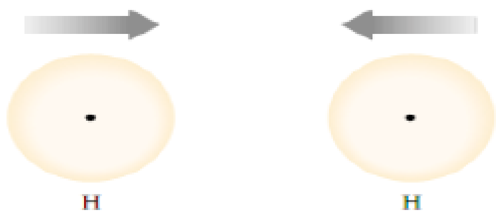
Ionic bonding cannot result from a reaction between two nonmetals, because their electronegativity difference is not great enough for electron transfer to take place. Instead, reactions between two nonmetals result in *covalent bonding.*

A **covalent bond** is formed when two atoms share one or more pairs of electrons. Covalent bonding occurs when the electronegativity difference, Δ(EN), between elements (atoms) is zero or relatively small.

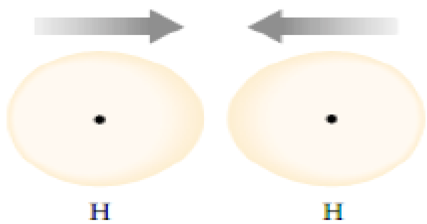
In predominantly covalent compounds the bonds between atoms *within* a molecule (*intra*molecular bonds) are relatively strong, but the forces of attraction *between* molecules (*inter*molecular forces) are relatively weak. As a result, covalent compounds have lower melting and boiling points than ionic compounds.

FORMATION OF COVALENT BONDS

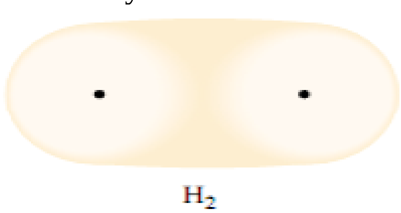
Let us look at the simplest case of covalent bonding, the reaction of two hydrogen atoms to form the diatomic molecule H_2. As you recall, an isolated hydrogen atom has the ground state electron configuration 1*s*1, with the probability density for this one electron spherically distributed about the hydrogen nucleus.



As two hydrogen atoms approach each other, the electron of each hydrogen atom is attracted by the nucleus of the *other* hydrogen atom as well as by its own nucleus.

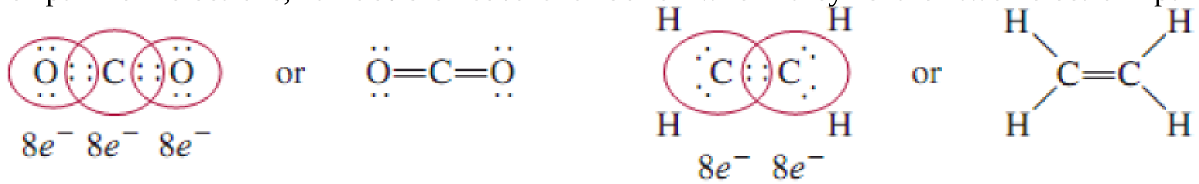


If these two electrons have opposite spins so that they can occupy the same region (orbital), both electrons can now preferentially occupy the region *between* the two nuclei, because they are attracted by both nuclei.

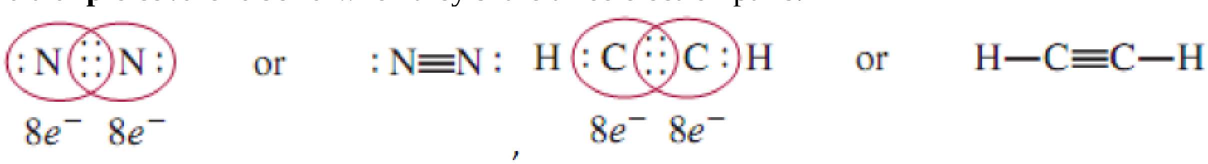


The electrons are *shared* between the two hydrogen atoms, and a single covalent bond is formed. We say that the 1*s* orbitals *overlap* so that both electrons are now in the orbitals of both hydrogen atoms. The closer together the atoms come, the more nearly this is true. In that sense, each hydrogen atom then has the helium configuration $1s^2$.

Other pairs of nonmetal atoms share electron pairs to form covalent bonds. The result of this sharing is that each atom attains a more stable electron configuration—frequently the same as that of the nearest noble gas. This results in a more stable arrangement for the bonded atoms. (This is discussed in Section 7-5.) Most covalent bonds involve sharing of two, four, or six electrons—that is, one, two, or three *pairs* of electrons. Two atoms form a **single covalent bond** when they share one pair of electrons, a **double covalent bond** when they share two electron pairs,



and a **triple covalent bond** when they share three electron pairs.



These are usually called simply *single, double,* and *triple* bonds. Covalent bonds that involve sharing of one and three electrons are known, but are relatively rare.

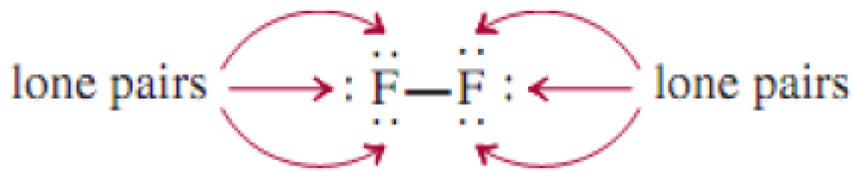
The structures we use to represent covalent compounds, such as H_2 and F_2, are called *Lewis structures*. ***A Lewis structure*** is a representation of covalent bonding in which shared electron pairs are shown either as lines or as pairs of dots between two atoms, and lone pairs are shown as pairs of dots on individual atoms. Only valence electrons are shown in a Lewis structure. Thus, the formation of H_2 from H atoms could be represented as

$$\mathrm{H}\cdot + \cdot\mathrm{H} \longrightarrow \mathrm{H}:\mathrm{H} \qquad \text{or} \qquad \mathrm{H}\text{—}\mathrm{H}$$

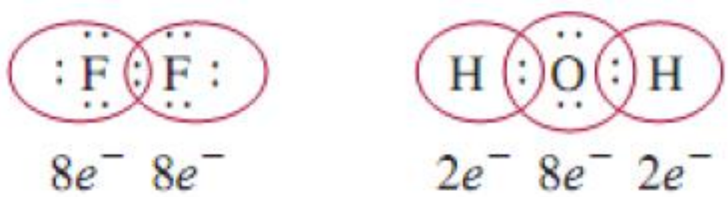
Where the dash represents a single bond.

Similarly, the combination of two fluorine atoms to form a molecule of fluorine, F_2, can be shown as

Note that only two valence electrons participate in the formation of F_2. The other, nonbonding electrons, are called *lone pairs*—pairs of valence electrons that are not involved in covalent bond formation. Thus, each F in F_2 has three lone pairs of electrons:



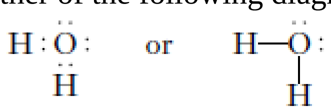
The formation of a hydrogen fluoride, HF, molecule from a hydrogen atom and a fluorine atom can be shown as



The formation of these molecules illustrates the **octet rule,** formulated by Lewis: An atom other than hydrogen tends to form bonds until it is surrounded by eight valence electrons. In other words, *a covalent bond forms when there are not enough electrons for each individual atom to have a complete octet.* By sharing electrons in a covalent bond, the individual atoms can complete their octets. The requirement for hydrogen is that it attains the electron configuration of helium, or a total of two electrons.

LEWIS FORMULAS FOR MOLECULES AND POLYATOMIC IONS

Lewis formulas are used to show the *valence electrons*. A water molecule can be represented by either of the following diagrams.



dot formula dash formula

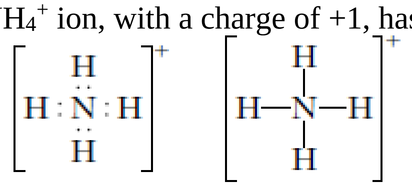
An H_2O molecule has two shared electron pairs, that is, two single covalent bonds. The O atom also has two unshared pairs.

In *dash formulas,* a shared pair of electrons is indicated by a dash. There are two *double* bonds in carbon dioxide, and its Lewis formula is:

dot formula dash formula

A CO_2 molecule has four shared electron pairs in two double bonds. The central atom (C) has no unshared pairs.

The covalent bonds in a polyatomic ion can be represented in the same way. The Lewis formula for the ammonium ion, NH_4^+, shows only eight electrons, even though the N atom has five electrons in its valence shell and each H atom has one, for a total of 5 +4(1) = 9 electrons. The NH_4^+ ion, with a charge of +1, has one less electron than the original atoms.



dot formula dash formula

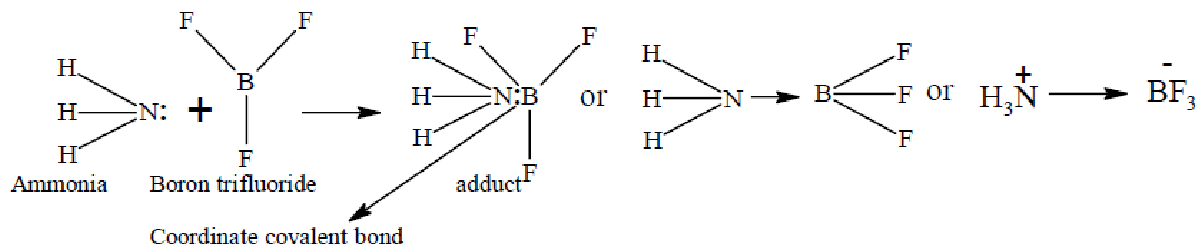
The writing of Lewis formulas is an electron bookkeeping method that is useful as a first approximation to suggest bonding schemes. It is important to remember that Lewis dot formulas only show the number of valence electrons, the number and kinds of bonds, and the order in which the atoms are connected. *They are not intended to show the three dimensional shapes of molecules and polyatomic ions.*

Coordinate covalent(Dative) Bond

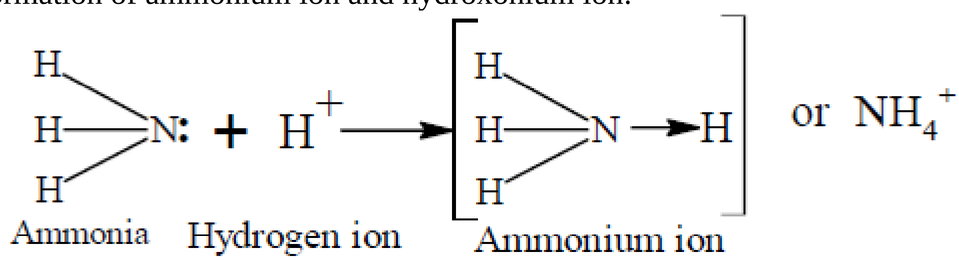
This is a special type of covalent bond in which two atoms are bonded in such a way that one member of the pair is supplying both electrons that are shared. The bond is represented by an arrow pointing from the atom supplying the lone pair of electrons to the other atom or a short bar between the two atoms with a positive charge on the donor and a negative charge on the acceptor:

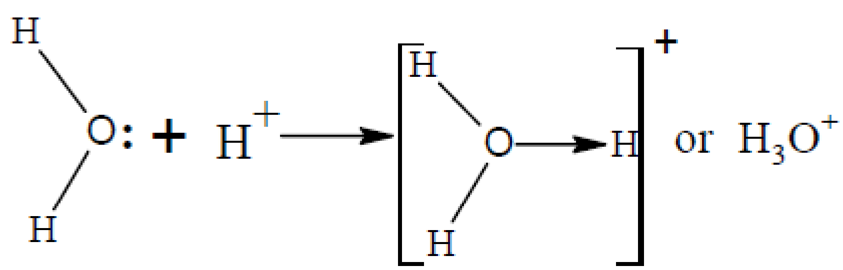
$$A: + B \rightarrow A:B \qquad \text{or} \qquad A \rightarrow B \qquad \text{or} \qquad A^{+}\text{—}B^{-}$$

This type of bond is used in forming an adduct between two covalent compounds e.g.ammonia boron trifluoride adduct:

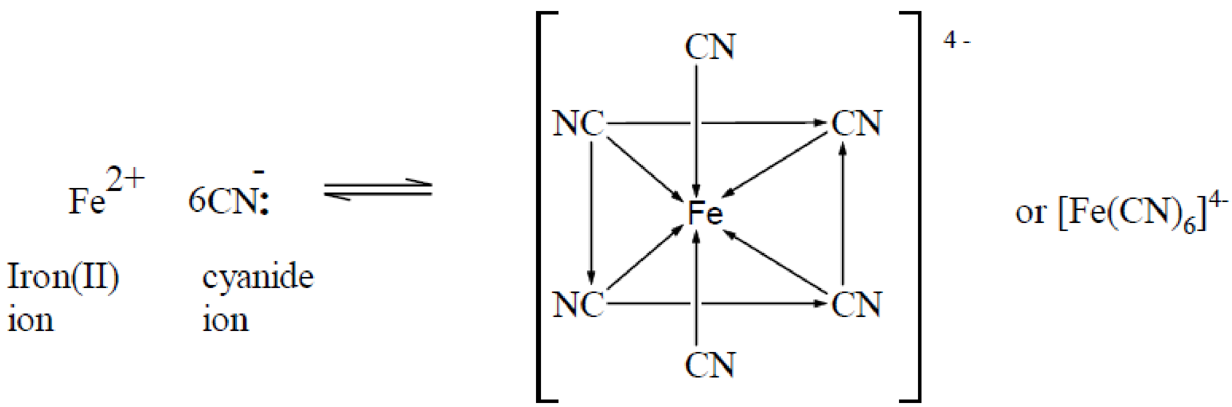


Coordinate covalent bond is also used in joining protons to neutral covalent molecules as in the formation of ammonium ion and hydroxonium ion:





Coordinate covalent bonds also join electron pair donors(ligand) to transition metal ions(with low lying empty orbitals) to form complexes(compound or ions) e.g. formation of hexacyanoferrate ion. The properties of coordinate covalent compound are similar to those of pure covalent compounds.



Hexacyanoferrate(II) ion

(a complex ion)

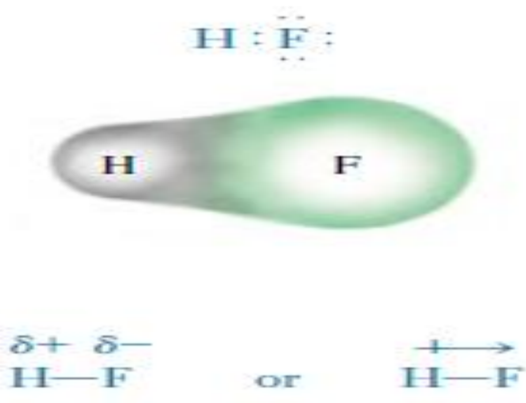
POLAR AND NONPOLAR COVALENT BONDS

Covalent bonds may be either *polar* or *nonpolar*. In a **nonpolar bond** such as that in the hydrogen molecule, H_2, (HSH or HXH) the electron pair is *shared equally* between the two hydrogen nuclei. We defined electronegativity as the tendency of an atom to attract electrons to itself in a chemical bond. Both H atoms have the same electronegativity. This means that the shared electrons are equally attracted to both hydrogen nuclei and therefore spend equal amounts of time near each nucleus. In this nonpolar covalent bond, the **electron density** is symmetrical about a plane that is perpendicular to a line between the two nuclei. This is true for all homonuclear *diatomic molecules*, such as H_2, O_2, N_2, F_2, and Cl_2, because the two identical atoms have identical electronegativity.

We can generalize: The covalent bonds in all homonuclear diatomic molecules must be nonpolar. Let us now consider *heteronuclear diatomic molecules*. Start with the fact that hydrogen fluoride, HF, is a gas at room temperature. This tells us that it is a covalent compound. We also know that the HXF bond has some degree of polarity because H and F are not identical atoms and therefore do not attract the electrons equally. But how polar will this bond be?

The electronegativity of hydrogen is 2.1, and that of fluorine is 4.0. Clearly, the F atom, with its higher electronegativity, attracts the shared electron pair much more strongly than does the H atom. We can represent the structure of HF as shown in the margin. The electron density is distorted in the direction of the more electronegative F atom. This small shift of electron density leaves H somewhat positive.

Covalent bonds, such as the one in HF, in which the *electron pairs are shared unequally* are called **polar covalent bonds.** Two kinds of notation used to indicate polar bonds are shown below:



The word "dipole" means "two poles." Here it refers to the positive and negative poles that result from the separation of charge within a molecule.

The $\delta -$ over the F atom indicates a "partial negative charge." This means that the F end of the molecule is somewhat more negative than the H end. The $\delta +$ over the H atom indicates a "partial positive charge," or that the H end of the molecule is positive *with respect to* the F end. We are *not* saying that H has a charge of +1 or that F has a charge of -1! A second way to indicate the polarity is to draw an arrow so that the head points toward the negative end (F) of the bond and the crossed tail indicates the positive end (H). The separation of charge in a polar covalent bond creates an electric **dipole.** We expect the dipoles in the covalent molecules HF, HCl, HBr, and HI to be different because F, Cl, Br, and I have different electronegativities. This tells us that atoms of these elements have different tendencies to attract an electron pair that they share with hydrogen. We indicate this difference as shown here, where Δ(EN) is the difference in electronegativity between two atoms that are bonded together.

	Most polar			Least polar
	H—F	H—Cl	H—Br	H—I
EN:	2.1 4.0	2.1 3.0	2.1 2.8	2.1 2.5
Δ(EN)	1.9	0.9	0.7	0.4

Exercises

1. What type of force is responsible for chemical bonding?
2. Explain the differences between ionic bonding and covalent bonding.
3. What kind of bonding (ionic or covalent) would you predict for the products resulting from the following combinations of elements?
   (a) $Na + Cl_2$; (c) $N_2 + O_2$,
   (b) $C + O_2$; (d) $S + O_2$.

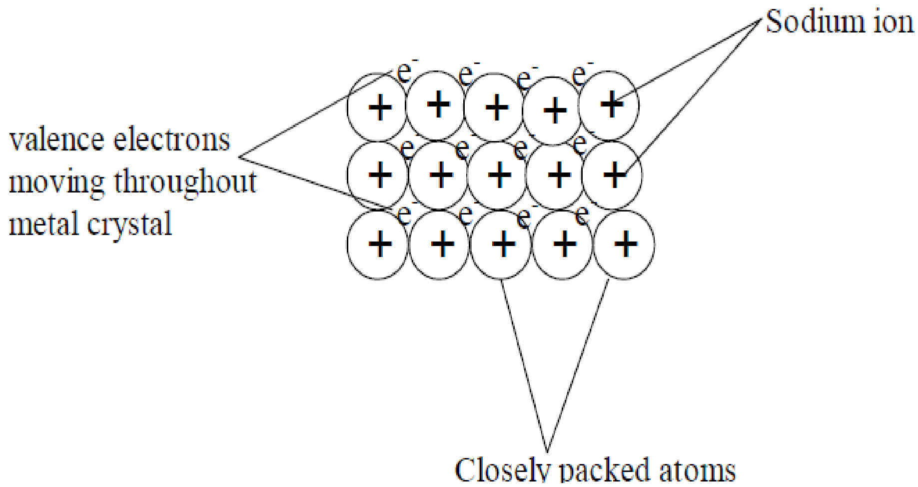
4. Predict whether the bonding between the following pairs of elements would be primarily ionic or covalent. Justify your answers. (a) Rb and Cl; (b) N and O; (c) Ca and F; (d) P and S; (e) C and F; (f) K and O.
5. Classify the following compounds as ionic or covalent: (a) $Ca(NO_3)_2$; (b) H_2S; (c) KNO_3; (d) $CaCl_2$; (e) H_2CO_3;
   (f) PCl_3; (g) Li_2O; (h) N_2H_4; (i) $SOCl_2$.
6. Why are solid ionic compounds rather poor conductors of electricity? Why does conductivity increase when an ionic compound is melted or dissolved in water?
7. Write the formula for the ionic compound that forms between each of the following pairs of elements: (a) Ca and Br_2; (b) Ba and Cl_2; (c) Na and Cl_2.
8. Write the formula for the ionic compound that forms between each of the following pairs of elements: (a) Cs and F_2; (b) Sr and S; (c) Na and Se.
9. When a *d*-transition metal ionizes, it loses its outer *s* electrons before it loses any *d* electrons. Using [nobel gas] $(n-1)\,d^x$ representations, write the outer-electron configurations for the following ions: (a) Cr^{3+}; (b) Mn^{2+}; (c) Ag^{+}; (d) Fe^{3+}; (e) Cu^{2+}; (f) Sc^{3+}; (g) Fe^{2+}.
10. Which of the following do not accurately represent stable binary ionic compounds? Why? $BaCl_2$; NaS; AlF_4; SrS_2; Ca_2O_3; $NaBr_2$; $LiSe_2$.
11. Which of the following do not accurately represent stable binary ionic compounds? Why? MgI; $Al(OH)_2$; InF_2; CO_2; $RbCl_2$; CsSe; Be_2O.

Metallic bond

Metallic bonds join atoms of metals and alloys together. The formation of this type of bonds is favoured by *large atomic radius, low ionization energy and a large number of electrons in the valence shell.* Metallic bond is a chemical bond in which metals that consist of positive cores of atoms are held together by a surrounding electron cloud or “sea”of electrons.

A sodium metal crystal, for example can be regarded as an array of Na+ ions surrounded by sea of electrons. The valence (bonding) electrons are free to move throughout the crystal. i.e. they are delocalised over the entire metal crystal. The freedom of these electrons to move throughout the crystal is responsible for the electrical conductivity of a metal.

Each sodium atom, for example, has only one valence electron and this enters the electron cloud. The resulting bonding is weak because of the single electron from each atom and so sodium is a soft metal of low density and low melting point(98 ^{o}C)



In metallic magnesium, there are two valence electrons per atom; the bonding force is therefore roughly twice as strong as with sodium and magnesium is harder, denser and of higher melting point(650 ^{o}C) than sodium. The process continues through the period in the periodic table because there is increasing number of valence electrons from left to right of the periodic table. Metallic bonding *differs* from ionic bonding in that there are no ions present but *resembles* it in that the bonding forces are non-rigid and non-directional. *Metallic bonding* differs from *covalent bonding* in that metal atoms do not form separate (discrete) molecules but resembles it in that the atomic cores are bound together by being attracted to the electron cloud between them.

Intermolecular Forces

Intermolecular forces are attractive forces between molecules. In contrast to intermolecular forces, *intramolecular forces* hold atoms together in a molecule. *Intramolecular forces* stabilize individual molecules, whereas *intermolecular forces* are primarily responsible for the bulk properties of matter (for example, melting point and boiling point).

Generally, intermolecular forces are much weaker than intramolecular forces. The different types of intermolecular forces are; *Dipole-dipole, dipole-induced dipole, and dispersion forces* make up what chemists commonly refer to as *Vander Waals forces*,

Dipole-Dipole Forces

Dipole-dipole forces are attractive forces between polar molecules, that is, between molecules that possess dipole moments (A quantitative measure of the polarity of a bond is its dipole moment (μ), which is the product of the charge Q and the distance r between the charges).

$$\mu = Q \times r$$

Their origin is electrostatic, and they can be understood in terms of Coulomb's law. The larger the dipole moment, the greater the force. In liquids, polar molecules are not held as rigidly as in a solid, but they tend to align in a way that, on average, maximizes the attractive interaction.

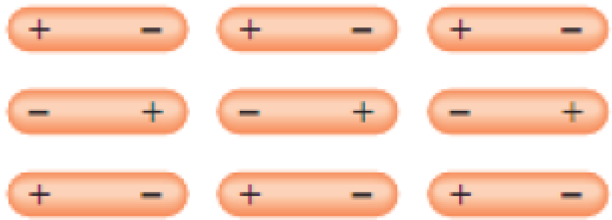
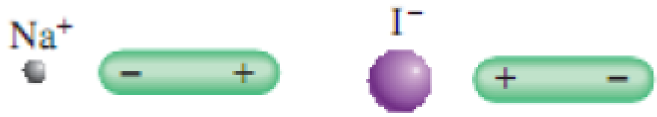


Figure showing the orientation of polar molecules in a solid.

Ion-Dipole Forces

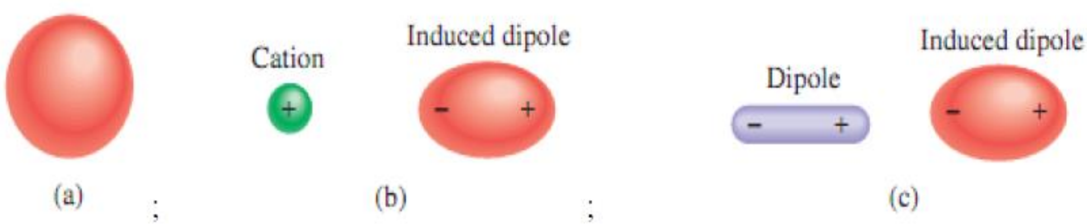
Ion-dipole forces attract an ion (either a cation or an anion) and a polar molecule to each other. The strength of this interaction depends on the charge and size of the ion and on the magnitude of the dipole moment and size of the molecule. The charges on cations are generally more concentrated, because cations are usually smaller than anions. Therefore, a cation interacts more strongly with dipoles than does an anion having a charge of the same magnitude.



Two types of ion-dipole interaction.

Dispersion Forces

If we place an ion or a polar molecule near an atom (or a nonpolar molecule), the electron distribution of the atom (or molecule) is distorted by the force exerted by the ion or the polar molecule, resulting in a kind of dipole. The dipole in the atom (or nonpolar molecule) is said to be an induced dipole because the separation of positive and negative charges in the atom (or nonpolar molecule) is due to the proximity of an ion or a polar molecule. The attractive interaction between an ion and the induced dipole is called *ion-induced dipole interaction*, and the attractive interaction between a polar molecule and the induced dipole is called *dipole-induced dipole interaction*.



(a) Spherical charge distribution in a helium atom. (b) Distortion caused by the approach of a cation. (c) Distortion caused by the approach of a dipole

The likelihood of a dipole moment being induced depends not only on the charge on the ion or the strength of the dipole but also on the *polarizability of the atom or molecule: that is, the ease with which the electron distribution in the atom (or molecule) can be distorted.* Generally, the larger the number of electrons and the more diffuse the electron cloud in the atom or molecule, the greater its polarizability. By diffuse cloud we mean an electron cloud that is spread over an appreciable volume, so that the electrons are not held tightly by the nucleus. Polarizability allows gases containing atoms or nonpolar molecules (e.g. He and N_2) to condense. In a helium atom, the electrons are moving at some distance from the nucleus. At any instant it is likely that the atom has a dipole moment created by the specific positions of the electrons. This dipole moment is called an instantaneous dipole because it lasts for just a tiny fraction of a second. In the next instant, the electrons are in different locations and the atom has a new instantaneous dipole, and so on. Averaged over time (that is, the time it takes to make a dipole moment measurement), however, the atom has no dipole moment because the instantaneous dipoles all cancel one another. In a collection of He atoms, an instantaneous dipole of one He atom can induce a dipole in each of its nearest neighbors.

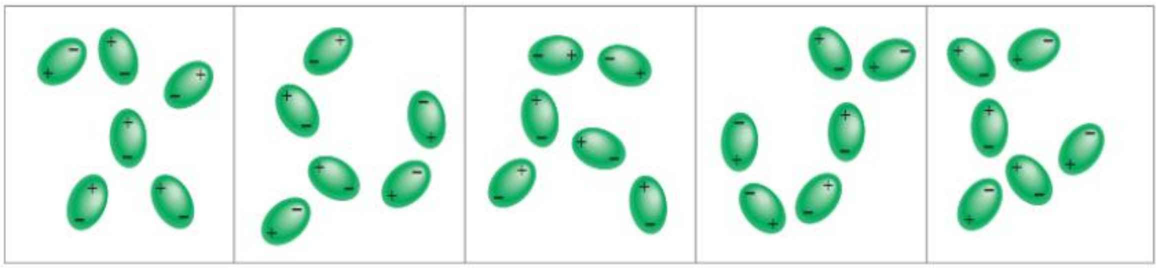


Figure showing the Induced dipoles interacting with each other. Such patterns exist only momentarily; new arrangements are formed in the next instant. This type of interaction is responsible for the condensation of nonpolar gases.

At the next moment, a different instantaneous dipole can create temporary dipoles in the surrounding He atoms. This kind of interaction produces dispersion forces, attractive forces that arise as a result of temporary dipoles induced in atoms or molecules. At very low temperatures (and reduced atomic speeds), dispersion forces are strong enough to hold He atoms together, causing the gas to condense. The attraction between nonpolar molecules can be explained similarly.

Dispersion forces, which are also called *London forces*, usually increase with molar mass because molecules with larger molar mass tend to have more electrons, and dispersion forces increase in strength with the number of electrons.

The Hydrogen Bond

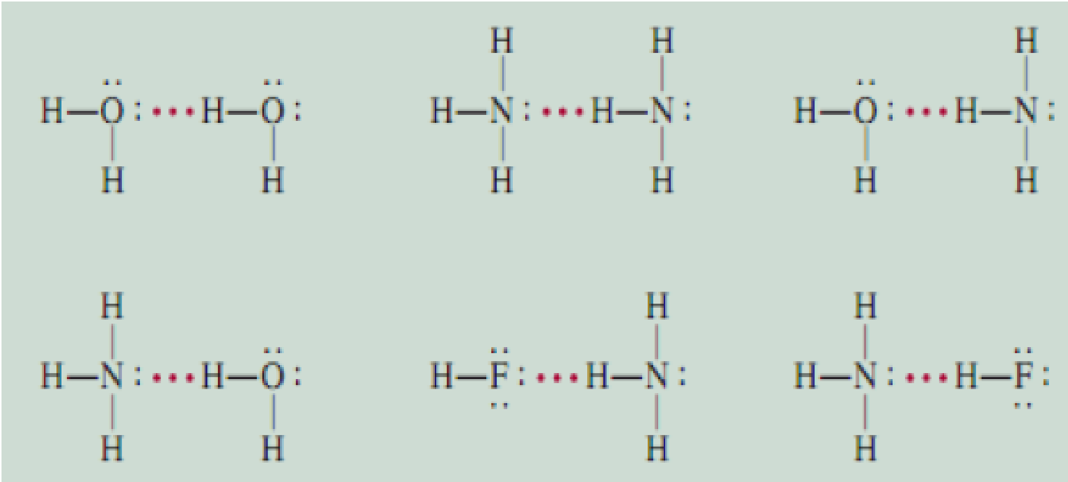
Normally, the boiling points of a series of similar compounds containing elements in the same periodic group increase with increasing molar mass. This increase in boiling point is due to the increase in dispersion forces for molecules with more electrons. Hydrogen compounds of Group 4A follow this trend. The lightest compound, CH_4, has the lowest boiling point, and the heaviest compound, SnH_4, has the highest boiling point. However, hydrogen compounds of the elements in Groups 5A, 6A, and 7A do not follow this trend. In each of these series, the lightest compound (NH_3, H_2O, and HF) has the highest boiling point, contrary to our expectations based on molar mass. This observation must mean that there are stronger intermolecular attractions in NH_3, H_2O, and HF, compared to other molecules in the same groups.

In fact, this particularly strong type of intermolecular attraction is called the *hydrogen bond*, which is a special type of dipole-dipole interaction between the hydrogen atom in a polar bond, such as N—H, O—H, or F—H, and an electronegative O, N, or F atom. The interaction is written:

$$\mathrm{A{-}H \cdots B} \quad \text{or} \quad \mathrm{A{-}H \cdots A}$$

A and B represent O, N, or F; A—H is one molecule or part of a molecule and B is a part of another molecule; and the dotted line represents the hydrogen bond. The three atoms usually lie in a straight line, but the angle AHB (or AHA) can deviate as much as 30° from linearity. Note that the O, N, and F atoms all possess at least one lone pair that can interact with the hydrogen atom in hydrogen bonding. N, O, F are the three most electronegative elements that take part in hydrogen bonding.

Examples of hydrogen bonding.

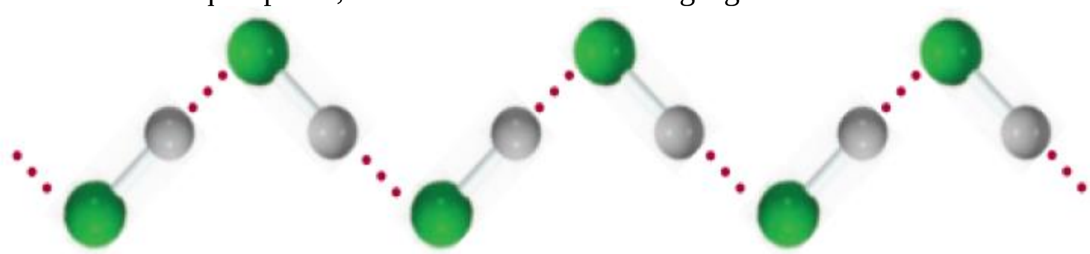


Hydrogen bonding in water, ammonia, and hydrogen fluoride: Solid lines represent covalent bonds, and dotted lines represent hydrogen bonds.

The small size of the electronegative atoms allows the atom to come sufficiently close forn the resultant attraction to have significant effects on properties such as melting and boiling points. The higher the electronegativity of the heteroatom X, the stronger the bond and the higher the melting and boiling points. Thus molecules with H—F bonds have higher boiling point than molecules with H—N bond. Also, the more the number of hydrogen bonds in a molecule, the higher the melting or boiling points, e.g. ammonia boils at higher temperature than all its alkyl derivatives, i.e.;

$$NH_3 > RNH_2 > R_2NH > R_3N$$

The strength of a hydrogen bond is determined by the coulombic interaction between the lone-pair electrons of the electronegative atom and the hydrogen nucleus. For example, fluorine is more electronegative than oxygen, and so we would expect a stronger hydrogen bond to exist in liquid HF than in H2O. In the liquid phase, the HF molecules form zigzag chains:



The boiling point of HF is lower than that of water because each H_2O takes part in four intermolecular hydrogen bonds. Therefore, the forces holding the molecules together are stronger in H_2O than in HF. The intermolecular forces are all attractive in nature. Though, those molecules also exert repulsive forces on one another. Thus, when two molecules approach each other, the repulsion between the electrons and between the nuclei in the molecules comes into play. The magnitude of the repulsive force rises very steeply as the distance separating the molecules in a condensed phase decreases. This is the reason that liquids and solids are so hard to compress. In these phases, the molecules are already in close contact with one another, and so they greatly resist being compressed further.

Properties of hydrogen bonds

1. Hydrogen bonds, like covalent bonds, are directional in nature
2. Hydrogen bonds are essentially an electrostatic bond.
3. Hydrogen bonds are stronger than Vander Waals forces but much weaker than the forces involved in ionic or covalent bonds.

Exercises

1. When s and p orbitals overlap, what kinds of bonds are formed? and when p and p orbitals overlap?
2. Differentiate between (a) ionic and covalent bond (b) polar and non-polar covalent bonds giving examples
3. Explain why

i. NH_3 is highly soluble in water

ii. NH_3 has higher boiling point than PH_3

iii. NF_3 is known but NF_5 is not?

4. What do you understand by intra-molecular and intermolecular hydrogen bonding? Explain with examples
5. Explain a bond pair and a lone pair of electrons
6. List 3 elements each for the following categories in the periodic table and state why they are called

i. d-block elements (ii) s –block element (iii) p-block elements (iv) group IB elements

7. In H-O-H what type of bond is formed here? Explain clearly.

STRUCTURE OF SOLIDS

Solids are classified into two groups; crystalline solids and amorphous solids. **Crystalline solids** have regular ordered arrays of components held together by uniform intermolecular forces while **the components of amorphous solids** are not arranged in regular arrays. With few exceptions, the particles that make up a solid material, whether ionic, molecular, covalent, or metallic are held in place by strong attractive forces between them.

The locations of the constituent particles of crystalline solids (ions, atoms or molecules) are usually represented by a **lattice**, and the location of each particle is a **lattice point**.

Crystalline solids or crystals have distinctive internal structures that in turn lead to distinctive flat surfaces or faces. The intermolecular forces holding the solid together are uniform, and the same amount of thermal energy is needed to break any interaction simultaneously.

Amorphous solids have two characteristic properties. When cleaved or broken, they produce fragments with **irregular, often curved** surfaces; and they have poorly defined patterns when exposed to x-rays because their components are not arranged in a regular fashion.

Almost any substance can solidify in amorphous form if the liquid phase is cooled rapidly enough. Some solids however, are intrinsically amorphous, because either their components cannot fit together well enough to form a stable crystalline lattice, or they contain impurities that disrupt the lattice.

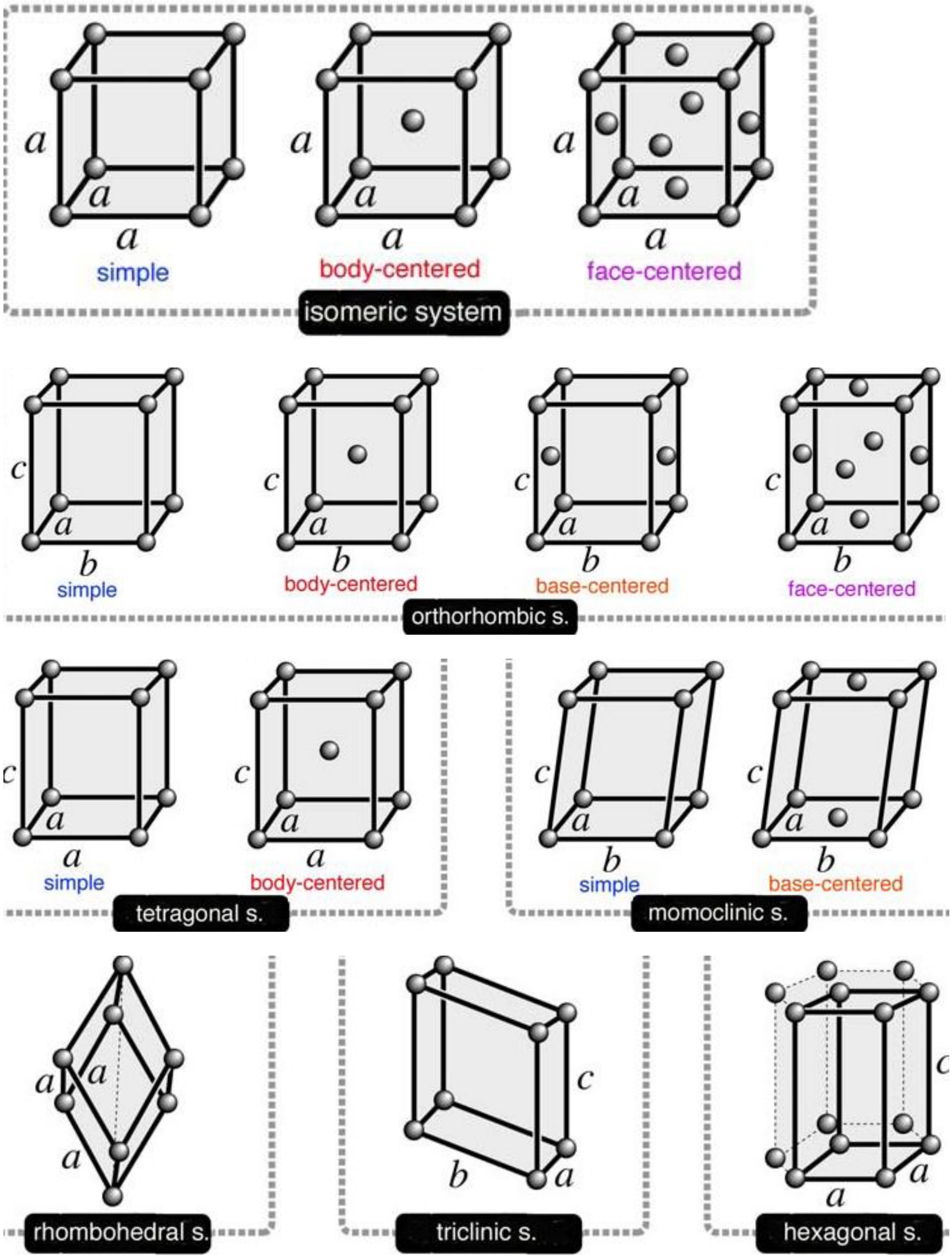
Some examples of functional amorphous solids and their uses

amorphous materials	uses
quartz glass	optical fiber
chalcogenide glass	selenium membranes for copy machines
amorphous silicon	solar cell
amorphous alloy	iron/cobalt metal (magnetic material)
polymer	polystyrene
amorphous carbon	carbon black (adsorbent)
gel	silica gel (adsorbent)

Crystalline solids or crystals

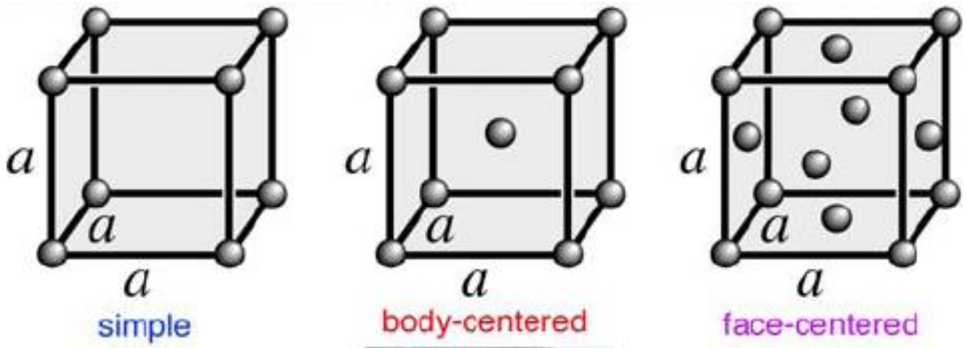
Crystalline solid consists of repeating patterns of its components in three dimensions (a crystal lattice). The basic repeating unit of a crystal lattice is called a unit cell.

In 1848, the French crystallographer Auguste Bravais (1811-1863) classified the crystal lattices based on their symmetries, and found that there are 14 kinds of crystal lattices as indicated in Figure below. These are called the **Bravais lattices**. These 14 Bravais lattices are classified into seven crystal systems. The three well known crystal lattices are the **cubic systems**: **simple cubic lattice**, **body-centered cubic lattice** and **face-centered cubic lattice**.



Crystals are classified into 14 Bravais lattices and 7 crystalline systems

UNIT CELLS: There are seven fundamentally different kinds of unit cells. We will focus on cubic unit cells, in which all sides have the same length and all angles are 90^o, but the concepts used here also apply to substances whose unit cells are not cubic. If the cubic unit cell consists of 8 component atoms, molecules or ions located at the corners of the cube, it is called **simple cubic**. If the unit cell also contains an identical component at the centre of the cube, then it is **body-centered cubic (bcc).** If there are components in the centre of each face, in addition to those at the corners of the cube, then the unit cell is **face - centered cubic (fcc).**



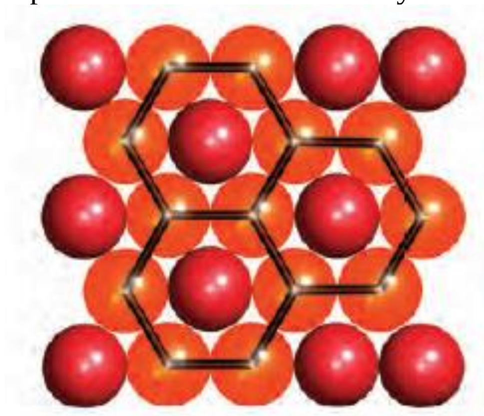
In the solid state, both metallic and ionic compounds possess ordered arrays of atoms or ions and form crystalline materials with lattice structures. Studies of their structures may conveniently be considered as related topics because both are concerned with the packing of spherical atoms or ions. However, differences in bonding result in quite distinct properties for metallic and ionic solids. In metals, the bonding is essentially covalent. The bonding electrons are delocalized over the whole crystal, giving rise to the high electrical conductivity that is characteristic of metals. Ionic bonding in the solid state arises from electrostatic interactions between charged species (ions), e.g. Na^+ and Cl^- in rock salt.

Crystals can be classified depending on their constituent particles, *i.e*., atoms, molecules or ions as shown on the table below:

unit particles		type of bond	typical properties	examples
atoms	network	covalent bonds with direction	hard high melting points insulator	diamond
	metal	covalent bonds without direction	variable hardness variable melting points conductor	silver iron
	rare gases	intermolecular forces	very low melting points	argon
molecules		dipole-dipole interaction (polar molecules)	soft low melting points insulator	ice dry ice
ions		ionic bonds	hard high melting points insulator	sodium chloride

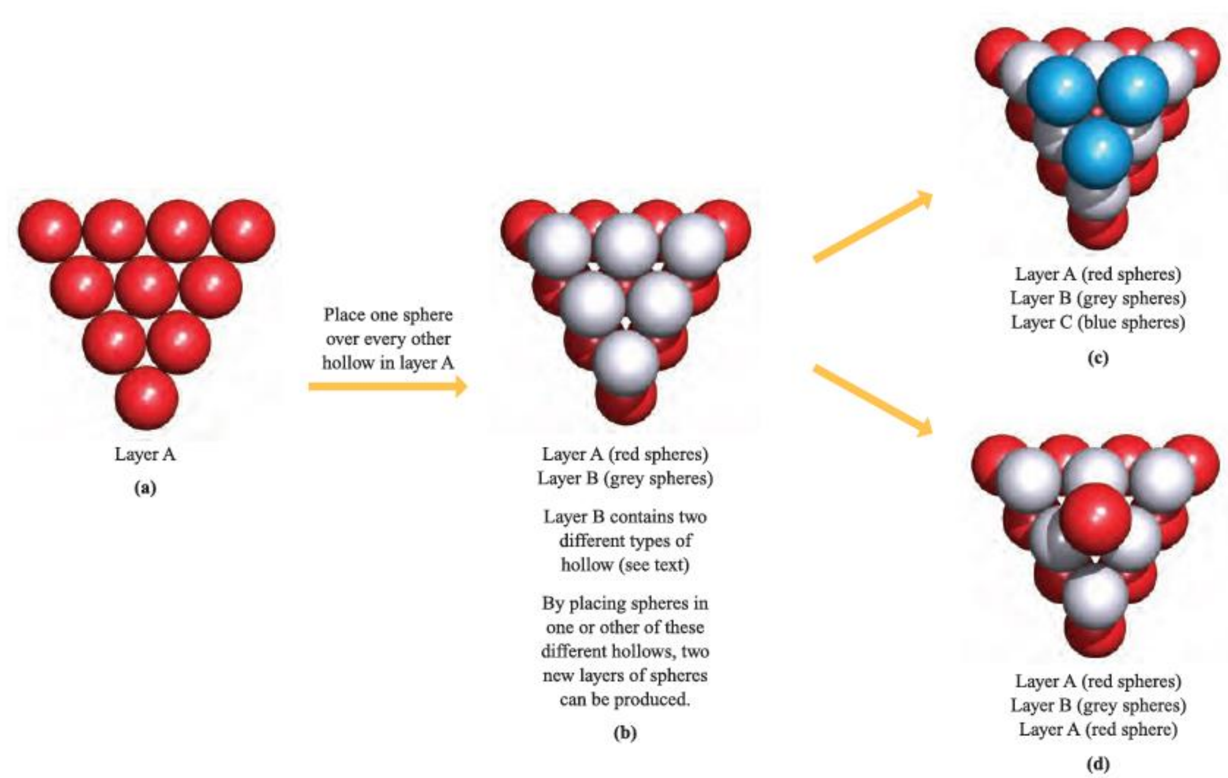
Cubic and hexagonal close-packing

Let us place a number of equal-sized spheres in a rectangular box, with the restriction that there must be a regular arrangement of spheres. The Figure below shows the most efficient way in which to cover the floor of the box. Such an arrangement is close-packed, and spheres that are not on the edges of the assembly are in contact with six other spheres within the layer. A motif of hexagons is produced within the assembly.



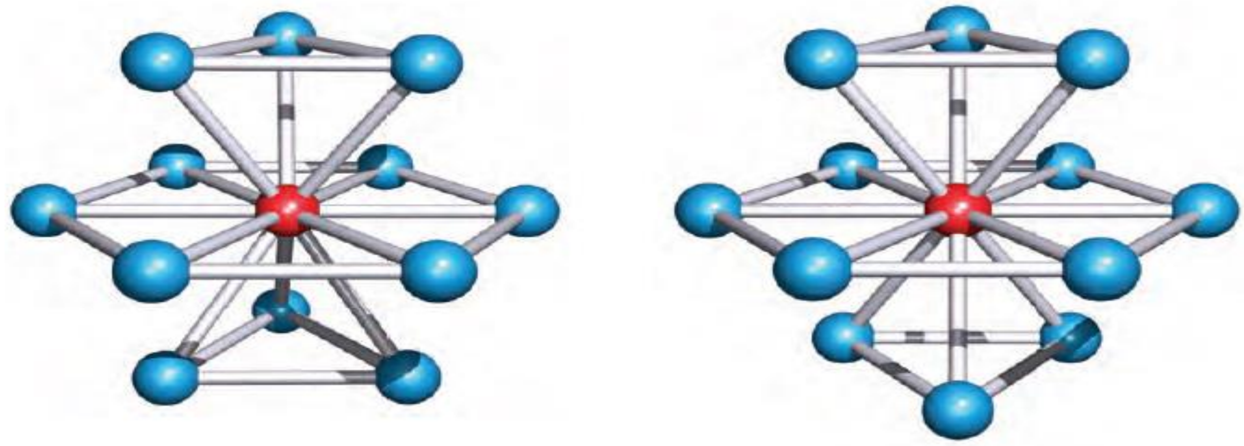
Part of one layer of a close-packed arrangement of equal-sized spheres. It contains hexagonal motifs.

The Figure **a** below shows part of the same close-packed arrangement of spheres; hollows lie between the spheres and we can build a second layer of spheres upon the first by placing spheres in these hollows. However, if we arrange the spheres in the second layer so that close-packing is again achieved, it is possible to occupy only every other hollow.



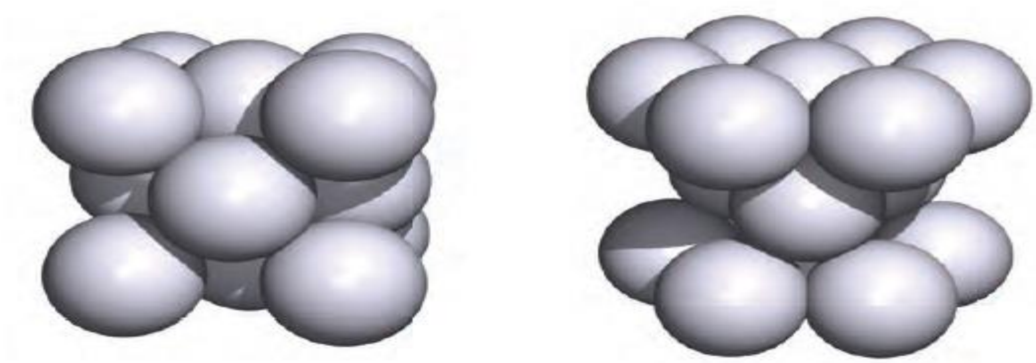
This is shown on going from Figure a to b. Now consider the hollows that are visible in layer B in b. There are two distinct types of hollows. Of the four hollows between the grey spheres in layer B, one lies over a red sphere in layer A, and three lie over hollows in layer A. The consequence of this is that when a third layer of spheres is constructed, two different close-packed arrangements are possible as shown in Figures c and d. The arrangements shown can, of course, be extended sideways, and the sequences of layers can be repeated such that the fourth layer of spheres is equivalent to the first, and so on.

The two close-packed arrangements are distinguished in that one contains two repeating layers, ABABAB. . . , while the second contains three repeating layers, ABCABC. . . (Figures d and c respectively). The ABABAB. . . and ABCABC. . . packing arrangements are called hexagonal close-packing (hcp) and cubic close-packing (ccp), respectively. In each structure, any given sphere is surrounded by (and touches) 12 other spheres and is said to have 12 nearest neighbours, to have a coordination number of 12, or to be 12-coordinate. Figures a and b below show representations of the ABABAB. . . and ABCABC. . . arrangements which illustrate how this coordination number arises; in these diagrams, 'ball-and-stick' representations of the lattice are used to allow the connectivity to be seen. This type of representation is commonly used but does not imply that the spheres do not touch one another.



(a) (b)

In both the (a) ABA and (b) ABC close-packed arrangements, the coordination number of each atom is 12.



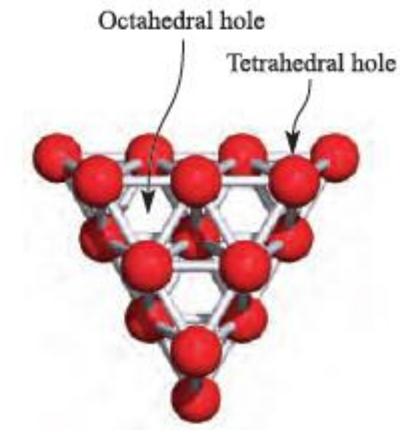
(a) (b)

Unit cells of (a) a cubic close-packed (face-centred cubic) lattice and (b) a hexagonal close-packed lattice.

The unit cells in immediate Figures a and b above characterize cubic (ccp) and hexagonal close-packing (hcp).

The unit cell: hexagonal and cubic close-packing

A unit cell is a fundamental concept in solid state chemistry, and is the smallest repeating unit of the structure which carries all the information necessary to construct unambiguously an infinite lattice. The smallest repeating unit in a solid state lattice **is a unit cell.** Close-packed structures contain octahedral and tetrahedral holes (or sites).



Polymorphism: phase changes in the solid state

It is generally convenient to consider the structures of metals in terms of the observed lattice type at 298K and atmospheric pressure, but these data do not tell the whole story. When subjected to changes in temperature and/or pressure, the structure of a metal may change; each form of the metal is a particular polymorph. For example, scandium undergoes a reversible transition from an hcp lattice (α-Sc) to a bcc lattice (β-Sc) at 1610 K. Some metals undergo more than one change: at atmospheric pressure, Mn undergoes transitions from the α - to β-form at 983 K, from the β - to γ-form at 1352 K, and from γ- to σ-Mn at 1416 K. Although α -Mn adopts a complex lattice, the β -polymorph has a somewhat simpler structure containing two 12-coordinate Mn environments, the γ -form possesses a distorted ccp structure, and the σ -polymorph adopts a bcc lattice. Phases that form at high temperatures may be quenched to lower temperatures (i.e. rapidly cooled with retention of structure), allowing the structure to be determined at ambient temperatures. Thermochemical data show that there is usually very little difference in energy between different polymorphs of an element.

If a substance exists in more than one crystalline form, **it is polymorphic. Polymorphism** is the existence of a substance in more than one crystalline form. Substances that exist in more than one crystalline form are called **polymorphs**.

CHEMISTRY OF SELECTED METALS

Group 1: The alkali metals

The alkali metals – lithium, sodium, potassium, rubidium, caesium and francium – are members of group 1 of the periodic table, and each has a ground state valence electronic configuration ns^1. The elements of group 1 are collectively known as **alkali metals** after the alkaline properties of their hydroxides such as NaOH. The atoms have the *(ns)*^{1} electron configuration and the M^+ ions are therefore easily formed. Alkali metals are the most electropositive of all elements, and their compounds among the most ionic. Discussions of these metals usually neglect the heaviest member of the group; only artificial isotopes of francium are known, the longest lived, $^{223}_{87}Fr$, having $t_{1/2}$ = 21.8 min.

Occurrence

Sodium and potassium are abundant in the Earth's biosphere (2.6 % and 2.4 % respectively) but do not occur naturally in the elemental state. The main sources of Na and K are rock salt (almost pure NaCl), natural brines and seawater, sylvite (KCl), sylvinite (KCl/NaCl) and carnallite ($KCl.MgCl_2.6H_2O$). Other Na- and K-containing minerals such as borax ($Na_2[B_4O_5](OH)_48H_2O$and Chile saltpetre ($NaNO_3$) are commercially important sources of other elements (e.g. B and N respectively). Unlike many inorganic chemicals, NaCl need not be manufactured since large natural deposits are available. Evaporation of seawater yields a mixture of salts, but since NaCl represents the major component of the mixture, its production in this manner is a viable operation. In contrast to Na and K, natural abundances of Li, Rb and Cs are small (% abundance Rb > Li > Cs); these metals occur as various silicate minerals, e.g. spodumene ($LiAlSi_2O_6$).

Extraction

Sodium, economically much the most important of the alkali metals, is manufactured by the Downs process in which molten NaCl) is electrolysed; $CaCl_2$ is added to reduce the operating temperature to about 870 K, since pure NaCl melts at 1073 K. The design of the electrolysis cell is critical to prevent reformation of NaCl by recombination of Na and Cl_2. Downs process is used for Cl_2 production.

At the cathode: $Na^+(l) + e^- \longrightarrow Na(l)$

At the anode: $2Cl^-(l) \longrightarrow Cl_2(g) + 2e^-$

Overall reaction: $2Na^+(l) + 2Cl^-(l) \longrightarrow 2Na(l) + Cl_2(g)$

Lithium is extracted from LiCl in a similar electrolytic process; LiCl is first obtained from spodumene by heating with CaO to give LiOH, which is then converted to the chloride. Potassium can be obtained electrolytically from KCl, but a more efficient method of extraction is the action of Na vapour on molten KCl in a counter-current fractionating tower. This yields an Na–K alloy which can be separated into its components by distillation. Similarly, Rb and Cs can be obtained from RbCl and CsCl, small quantities of which are produced as by-products from the extraction of Li from spodumene.

Small amounts of Na, K, Rb and Cs can be obtained by thermal decomposition of their azides; an application of NaN_3 is in car airbags. Lithium cannot be obtained from an analogous reaction because the products recombine, yielding the nitride, Li_3N

$$2NaN_3 \xrightarrow{570\,K,\ \textit{in vacuo}} 2Na + 3N_2$$

Major uses of the alkali metals and their compounds

Lithium has the lowest density (0.53 g cm^{-3}) of all known metals. It is used in the manufacture of alloys, and in certain glasses and ceramics. Lithium carbonate is used in the treatment of manic-depressive disorders, although large amounts of lithium salts damage the central nervous system.

Sodium, potassium and their compounds have many uses of which selected examples are given here. Sodium– potassium alloy is used as a heat-exchange coolant in nuclear reactors. A major use of Na–Pb alloy was in the production of the anti-knock agent $PbEt_4$, but the increasing demand for unleaded fuels renders this of decreasing importance.

The varied applications of compounds of Na include those in the paper, glass, detergent, chemical and metal industries. The major consumption of NaCl is in the manufacture of NaOH, Cl_2 and Na_2CO_3. A large fraction of salt is used for winter road de-icing.

However, in addition to the corrosive effects of NaCl, environmental concerns have focused on the side-effects on roadside vegetation and run-off into water sources.

Both Na and K are involved in various electrophysiological functions in higher animals. The $[Na^+]:[K^+]$ ratio is different in intra- and extra-cellular fluids, and the concentration gradients of these ions across cell membranes are the origin of the trans-membrane potential difference that, in nerve and muscle cells, is responsible for the transmission of nerve impulses. A balanced diet therefore includes both Na^+ and K^+ salts. Potassium is also an essential plant nutrient, and K^+ salts are widely used as fertilizers. Uses of Li and Na in batteries are highlighted in, and the use of KO_2 in breathing masks.
Many organic syntheses involve Li, Na or their compounds, and uses of the reagents $Na[BH_4]$ and $Li[AlH_4]$ are widespread. Alkali metals and some of their compounds also have uses in catalysts, e.g. the formation of MeOH from H_2 and CO where doping the catalyst with Cs makes it more effective.

Physical properties

The alkali metals illustrate, more clearly than any other group of elements, the influence of increase in atomic and ionic size on physical and chemical properties. Thus, the group 1 metals are often chosen to illustrate general principles. Some physical properties of the group 1 metals are given in Table 1.
TABLE 1

Property	Li	Na	K	Rb	Cs
Atomic number, Z	3	11	19	37	55
Ground state electronic configuration	$[He]2s^1$	$[Ne]3s^1$	$[Ar]4s^1$	$[Kr]5s^1$	$[Xe]6s^1$
Enthalpy of atomization, $\Delta_a H^o(298\,K)$ / kJ mol^{-1}	161	108	90	82	78
Dissociation enthalpy of M–M bond in M_2 (298 K) / kJ mol^{-1}	110	74	55	49	44
Melting point, mp / K	453.5	371	336	312	301.5
Boiling point, bp / K	1615	1156	1032	959	942
Standard enthalpy of fusion, $\Delta_{fus}H^o(mp)$ / kJ mol^{-1}	3.0	2.6	2.3	2.2	2.1
First ionization energy, IE_1 / kJ mol^{-1}	520.2	495.8	418.8	403.0	375.7
Second ionization energy, IE_2 / kJ mol^{-1}	7298	4562	3052	2633	2234
Metallic radius, r_{metal} / pm‡	152	186	227	248	265
Ionic radius, r_{ion} / pm*	76	102	138	149	170
Standard enthalpy of hydration of M^+, $\Delta_{hyd}H^o(298\,K)$ / kJ mol^{-1}	−519	−404	−321	−296	−271
Standard entropy of hydration of M^+, $\Delta_{hyd}S^o(298\,K)$ / J K^{-1} mol^{-1}	−140	−110	−70	−70	−60
Standard Gibbs energy of hydration of M^+, $\Delta_{hyd}G^o(298\,K)$ / kJ mol^{-1}	−477	−371	−300	−275	−253
Standard reduction potential, $E^o_{M^+/M}$ / V	−3.04	−2.71	−2.93	−2.98	−3.03
NMR active nuclei (% abundance, nuclear spin)	^{6}Li (7.5, $I = 1$); ^{7}Li (92.5, $I = \frac{3}{2}$)	^{23}Na (100, $I = \frac{3}{2}$)	^{39}K (93.3, $I = \frac{3}{2}$); ^{41}K (6.7, $I = \frac{3}{2}$)	^{85}Rb (72.2, $I = \frac{5}{2}$); ^{87}Rb (27.8, $I = \frac{3}{2}$)	^{133}Cs (100, $I = \frac{7}{2}$)

Some important points arising from these data are listed below.

- **(a)** .With increasing atomic number, the atoms become larger and the strength of metallic bonding decreases.
- **(b)** The effect of increasing size evidently outweighs that of increasing nuclear charge, since the ionization energies decrease from Li to Cs. The values of IE_2 for all the alkali metals are so high that the formation of M^{2+} ions under chemically reasonable conditions is not viable.
- **(c)** . Values of $E^o_{M+/M}$ are related to energy changes accompanying the processes:

$M(s) \rightarrow M(g)$ *atomization*
$M(g) \rightarrow M^+(g)$ *ionization*
$M^+(g) \rightarrow M^+(aq)$ *hydration*

and down group 1, differences in these energy changes almost cancel out, resulting in similar $E^o_{M+/M}$ values. The lower reactivity of Li towards H_2O is kinetic rather than thermodynamic in origin; Li is a harder and higher melting metal, is less rapidly dispersed, and reacts more slowly than its heavier congeners.

In general, the chemistry of the group 1 metals is dominated by compounds containing M^+ ions. However, a small number of compounds containing the M^+ ion (M=Na, K, Rb or Cs) are known.

Considerations of lattice energies calculated using an electrostatic model provide a satisfactory understanding for the fact that ionic compounds are central to the chemistry of Na, K, Rb and Cs. That Li shows a so-called 'anomalous' behaviour and exhibits a diagonal relationship to Mg can be explained in terms of similar energetic considerations.

Radioactive isotopes

In addition to the radioactivity of Fr, 0.02 % of naturally occurring K consists of ^{40}K, a β-particle and positron emitter with $t_{1/2}$ = 1.26 x10^9 yr. This provides the human body with a natural source of radioactivity, albeit at very low levels.

Although Li, Na and K are stored under a hydrocarbon solvent to prevent reaction with atmospheric O_2 and water vapour, they can be handled in air, provided undue exposure is avoided; Rb and Cs should be handled in an inert atmosphere.

Reaction with water

Lithium reacts quickly with water; Na reacts vigorously, and K, Rb and Cs react violently with the ignition of H_2 produced.

$$2Li + 2H_2O \longrightarrow 2LiOH + H_2$$

Sodium is commonly used as a drying agent for hydrocarbon and ether solvents.

Reaction with halogens

All the metals react with the halogens and H_2 when heated. The energetics of metal hydride formation are essentially like those of metal halide formation, being expressed in terms of a Born–Haber cycle.

$$2M + X_2 \longrightarrow 2MX \qquad X = \text{halogen}$$

$$2M + H_2 \longrightarrow 2MH$$

Reaction with Nitrogen

Lithium reacts spontaneously with N_2, and the reaction occurs at 298K to give red-brown, moisture-sensitive lithium nitride. Solid Li_3N has an interesting lattice structure and a high ionic conductivity. Attempts to prepare the binary nitrides of the later alkali metals were not successful until 2002. Na_3N (which is very moisture-sensitive) may be synthesized in a vacuum chamber by depositing atomic sodium and nitrogen onto a cooled sapphire substrate and then heating to room temperature. The structure of Na_3N contrasts sharply with that of Li_3N with Na_3N adopting an anti-ReO_3 structure in which the Na^+ ions are 2-coordinate and the N^{3-} ions are octahedrally sited.

$$6Li + N_2 \longrightarrow 2Li_3N$$

The alkali metals dissolve in mercury, Hg, to give amalgams. Sodium amalgam (which is a liquid only when the percentage of Na is low) is a useful reducing agent in inorganic and organic chemistry; it can be used in aqueous media because there is a large over potential for the discharge of H_2.

Oxides, peroxides, superoxides, suboxides and ozonides

When the group 1 metals are heated in an excess of air or in O_2, the principal products obtained depend on the metal: lithium oxide, Li_2O, sodium peroxide, Na_2O_2, and the superoxides KO_2, RbO_2 and CsO_2.

$$4Li + O_2 \longrightarrow 2Li_2O \qquad \textit{oxide formation}$$

$$2Na + O_2 \longrightarrow Na_2O_2 \qquad \textit{peroxide formation}$$

$$K + O_2 \longrightarrow KO_2 \qquad \textit{superoxide formation}$$

The oxides Na_2O, K_2O, Rb_2O and Cs_2O can be obtained impure by using a limited air supply, but are better prepared by thermal decomposition of the peroxides or superoxides.

All the oxides are strong bases, the basicity increasing from Li_2O to Cs_2O. A peroxide of lithium can be obtained by the action of H_2O_2 on an ethanolic solution of LiOH, but it decomposes on heating.

Sodium peroxide (widely used as an oxidizing agent) is manufactured by heating Na metal on Al trays in air; when pure, Na_2O_2 is colourless and the faint yellow colour usually observed is due to the presence of small amounts of NaO_2. The superoxides and peroxides contain the paramagnetic $[O_2]^-$ and diamagnetic $[O2]^{2-}$ ions respectively . Superoxides have magnetic moments of $\approx 1.73_{\mu B}$ consistent with one unpaired electron. Partial oxidation of Rb and Cs at low temperatures yields suboxides such as Rb_6O, Rb_9O_2, Cs_7O and $Cs_{11}O_3$. Their structures consist of octahedral units of metal ions with the oxygen residing at the centre; the octahedral are fused together by sharing faces. The alkali metal oxides, peroxides and superoxides react with water according to the equations below. One use of KO_2 is in breathing masks where it absorbs H_2O producing O_2 for respiration and KOH, which absorbs exhaled CO_2.

$$M_2O + H_2O \longrightarrow 2MOH$$

$$M_2O_2 + 2H_2O \longrightarrow 2MOH + H_2O_2$$

$$2MO_2 + 2H_2O \longrightarrow 2MOH + H_2O_2 + O_2$$

$$KOH + CO_2 \longrightarrow KHCO_3$$

Sodium peroxide reacts with CO_2 to give Na_2CO_3, rendering it suitable for use in air purification in confined spaces (e.g. in submarines); KO_2 acts similarly but more effectively.

Although all the group 1 peroxides decompose on heating according to equation below, their thermal stabilities depend on cation size; Li_2O_2 is the least stable peroxide, while Cs_2O_2 is the most stable. The stabilities of the superoxides (with respect to decomposition to M_2O_2 and O_2) follow a similar trend.

$$M_2O_2(s) \longrightarrow M_2O(s) + \tfrac{1}{2}O_2(g)$$

Ozonides,MO_3, containing the paramagnetic, bent $[O_3]^-$ ion, are known for all the alkali metals. The salts KO_3, RbO_3 and CsO_3 can be prepared from the peroxides or superoxides by reaction with ozone, but this method fails, or gives low yields, for LiO_3 and NaO_3. These ozonides have recently been prepared in liquid ammonia by the interaction of CsO3 with an ion-exchange resin loaded with either Li^+ or Na^+ ions. The ozonides are violently explosive. Hydroxides

NaOH is used throughout organic and inorganic chemistry wherever a cheap alkali is needed. Solid NaOH (mp 591 K) is often handled as flakes or pellets, and dissolves in water with considerable evolution of heat. Potassium hydroxide (mp 633 K) closely resembles NaOH in preparation and properties. It is more soluble than NaOH in EtOH, in which it produces a low concentration of ethoxide ions; this gives rise to the use of ethanolic KOH in organic synthesis.

$$C_2H_5OH + [OH]^- \rightleftharpoons [C_2H_5O]^- + H_2O$$

The reactions with CO are of interest since they give metal formats (methanoates):

$$NaOH + CO \xrightarrow{450\,K} HCO_2Na$$

Non-metals that do not form stable hydrides, and amphoteric metals, react with aqueous MOH to yield H_2 and oxoanions, e.g. reaction

$$2Al + 2NaOH + 6H_2O \longrightarrow 2Na[Al(OH)_4] + 3H_2$$

Salts of oxoacids: carbonates and hydrogencarbonates

The properties of alkali metal salts of most oxoacids depend on the oxoanion present and not on the cation; thus we tend to discuss salts of oxoacids under the appropriate acid. However, we single out the carbonates and hydrogencarbonates because of their importance. Whereas Li_2CO_3 is sparingly soluble in water, the remaining carbonates of the group 1 metals are very soluble.

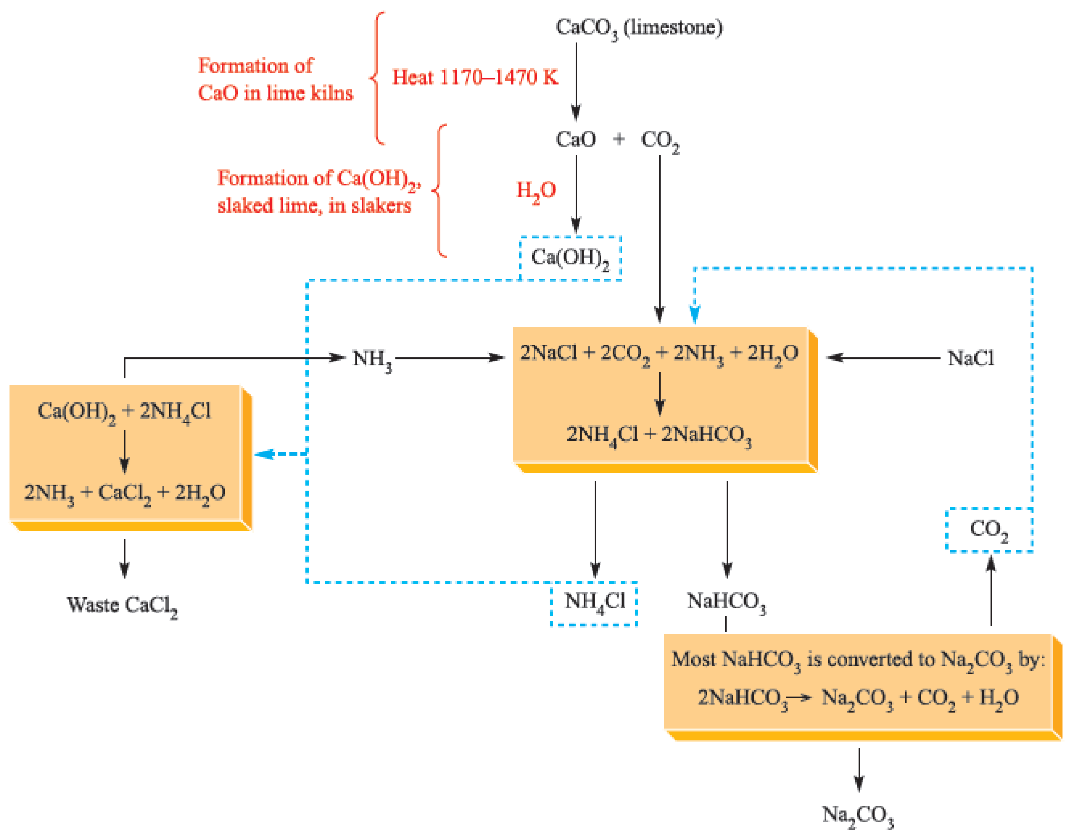
In many countries, sodium carbonate (soda ash) and sodium hydrogencarbonate (commonly called sodium bicarbonate) are manufactured by the Solvay process, but this is being superseded where natural sources of the mineral trona, $Na_2CO_3.NaHCO_3.2H_2O$, are available.

Sodium hydrogencarbonate, although a direct product in the Solvay process, is also manufactured by passing CO_2 through aqueous Na_2CO_3 or by dissolving trona in H_2O saturated with CO_2. Its uses include those as a foaming agent, a food additive (e.g. baking powder) and an effervescent in pharmaceutical products. The Solvay company has now developed a process for using $NaHCO_3$ in pollution control, e.g. by neutralizing SO_2 or HCl in industrial and other waste emissions.

There are some notable differences between Na^+ and other alkali metal $[CO_3]^{2-}$ and $[HCO_3]^-$ salts. Whereas $NaHCO_3$ can be separated in the Solvay process by precipitation, the same is not true of $KHCO_3$. Hence, K_2CO_3 is produced, not via $KHCO_3$, but by the reaction of KOH with CO_2; K_2CO_3 has uses in the manufacture of certain glasses and ceramics. Among its applications, $KHCO_3$ is used as a buffering agent in water treatment and wine production. Lithium carbonate is only sparingly soluble in water; '$LiHCO_3$' has not been isolated. The thermal stabilities of the group 1 metal carbonates with respect to reaction below increase down the group as rM^+ increases, lattice energy being a crucial factor. Such a trend in stability is common to all series of oxo-salts of the alkali metals.

$$M_2CO_3 \xrightarrow{\Delta} M_2O + CO_2$$

The solid state structures of $NaHCO_3$ and $KHCO_3$ exhibit hydrogen bonding. In $KHCO_3$, the anions associate in pairs whereas in $NaHCO_3$, infinite chains are present. In each case, the hydrogen bonds are asymmetrical. Sodium silicates are also of great commercial importance.



Schematic representation of the Solvay process for the manufacture of Na_2CO_3 and $NaHCO_3$ from $CaCO_3$, NH_3 and NaCl. The recycling parts of the process are shown with blue, broken lines.

The group 2 metals (alkaline-earths metals)

The elements known commonly as **alkaline earths metals** have atoms with the *(ns)*^{2} configuration and almost always have the +2 oxidation state in their compounds. The relationships among the elements in group 2 – beryllium, Be, magnesium, Mg, calcium,Ca, strontium,Sr, barium, Ba and radium, Ra – are very like those among the alkali metals. However, Be stands apart from the other group 2 metals to a greater extent than does Li from its homologues. For example, whereas Li^+ and Na^+ salts (with a common counter-ion) usually crystallize with the same lattice type, this is not true for Be(II) and Mg(II) compounds. Beryllium compounds tend either to be covalent or to contain the hydrated $[Be(H_2O)_4]^{2+}$ ion. The high values of the enthalpy of atomization and ionization energies of the Be atom, and the small size and consequent high charge density of a naked Be^{2+} ion, militate against the formation of naked Be^{2+}. Further, the restriction of the valence shell of Be to an octet of electrons excludes the formation of more than four localized 2c-2e bonds by a Be atom. It is noteworthy that Be is the only group 2 metal not to form a stable complex with $[EDTA]^{4-}$.The elements Ca, Sr, Ba and Ra are collectively known as the alkaline earth metals. We shall have little to say about radium; it is radioactive and is formed as $^{226}{}_{88}$Ra (a-emitter, $t_{1/2}$=1622 yr) in the 238 ^{92}U decay series. Uses of radium-226 in cancer treatment have generally been superseded by other radioisotopes.

Occurrence, extraction and uses

Occurrence

Beryllium occurs principally as the silicate mineral beryl, $Be_3Al_2[Si_6O_{18}]$; it is also found in many natural minerals, and precious forms include emerald and aquamarine. Magnesium and calcium are the eighth and fifth most abundant elements, respectively, in the Earth's crust, and Mg, the third most abundant in the sea. The elements Mg, Ca, Sr and Ba are widely distributed in minerals and as dissolved salts in seawater; some important minerals are dolomite ($CaCO_3.MgCO_3$), magnesite ($MgCO_3$), olivine ((Mg,Fe)2SiO_4), carnallite ($KCl.MgCl_2$.6H2O), $CaCO_3$ (in the forms of chalk, limestone and marble), gypsum ($CaSO_4.2H_2O$), celestite (SrSO4), strontianite ($SrCO_3$) and barytes ($BaSO_4$). The natural abundances of Be, Sr and Ba are far less than those of Mg and Ca.

Extraction

Of the group 2 metals, only Mg is manufactured on a large scale. Dolomite is thermally decomposed to a mixture of MgO and CaO, and MgO is reduced by ferrosilicon in Ni vessels; Mg is removed by distillation in vacuo.

$$2MgO + 2CaO + FeSi \xrightarrow{1450\,K} 2Mg + Ca_2SiO_4 + Fe$$

Extraction of Mg by electrolysis of fused $MgCl_2$ is also important and is applied to the extraction of the metal from seawater. The first step is precipitation of $Mg(OH)_2$ by addition of $Ca(OH)_2$ (slaked lime), produced from $CaCO_3$ (available as various calcareous deposits). Neutralization with hydrochloric acid and evaporation of water gives $MgCl_2.xH_2O$, which, after heating at 990 K, yields the anhydrous chloride. Electrolysis of molten $MgCl_2$ and solidification of Mg completes the process.

$$2HCl + Mg(OH)_2 \longrightarrow MgCl_2 + 2H_2O$$

At the cathode: $Mg^{2+}(l) + 2e^- \longrightarrow Mg(l)$

At the anode: $2Cl^-(l) \longrightarrow Cl_2(g) + 2e^-$

Beryllium is obtained from beryl by first heating with Na_2SiF_6, extracting the water-soluble BeF_2 formed, and precipitating $Be(OH)_2$. Beryllium is then produced either by reduction of BeF_2, or by electrolysis of $BeCl_2$ fused with NaCl.

$$BeF_2 + Mg \xrightarrow{1550\,K} Be + MgF_2$$

The production of Ca is by electrolysis of fused $CaCl_2$ and CaF_2; Sr and Ba are extracted by reduction of the corresponding oxides by Al, or by electrolysis of MCl_2 (M= Sr, Ba).

Major uses of the group 2 metals and their compounds

Caution! Beryllium and soluble barium compounds are extremely toxic. Beryllium is one of the lightest metals known, is nonmagnetic, and has a high thermal conductivity and a very high melting point (1560 K); these properties, combined with inertness towards aerial oxidation, render it of industrial importance.

Berilium is used in the manufacture of body parts in high-speed aircraft and missiles, and in communication satellites. Because of its low electron density, Be is a poor absorber of electromagnetic radiation and, as a result, is used in X-ray tube windows. Its high melting point and low cross-section for neutron capture make Be useful in the nuclear energy industry.

The presence of Mg in Mg/Al alloys imparts greater mechanical strength and resistance to corrosion, and improves fabrication properties; Mg/Al alloys are used in aircraft and automobile body parts and lightweight tools. Miscellaneous uses include flares, fireworks and photographic flashlights, and medical applications such as indigestion powders (milk of magnesia, Mg $(OH)_2$) and a purgative (Epsom salts, MgSO4.7H_2O). Both Mg^{2+} and Ca^{2+} ions are catalysts for diphosphate–triphosphate transformations in biological systems; Mg^{2+} is an essential constituent of chlorophylls in green plants.

Uses of compounds of calcium far outnumber those of the metal, with the world production of CaO, $Ca(OH)_2$, CaO.MgO, $Ca(OH)_2$.MgO and $Ca(OH)_2.Mg(OH)_2$ being ≈118 000Mt in 2000. Calcium oxide (quicklime or lime) is produced by calcining limestone and a major use is as a component in building mortar. Dry sand and CaO mixtures can be stored and transported; on adding water, and as CO_2 is absorbed, the mortar sets as solid $CaCO_3$.The sand in themortar is a binding agent.

$$\underset{\text{Quicklime}}{CaO(s)} + H_2O(l) \longrightarrow \underset{\text{Slaked lime}}{Ca(OH)_2(s)} \quad \Delta_r H^o = -65\ kJ\ mol^{-1}$$

$$Ca(OH)_2(s) + CO_2(g) \longrightarrow CaCO_3(s) + H_2O(l)$$

Other important uses of lime are in the steel industry, pulp and paper manufacturing, and extraction of Mg. Calcium carbonate is in huge demand in, for example, steel, glass, cement and concrete manufacturing, and the Solvay process. Recent applications of $CaCO_3$ and $Ca(OH)_2$ with environmental significance are in desulfurization processes. Large quantities of $Ca(OH)_2$ are used to manufacture bleaching powder, $Ca(OCl)_2.Ca(OH)_2.CaCl_2.2H_2O$ and in water treatment.

Calcium fluoride occurs naturally as the mineral fluorspar, and is commercially important as the raw material for the manufacture of HF and F_2. Smaller amounts of CaF_2 are used as a flux in the steel industry, for welding electrode coatings, and in glass manufacture; prisms and cell windows made from CaF_2 are used in spectrophotometers.

$$CaF_2 + \underset{\text{conc}}{2H_2SO_4} \longrightarrow 2HF + Ca(HSO_4)_2$$

The two mineral sources for strontium are the sulfate (celestite) and carbonate (strontianite). In 2001, 75% of strontium used in the US went into the manufacture of faceplate glass in colour television cathode-ray tubes in order to stop X-ray emissions. It is present as SrO and has the added advantage of enhancing television picture quality. Other uses of strontium include ferrite ceramic magnets and pyrotechnics.

Barite (or barytes) is the mineral form of $BaSO_4$. World production in 2001 was ≈ 6600 Mt, with Chile supplying over half this total. The major use of barite is as a weighting material in oil- and gas-well drilling fluids. On a much smaller scale of application, the ability of $BaSO_4$ to stop the passage of X-rays leads to its use as a ‘barium meal’ in radiology for imaging the alimentary tract. Uses of Ba as a ‘getter’ in vacuum tubes arise from its high reactivity with gases including O_2 and N_2.

Physical properties

Selected physical properties of the group 2 elements are listed in the Table below. The intense radioactivity of Ra makes it impossible to obtain all the data for this element.

Property	Be	Mg	Ca	Sr	Ba	Ra
Atomic number, Z	4	12	20	38	56	88
Ground state electronic configuration	$[He]2s^2$	$[Ne]3s^2$	$[Ar]4s^2$	$[Kr]5s^2$	$[Xe]6s^2$	$[Rn]7s^2$
Enthalpy of atomization, $\Delta_a H^o(298\ K)$ / kJ mol^{-1}	324	146	178	164	178	130
Melting point, mp / K	1560	923	1115	1040	1000	973
Boiling point, bp / K	≈3040	1380	1757	1657	1913	1413
Standard enthalpy of fusion, $\Delta_{fus} H^o(mp)$ / kJ mol^{-1}	7.9	8.5	8.5	7.4	7.1	–
First ionization energy, IE_1 / kJ mol^{-1}	899.5	737.7	589.8	549.5	502.8	509.3
Second ionization energy, IE_2 / kJ mol^{-1}	1757	1451	1145	1064	965.2	979.0
Third ionization energy, IE_3 / kJ mol^{-1}	14850	7733	4912	4138	3619	3300
Metallic radius, r_{metal} / pm‡	112	160	197	215	224	–
Ionic radius, r_{ion} / pm*	27	72	100	126	142	148
Standard enthalpy of hydration of M^{2+}, $\Delta_{hyd} H^o(298\ K)$ / kJ mol^{-1}	−2500	−1931	−1586	−1456	−1316	–
Standard entropy of hydration of M^{2+}, $\Delta_{hyd} S^o(298\ K)$ / J K^{-1} mol^{-1}	−300	−320	−230	−220	−200	–
Standard Gibbs energy of hydration of M^{2+}, $\Delta_{hyd} G^o(298\ K)$ / kJ mol^{-1}	−2410	−1836	−1517	−1390	−1256	–
Standard reduction potential, $E^o_{M^{2+}/M}$ / V	−1.85	−2.37	−2.87	−2.89	−2.90	−2.92

Some general points to note from the Table are as follows.

. The general trend in decreasing values of IE_1 and IE2 down the group (see Section 1.10) is broken by the increase in going from Ba to Ra, attributed to the thermodynamic 6s inert pair effect.

(a) High values of IE_3 preclude the formation of M^{3+} ions.

(b) Quoting a value of rion for beryllium assumes that the Be^{2+} ion is present in BeF_2 and BeO, a questionable assumption.
(c) There are no simple explanations for the irregular group variations in properties such as melting points and Δ_aH^o.
(d) Values of Eo for the M^{2+}/M couple are fairly constant (with the exception of Be), and can be explained in a similar way as for the group 1 metals.

Reactivity
Beryllium and magnesium are passivated and are kinetically inert to O_2 and H_2O at ambient temperatures. However, Mg amalgam liberates H_2 from water, since no coating of oxide forms on its surface; Mg metal reacts with steam or hot water.

$$2Be + O_2 \longrightarrow \underset{\text{protective oxide coating on metal}}{2BeO}$$

$$Mg + \underset{\text{steam}}{2H_2O} \longrightarrow Mg(OH)_2 + H_2$$

Beryllium and magnesium dissolve readily in non-oxidizing acids; magnesium is attacked by nitric acid, whereas beryllium reacts with dilute HNO_3 but is passivated by concentrated nitric acid. Magnesium does not react with aqueous alkali, whereas Be forms an amphoteric hydroxide. The metals Ca, Sr and Ba exhibit similar chemical behaviours, generally resembling, but being slightly less reactive than, Na.

They react with water and acids liberating H_2, and the similarity with Na extends to dissolution in liquid NH_3 to give blue solutions containing solvated electrons.

From these solutions, it is possible to isolate hexaammines, $[M(NH_3)_6]$ (M=Ca, Sr, Ba), but these slowly decompose to amides.

$$[M(NH_3)_6] \longrightarrow M(NH_2)_2 + 4NH_3 + H_2 \qquad M = Ca, Sr, Ba$$

When heated, all the group 2 metals combine with O_2, N_2, sulfur or halogens.

$$2M + O_2 \xrightarrow{\Delta} 2MO$$

$$3M + N_2 \xrightarrow{\Delta} M_3N_2$$

$$8M + S_8 \xrightarrow{\Delta} 8MS$$

$$M + X_2 \xrightarrow{\Delta} MX_2 \qquad X = F, Cl, Br, I$$

Differences between the first and later members of group 2 are illustrated by the formation of hydrides and carbides. When heated with H_2, Ca, Sr and Ba form saline hydrides, MH_2, but Mg reacts only under high pressure. In contrast, BeH_2 (which is polymeric) is prepared from beryllium alkyls. Beryllium combines with carbon at high temperatures to give Be_2C which possesses an antifluorite lattice. The other group 2 metals form carbides MC_2 which contain the $[C{\equiv}C]^{2-}$ ion, and adopt NaCl lattices that are elongated along one axis. Whereas Be_2C reacts with water according to the equation below, the carbides of the later metals hydrolyse to yield C_2H_2. CaH_2 is used as a drying agent but its reaction with water is highly exothermic.

$$Be_2C + 4H_2O \longrightarrow 2Be(OH)_2 + CH_4$$

$$MC_2 + 2H_2O \longrightarrow M(OH)_2 + C_2H_2 \qquad M = Mg, Ca, Sr, Ba$$

The carbide Mg_2C_3 (which contains the linear $[C_3]^{4-}$ ion, isoelectronic with CO_2) is formed by heating MgC_2, or by reaction of Mg dust with pentane vapour at 950 K. Reaction of Mg_2C_3 with water produces MeC≡CH.

$$\left[C{=}C{=}C \right]^{4-}$$

Halides

Beryllium halides

Anhydrous beryllium halides are covalent. The fluoride, BeF_2, is obtained as a glass (sublimation point 1073 K) from the thermal decomposition of $[NH_4]_2[BeF_4]$, itself prepared from BeO and NH_3 in an excess of aqueous HF. Molten BeF_2 is virtually a non-conductor of electricity, and the fact that solid BeF_2 adopts a β-cristobalite lattice is consistent with its being a covalent solid. Beryllium difluoride is very soluble in water, the formation of $[Be(H_2O)_4]^{2+}$ being thermodynamically favourable.

Anhydrous $BeCl_2$ (mp 688 K, bp 793 K) can be prepared by the reaction below. This is a standard method of preparing a metal chloride that cannot be made by dehydration of hydrates obtained from aqueous media. In the case of Be, $[Be(H_2O)_4]^{2+}$is formed and attempted dehydration of $[Be(H_2O)_4]Cl_2$ yields the hydroxide, not the chloride.

$$2BeO + CCl_4 \xrightarrow{1070\ K} 2BeCl_2 + CO_2$$

$$[Be(H_2O)_4]Cl_2 \xrightarrow{\Delta} Be(OH)_2 + 2H_2O + 2HCl$$

Halides of Mg, Ca, Sr and Ba

The fluorides of Mg(II), Ca(II), Sr(II) and Ba(II) are ionic, have high melting points, and are sparingly soluble in water, the solubility increasing slightly with increasing cation size (Ksp for MgF_2, CaF_2 , SrF_2 and BaF_2 = 7.42 x 10^{-11}, 1.46 x10^{-10}, 4.33 x10^{-9} and 1.84 x10^{-7} respectively). In contrast to the behaviour of BeF_2, none of the later metal fluorides behaves as a Lewis acid.

Magnesium chloride, bromide and iodide crystallize from aqueous solution as hydrates which undergo partial hydrolysis when heated. The anhydrous salts are, therefore, prepared by reaction.

$$Mg + X_2 \longrightarrow MgX_2 \qquad X = Cl, Br, I$$

A hygroscopic solid absorbs water from the surrounding air but does not become a liquid.

Anhydrous MCl_2, MBr_2 and MI_2 (M = Ca, Sr and Ba) can be prepared by dehydration of the hydrated salts. These anhydrous halides are hygroscopic and $CaCl_2$ (manufactured as a by-product in the Solvay process) is used as a laboratory drying agent and for road de-icing.

Oxides and hydroxides

Beryllium oxide, BeO, is formed by ignition of Be or its compounds in O_2. It is an insoluble white solid. The oxides of the other group 2 metals are usually prepared by thermal decomposition of the corresponding carbonate, for which temperature T refers to $P(CO_2)$ = 1bar.

$$MCO_3 \xrightarrow{T\,K} MO + CO_2 \qquad \left\{ \begin{array}{lr} M = Mg & T = 813\,K \\ \quad Ca & 1173\,K \\ \quad Sr & 1563\,K \\ \quad Ba & 1633\,K \end{array} \right.$$

The action of water on MgO slowly converts it to $Mg(OH)_2$ which is sparingly soluble. Oxides of Ca, Sr and Ba react rapidly and exothermically with water, and absorb CO_2 from the atmosphere. The conversion of CaO to calcium carbide and its subsequent hydrolysis is industrially important, although, as an organic precursor, ethyne is being superseded by ethene.

$$CaO + 3C \xrightarrow{2300\,K} CaC_2 + CO$$

$$CaC_2 + 2H_2O \longrightarrow Ca(OH)_2 + C_2H_2$$

APPLICATIONS

When one looks for a commercially viable refractory oxide, MgO (magnesia) is high on the list: it has a very high melting point (3073 K), can withstand heating above 2300K for long, periods, and is relatively inexpensive. Magnesia is fabricated into bricks for lining furnaces in steelmaking. Incorporating chromium ore into the refractory bricks increases their resistance to thermal shock. Magnesia bricks are also widely used in night-storage radiators: MgO conducts heat extremely well, but also has the ability to store it. In a radiator, the bricks absorb heat which is generated by electrically heated filaments during periods of 'off-peak' consumer rates, and then radiate the thermal energy over relatively long periods.

Group 2 metal peroxides

The **group 2 metal peroxides** MO_2, are known for M = Mg, Ca, Sr and Ba. Attempts to prepare BeO_2 have so far failed, and there is no experimental evidence for any beryllium peroxide compound. As for the group 2 metal peroxides, the stability with respect to the decomposition reaction increases with the size of theM^{2+} ion. This trend arises from the difference between the lattice energies of MO and MO_2.

$$MO_2 \longrightarrow MO + \tfrac{1}{2}O_2 \quad (M = Mg, Ca, Sr, Ba)$$

e.g. $MgO_2 \rightarrow MO + {}^{1}/_{2}\, O_2$

All the peroxides are strong oxidizing agents. Magnesium peroxide (used in toothpastes) is manufactured by reacting $MgCO_3$ or MgO with H_2O_2. Calcium peroxide is prepared by cautious dehydration of $CaO_2.8H_2O$, itself made by reaction.

$$Ca(OH)_2 + H_2O_2 + 6H_2O \longrightarrow CaO_2{\cdot}8H_2O$$

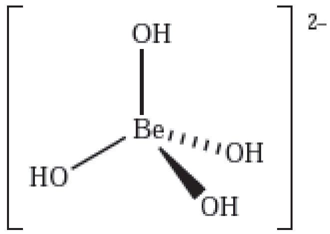
The reactions of SrO and BaO with O_2 (600 K, 200 bar pressure, and 850 K, respectively) yield SrO_2 and BaO_2. Pure BaO_2 has not been isolated and the commercially available material contains BaO and $Ba(OH)_2$. Reactions of the peroxides with acids generate H_2O_2.

$$SrO_2 + 2HCl \longrightarrow SrCl_2 + H_2O_2$$

Hydroxides of Group 2 metals

Beryllium hydroxide is amphoteric and this sets it apart from the hydroxides of the other group 2 metals which are basic.

In the presence of excess $[OH]^-$, $Be(OH)_2$ behaves as a Lewis acid, forming the tetrahedral complex ion, but $Be(OH)_2$ also reacts with acids, e.g.



$$Be(OH)_2 + 2[OH]^- \longrightarrow [Be(OH)_4]^{2-}$$
$$Be(OH)_2 + H_2SO_4 \longrightarrow BeSO_4 + 2H_2O$$

Magnesium hydroxide acts as a weak base, whereas $Ca(OH)_2$, $Sr(OH)_2$ and $Ba(OH)_2$ are strong bases. Soda lime is a mixture of NaOH and $Ca(OH)_2$ and is manufactured from CaO and aqueous NaOH; it is easier to handle than NaOH and is commercially available, being used, for example, as an absorbent for CO_2, and in qualitative tests for $[NH_4]^+$ salts, amides, imides and related compounds which evolve NH_3 when heated with soda lime.

The carbonates of Mg and the later metals are sparingly soluble in water; their thermal stabilities increase with cation size, and this trend can be rationalized in terms of lattice energies. The metal carbonates are much more soluble in a solution of CO_2 than in water due to the formation of $[HCO_3]^-$. However, salts of the type $M(HCO_3)_2$' have not been isolated. Hard water contains Mg^{2+} and Ca^{2+} ions which complex with the stearate ions in soaps, producing insoluble 'scum' in household baths and basins. Temporary hardness is due to the presence of hydrogencarbonate salts and can be overcome by boiling (which shifts equilibrium to the right-hand side causing $CaCO_3$, or similarly $MgCO_3$, to precipitate) or by adding an appropriate amount of $Ca(OH)_2$ (again causing precipitation).

$$Ca(HCO_3)_2(aq) \rightleftharpoons CaCO_3(s) + CO_2(g) + H_2O(l)$$
$$Ca(HCO_3)_2(aq) + Ca(OH)_2(aq) \longrightarrow 2CaCO_3(s) + 2H_2O(l)$$

Permanent hardness is caused by other Mg^{2+} and Ca^{2+} salts (e.g. sulfates). The process of water softening involves passing the hard water through a cation-exchange resin.

Barium sulfate is a sparingly soluble salt ($K_{sp} = 1.07 \times 10^{-10}$) and the formation of a white precipitate of $BaSO_4$ is used as a qualitative test for the presence of sulphate ions in aqueous solution.

$$BaCl_2(aq) + [SO_4]^{2-}(aq) \longrightarrow BaSO_4(s) + 2Cl^-(aq)$$

NOTE: A hydrate X. nH_2O in which n = 1/ 2 is called a hemihydrate; if n = $1\frac{1}{2}$, it is a sesquihydrate.

Diagonal relationships between Li and Mg, and between Be and Al

The properties of Li and its compounds are often considered to be anomalous when compared with those of the later group 1 metals, and that a diagonal relationship exists between Li and Mg. A similar diagonal relationship between Be and Al also exist.

From a comparison of the properties of Li with those of Na and K, or of Li with Mg, it can be seen that Li resembles Mg more closely than it does the later members of group 1. One crucial factor is that the charge densities of Li^+ and Mg^{2+} are similar because the increase in charge is offset by an increase in ion size. Likewise, the charge densities of Be^{2+} and Al^{3+} are similar.

These diagonal relationships result in similarities between the chemistries of Li and Mg, and between Be and Al, and set the first members of each group apart from their heavier congeners. Small cations such as Li^+, Mg^{2+}, Be^{2+} and Al^{3+} possess high charge densities and each has a high polarizing power.

Lithium and magnesium

Some of the chemical properties of Li that make it diagonally related to Mg rather than vertically related to the other alkali metals are summarized below.

- **(a)** Lithium readily combines with N_2 to give the nitride, Li_3N; Mg reacts with N_2 to give Mg_3N_2.
- **(b)** Lithium combines with O_2 to give the oxide Li_2O rather than a peroxide or superoxide; Mg forms MgO. The peroxides of both metals can be formed by reacting LiOH or $Mg(OH)_2$ with H_2O_2.
- **(c)** Lithium and magnesium carbonates decompose readily on heating to give Li_2O and CO_2, and MgO and CO_2 respectively; down the group, the carbonates of the group 1 metals become increasingly stable with respect thermal decomposition.
- **(d)** Lithium and magnesium nitrates decompose on heating according to the equations 1 and 2 below, whereas $NaNO_3$ and the later alkali metal nitrates decompose according to third equation.

$$4LiNO_3 \xrightarrow{\Delta} 2Li_2O + 2N_2O_4 + O_2$$

$$2Mg(NO_3)_2 \xrightarrow{\Delta} 2MgO + 2N_2O_4 + O_2$$

$$2MNO_3 \xrightarrow{\Delta} 2MNO_2 + O_2 \qquad (M = Na, K, Rb, Cs)$$

- **(e)** The Li^+ and Mg^{2+} ions are more strongly hydrated in aqueous solution than are the ions of the later group 1 and 2 metals.
- **(f)** LiF and MgF_2 are sparingly soluble in water; the later group 1 fluorides are soluble.
- **(g)** LiOH is much less soluble in water than the other alkali metal hydroxides; $Mg(OH)_2$ is sparingly soluble.
- **(h)** $LiClO_4$ is much more soluble in water than the other alkali metal perchlorates; $Mg(ClO_4)_2$ and the later group 2 metal perchlorates are very soluble.

Beryllium and aluminium

Representative chemical properties of Be that make it diagonally related to Al rather than vertically related to the later group 2 metals are given below.

- **(a)** The Be^{2+} ion is hydrated in aqueous solution, forming $[Be(H_2O)_4]^{2+}$ in which the Be^{2+} centre significantly polarizes the already polar O-H bonds, leading to loss of H^+ ; similarly, the highly polarizing Al^{3+} makes $[Al(H_2O)_6]^{3+}$ acidic (p_{Ka} = 5.0,).
- **(b)** Be and Al both react with aqueous alkali, liberating H_2; Mg does not react with aqueous alkali.
- **(c)** $Be(OH)_2$ and $Al(OH)_3$ are amphoteric, reacting with both acids and bases; the hydroxides of the later group 2 metals are basic.
- **(d)** $BeCl_2$ and $AlCl_3$ fume in moist air, reacting to give HCl.
- **(e)** Both Be and Al form complex halides, hence the ability of the chlorides to act as Friedel–Crafts catalysts.

SUMMARY OF THE CHEMISTRY OF LITHIUM

	Li	Na K Rb Cs
Element	Hard metal	Soft metals →
Hydroxide	Not a strong base	Strong bases →
Fluoride	Only slightly soluble in water	Readily soluble in water →
Chloride	Slightly hydrolysed in hot solution	Not hydrolysed →
Bromide and iodide	Soluble in many organic solvents	Insoluble in most organic solvents →
Carbonate	Evolves carbon dioxide on heating	Stable to heat →

SELECTED CHEMISTRY OF NON-METALS

THE HALOGENS

The group VIIA (group 17) elements are known as the *halogens,* a word that has its origin in the Greek words *halos* meaning " salt " and *genes* meaning " born " or " formed. " In other words, the halogens are the " salt formers," which at least partially describes their chemical behaviour. The halogens are reactive elements with the result that they are always found combined. The group VIIA (group 17) elements include fluorine, F, chlorine, Cl, bromine, Br, iodine, I and astatine, At. In fact, many halogen compounds are quite stable, so they are not easy to convert to the elements. As a result, the free halogens were not obtained until comparatively modern times. The halogen group (17) is the most electronegative in the periodic table, and all elements readily form halide ions X^-.

Chlorine was prepared in 1774 by Scheele by the reaction of HCl with MnO_2, but it was Sir Humphrey Davy who suggested the name based on the Greek *chloros*, meaning "greenish-yellow." Bromine was discovered in 1826 by Balard, and the name of the element comes from the Greek word *bromos*, meaning "stench." Iodine was obtained from kelp in 1812 by Courtois, and its name is also from the Greek word *ioeides*, which means " violet. " Although elemental fluorine was not available centuries ago, a common mineral that contains fluorine is *fluorspar* or *fluorite,* CaF_2. When heated with sulfuric acid, hydrogen fluoride is produced, and it was known that long ago that the gas would etch glass. Fluorine is such a strong oxidizing agent that no practical means was available to obtain the element from its compounds until electrochemical processes were developed, although it has been prepared by chemical means in recent years. As a result it was 1886 before Moissan succeeded in preparing fluorine. Fluorine is limited to an octet of valence electrons. It is the most electronegative and reactive of all elements and often (as with oxygen) brings out the highest oxidation state in other elements: examples where no corresponding oxide is known include PtF_6 and AuF_5 All isotopes of astatine, which was obtained in

1940, are radioactive, so there is little chemistry to consider for the element. Astatine is the heaviest member of group 17 and is known only in the form of radioactive isotopes, all of which have short half-lives. The longest lived isotope is ^{210}At (t1/2 = 8.1 h). Several isotopes are present naturally as transient products of the decay of uranium and thorium minerals; ^{218}At is formed from the β-decay of ^{218}Po. Other isotopes are artificially prepared, e.g. ^{211}At (an α-emitter) from the nuclear reaction $^{209}_{83}Bi$ (α, 2n) $^{211}_{85}At$, and may be separated by vacuum distillation. In general, At is chemically similar to iodine. Tracer studies (which are the only sources of information about the element) show that At_2 is less volatile than I_2, is soluble in organic solvents, and is reduced by SO_2 to At^- which can be coprecipitated with AgI or TlI. Hypochlorite, ClO^-, or peroxodisulphate, $S_2O_8^{2-}$, oxidizes astatine to an anion that is carried by
IO_3^- (e.g. coprecipitation with $AgIO_3$) and is therefore probably AtO_3^- AtO_3^-. Less powerful oxidizing agents such as Br_2 also oxidize astatine, probably to AtO^- or AtO_2^-.

Most of the differences between fluorine and the later halogens can be attributed to the:

- **(a)** inability of F to exhibit any oxidation state other than -1in its compounds;
- **(b)** relatively small size of the F atom and F^- ion;
- **(c)** low dissociation energy of F_2;
- **(d)** higher oxidizing power of F_2.

Occurrence and Extraction

There are numerous minerals that contain halogens. Minerals that contain fluorine include fluorite, CaF_2, *cryolite*, Na_3AlF_6 , and *fluoroapatite* , $Ca_5(PO_4)_3F$. Fluoroapatite is found with calcium phosphate, which is very important in the production of fertilizers. Fluorite is found in Southeastern Illinois and Northwestern Kentucky, and cryolite is found in Greenland, although it is also produced synthetically because of its use in the electrochemical production of aluminum.

There are many natural sources of chlorine compounds, which is not surprising considering that it is the 20th most abundant element. Salt and salt water are widely available; the Great Salt Lake contains 23 % salt and the Dead Sea contain about 30 %. Because salt is so abundant, most minerals that contain chlorine are not important sources for economic reasons. Bromine is found in some salt brine and in the sea, as are some iodine compounds.

The Elements

Although the most common source of fluorine is CaF_2, the element is not prepared directly from fluorite.

Although producing fluorine by chemical means had not been accomplished for many years, it was accomplished in 1986 by making use of the reaction:

$$K_2MnF_6 + 2\ SbF_5 \rightarrow 2\ KSbF_6 + MnF_3 + \frac{1}{2}\ F_2$$

The reason why this reaction produces fluorine is that MnF_4 is thermodynamically unstable. Therefore, if the very strong Lewis acid SbF_5 removes two fluoride ions from MnF_6^{2-}, the result is MnF_4, and it decomposes to produce fluorine and MnF_3.

The production of chlorine is based on the electrolysis reaction:

$$2\ Na^+ + 2\ Cl^- + 2\ H_2O \xrightarrow{\text{Elect.}} 2\ Na^+ + 2\ OH^- + Cl_2 + H_2$$

This process not only produces chlorine, it is also a way in which enormous quantities of sodium hydroxide are produced with hydrogen being the other product. Two types of cells are in use. The first and by far the most important employs a diaphragm to separate the anode and cathode compartments.

A second type of cell utilizes a mercury cathode with which the sodium forms an amalgam.

$$Na^+ + 2\ Cl^- \xrightarrow[\text{Elect.}]{\text{Hg}} Na/Hg + Cl_2$$

Another but less common industrial reaction is based on the oxidation of HCl in which oxides of nitrogen function as a catalyst.

$$4\ HCl + O_2 \rightarrow 2\ Cl_2 + 2\ H_2O$$

Although more important as a means of producing sodium, the electrolysis of molten NaCl is carried out on an industrial scale.

$$2\ NaCl(l) \xrightarrow{\text{Elect.}} 2\ Na(l) + Cl_2(g)$$

Chlorine can be prepared in a laboratory by the oxidation of Cl^- by means of a suitable oxidizing agent. A procedure used by Scheele in 1774 involves heating a solution of HCl with MnO_2.

$$MnO_2 + 4\ HCl \rightarrow 2\ H_2O + MnCl_2 + Cl_2$$

Because chlorine will oxidize Br^- to give Br_2, sea water is treated with chlorine to produce bromine, which is then removed by blowing air through the mixture. Numerous reactions in which a bromide is oxidized can be carried out on a laboratory scale. Iodine is obtained primarily from sea water in a process similar to that for producing bromine. However, because I^- is easier to oxidize than Br^-, there is considerable latitude in the choice of oxidizing agents. Astatine is produced by making use of a radiochemical technique in which ^{209}Bi is the target and is subjected to α particle.

$$^{209}Bi + {}^{4}He^{2+} \rightarrow 2\ n + {}^{211}At$$

Fluorine occurs only as ^{19}F, but chlorine occurs as 75% ^{35}Cl and 25% ^{37}Cl. Bromine occurs about equally as ^{79}Br and ^{81}Br, but iodine occurs only as ^{127}I. The oxidizing strength of the halogens decreases in the series $F_2 > Cl_2 > Br_2 > I_2$. As expected, the halogens will react with many substances, and they produce a very large number of covalent and ionic compounds. The reaction of fl uorine with water liberates oxygen:

$$2\ F_2 + 2\ H_2O \rightarrow O_2 + 4\ H^+ + 4\ F^-$$

Characteristic properties of the halogens

All halogens have high electronegativity and accept electrons readily.

Properties of halogen

1. Oxidizing power of halogen

All halogens are strong oxidizing agents. Furthermore, free halogen atom also has high electron affinity. They all accepts electron readily. Comparatively, the oxidizing power of fluorine is even stronger than potassium manganate (VII).

Reaction with sodium

$$Cl_{2(g)} + 2Na_{(s)} \rightarrow 2NaCl_{(s)}$$
$$Br_{2(l)} + 2Na_{(s)} \rightarrow 2NaBr_{(s)}$$
$$I_{2(s)} + 2Na_{(s)} \rightarrow 2NaI_{(s)}$$

Reaction with iron(II) ion

$$Cl_{2(g)} + 2Fe^{2+}_{(aq)} \rightarrow 2Cl^-_{(aq)} + 2Fe^{3+}_{(aq)}$$
$$Br_{2(l)} + 2Fe^{2+}_{(aq)} \rightarrow 2Br^-_{(aq)} + 2Fe^{3+}_{(aq)}$$
$$I_{2(l)} + 2Fe^{2+}_{(aq)} \rightarrow 2I^-_{(aq)} + 2Fe^{3+}_{(aq)}$$

Reaction with phosphorus

Most phosphorus form two kinds of halides PX_3 (X = F, Cl, Br, I) and PX_5 (X = F, Cl, Br).

$2P(s) + 3F_2(g) \rightarrow 2PF_3(g)$
$2P(s) + 5F_2(g) \rightarrow 2PF_5(g)$
$2P(s) + 3Cl_2(g) \rightarrow 2PCl_3(l)$
$2P(s) + 5Cl_2(g) \rightarrow 2PCl_5(s)$
$2P(s) + 3Br_2(l) \rightarrow 2PBr_3(l)$
$2P(s) + 5Br_2(l) \rightarrow 2PBr_5(s)$
$2P(s) + 3I_2(s) \rightarrow 2PI_3(s)$

2. Disproportionation of halogen in alkalis

Disproportionation is a reaction in which a single species/ion/substance is simultaneously oxidized and reduced. Except fluorine, all halogens disproportionate in alkali. However, different halogens disproportionate differently in alkali medium at different temperatures.

Reaction of fluorine with alkali

$\underset{0}{2F_{2(g)}} + 2OH^-_{(aq)} \rightarrow \underset{-1}{OF_{2(aq)}} + \underset{-1}{2F^-_{(aq)}} + H_2O_{(l)}$ at room temp. (not a disproportionation reaction)

$\underset{0}{2F_{2(g)}} + 4OH^-_{(aq)} \rightarrow O_{2(g)} + \underset{-1}{4F^-_{(aq)}} + H_2O_{(l)}$ at about 70°C

$F_2(g)$ is extremely electronegative, it only behaves as an oxidizing agent but not a reducing agent.

Disproportionation of chlorine in alkali

$\underset{0}{Cl_{2(g)}} + 2OH^-_{(aq)} \rightarrow \underset{+1}{ClO^-_{(aq)}} + \underset{-1}{Cl^-_{(aq)}} + H_2O_{(l)}$ at room temp.

$\underset{0}{3Cl_{2(g)}} + 6OH^-_{(aq)} \rightarrow \underset{+5}{ClO_3^-{}_{(aq)}} + \underset{-1}{5Cl^-_{(aq)}} + 3H_2O_{(l)}$ at about 70°C

This may be considered as thermal decomposition of $ClO^-(aq)$

$\underset{+1}{3ClO^-_{(aq)}} \rightarrow \underset{-1}{2Cl^-_{(aq)}} + \underset{+5}{ClO_3^-{}_{(aq)}}$

Disproportionation of bromine in alkali

$\underset{0}{3Br_{2(l)}} + 6OH^-_{(aq)} \rightarrow \underset{+5}{BrO_3^-{}_{(aq)}} + \underset{-1}{5Br^-_{(aq)}} + 3H_2O_{(l)}$ at room temp.

Disproportionation of iodine in alkali

$\underset{0}{3I_{2(s)}} + 6OH^-_{(aq)} \rightarrow \underset{+5}{IO_3^-{}_{(aq)}} + \underset{-1}{5I^-_{(aq)}} + 3H_2O_{(l)}$ at room temp.

The difference in mode of disproportionation can be explained by the difference in electronegativity of halogen atom. As the most electronegative element, F tends to attain an oxidation state of -1 only.

With decreasing electronegativity on moving down the group, the element disproportionates and gets a more positive oxidation state more readily.

Example: Explain what is meant by 'disproportionation', and write an equation involving a compound of chlorine to illustrate your answer.
Some disproportionation reactions involving a compound of chlorine:
$H_2O + Cl_2 \rightarrow HCl + HOCl$
$2NaOH + Cl_2 \rightarrow NaCl + NaOCl + H_2O$
$4KClO_3 \rightarrow 3KClO_4 + KCl$
$3HOCl + 3H_2O \rightarrow 3H_3O^+ + ClO_3^- + 2Cl^-$
$3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$

Interhalogen Molecules and Ions of Halogens
One of the interesting attributes of the halogens is the fact that atoms of different types readily form bonds. When chlorine and fluorine are combined and heated, they react to form ClF, a compound known as an *interhalogen*.
$Cl_2 + F_2 \xrightarrow{250\,°C} 2\,ClF$
Because the halogen atoms have seven valence-shell electrons, the outer shell is filled when two atoms form a bond, but when an unshared pair is separated, two additional bonding sites are available. As a result, if ClF were to react with additional fluorine, the expected result would be ClF_3. If another unshared pair of electrons were uncoupled, there would be five bonding sites and the product would be ClF_5. A general formula for compounds containing two different halogens is XX'_n , where X ' is the lighter halogen and *n* is an odd number. The maximum value for *n* is 7, but the only interhalogen having that formula is IF_7. When *n* = 5, the known compounds are ClF_5, BrF_5, and IF_5, and none of the pentachlorides are stable. This is undoubtedly due to the effect of the larger size of the chlorine atom and the fact that it is a weaker oxidizing agent than fluorine. Bonds between other halogens and fl uorine are also more polar, so there is some stability imparted by partial charges on the atoms. As a result of these factors, the only XX'_7 compound is IF_7.

The simplest interhalogen molecules are diatomic molecules, XX ' . All of the molecules having X ' — F are known, and they are generally prepared by combination of the elements. The preparation of ClF was shown earlier, and the preparation of BrF is also from the elements at 10°C when the elements are diluted with nitrogen.
$Br_2 + F_2 \xrightarrow{10\,°C} 2\,BrF$
Disproportionation of BrF occurs with the formation of Br_2 and BrF_3 or BrF_5. Little is known about IF because it so unstable and disproportionates to I_2 and IF_5. Because of the favorable hard-soft interaction of Ag^- and I^-, IF can be obtained by the reaction
$AgF + I_2 \rightarrow IF + AgI$
The halogen chlorides are of course limited to those of Br and I, but although BrCl exists in equilibrium with Cl_2 and Br_2, it is so unstable that it is not obtained in a pure form. Iodine monochloride is a much more stable compound that exists in two forms. The α form consists of ruby-red needles, and the *β* form is a reddish-brown solid. The forms differ in the way that ICl molecules are linked by intermolecular forces in the solid state.
Reactions of the XX' interhalogens are numerous, but they generally constitute reactions in which halogenation occurs, those in which the interhalogen is a nonaqueous solvent, or Lewis acid-base reactions. In keeping with their being more numerous and important, the discussion will deal primarily with the fluorides. In contact with water, the XX' molecules react according to the reaction
$XX' + H_2O \rightarrow H^+ + X'^- + HOX$
The halogen having lower electronegativity is found as the hypohalous acid, because it is the atom that has +1 oxidation state. The monofl uorides are strong fluorinating agents and undergo many reactions typical of the halogens. For example, sulfuryl chloride results from the reaction of chlorine with SO_2, and a mixed halogen compound is obtained by the reaction with ClF.

$ClF + SO_2 \rightarrow ClSO_2F$

Oxidation-addition (oxad) reactions of this type also occur with other compounds as shown in the following examples:

$ClF + CO \rightarrow ClCOF$

$ClF + SF_4 \rightarrow SF_5Cl$

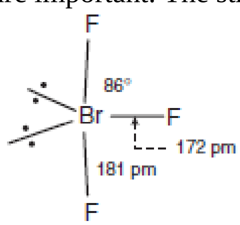
In the reaction with a strong Lewis acid such as AsF_5, ClF behaves as a Lewis base with the fluorine atom being the electron pair donor resulting in transfer of the F^- ion.

$2\ ClF + AsF_5 \rightarrow Cl_2F^+[AsF_6]^-$

ClF can also react as a Lewis base by accepting a fl uoride ion, as illustrated by the reaction

$ClF + KF \rightarrow K^+[ClF_2]^-$

Only three of the possible XX'$_3$ interhalogens (ClF_3, BrF_3, and IF_3, which is stable only at low temperature) are important. The structure of BrF_3 can be shown as



The structure of ClF_3 is similar except for the bond angle being 87.5° and the equatorial and axial bond lengths being 160 and 170 pm, respectively. ClF_3 is produced in large quantities for use as a fluorinating agent, even though it reacts violently in many cases.

$Cl_2 + 3\ F_2 \xrightarrow{250\ °C} 2\ ClF_3$

One use of ClF_3 is in the preparation of UF_6 by the reaction

$3\ ClF_3 + U(s) \rightarrow UF_6(l) + 3\ ClF(g)$

Many organic compounds sometimes react explosively with ClF_3, and it is such a strong fl uorinating agent that it can be used to prepare other interhalogens:

$2\ ClF_3 + Br_2 \rightarrow 2\ BrF_3 + Cl_2$

$10\ ClF_3 + 3\ I_2 \rightarrow 6\ IF_5 + 5\ Cl_2$

$Br_2 + 3\ F_2 \xrightarrow{200\ °C} 2\ BrF_3$

As in the case of ClF, the trifluoride can behave as both a fluoride ion donor and acceptor, as illustrated by the reactions carried out in liquid ClF_3:

$ClF_3 + AsF_5 \rightarrow ClF_2^+[AsF_6]^-$

$ClF_3 + NOF \rightarrow NO^+[ClF_4]^-$

Bromine trifl uoride has also been used extensively as a nonaqueous solvent that behaves in some reactions as if it ionized slightly according to the equation

$2\ BrF_3 \rightleftharpoons BrF_2^+ + BrF_4^-$

It is interesting to note that the electrical conductivity of liquid BrF_3 is six orders of magnitude higher than that of liquid ClF_3, in which autoionization is assumed not to occur. In liquid BrF_3, SbF_5 is an acid because it generates the BrF_2^+ ion.

$SbF_5 + BrF_3 \rightleftharpoons BrF_2^+ + SbF_6^-$

Because $KBrF_4$ contains the BrF_4^- ion, it is a base. Therefore, the reaction

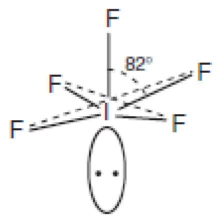
$BrF_2^+SbF_6^- + KBrF_4 \rightarrow 2\ BrF_3 + KSbF_6$

is a neutralization reaction in liquid BrF_3. Because ClF_3 reacts so violently with many substances, BrF_3, which is a slightly milder fl uorinating agent, has been widely used to prepare metal fl uorides from the oxides or the metals.

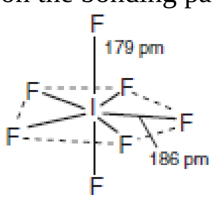
Of the compounds having the formula XX'$_5$, only the fl uorides of Cl, Br, and I have been thoroughly studied. The chlorine compound is prepared by heating chlorine with an excess of fl uorine at high temperature and pressure.

$$Cl_2 + 5\,F_2 \rightarrow 2\,ClF_5$$

A similar reaction but with less stringent conditions is used to produce BrF_5. IF_5 is prepared from the elements at room temperature. The only heptafl uoride is IF_7, which is prepared from the elements using an excess of fluorine and a temperature of 250 to 300°C. All of these penta- and heptafl uorides are extremely strong fl uorinating agents. There are 12 electrons surrounding the central atom, which results in a square-base pyramid structure. The structure of the pentafl uoro compounds is represented by IF_5 , which can be shown as



in which the iodine resides below the base of the pyramid as a result of the repulsion of the unshared pair on the bonding pairs. The structure of IF_7 is a pentagonal bipyramid.



All three of the pentafluorides react violently if not explosively with many substances. In the case of ClF_5, the reaction with water is

$$ClF_5 + 2\,H_2O \rightarrow 4\,HF + FClO_2$$

BrF_5 reacts with water to produce HF and $HBrO_3$ if the water is diluted with a less reactive solvent such as acetonitrile. IF_5 reacts with water to produce HF and HIO_3.

$$IF_5 + 3\,H_2O \rightarrow 5\,HF + HIO_3$$

The low conductivity of liquid IF_5 is believed to be as a result of slight ionization,

$$2\,IF_5 \rightleftharpoons IF_4^{\,+} + IF_6^{\,-}$$

and compounds that contain each of the product ions have been isolated. IF_5 boils at approximately 101°C, and at that temperature it reacts with KF to give IF_6^- .

$$KF + IF_5 \rightarrow KIF_6$$

Although IF_5 is quite stable, when heated to 500°C it disproportionates to give I_2 and IF_7.

Because of the size of the iodine atom and the fact that it is easier to oxidize to the +7 state, IF_7 is the only XX'$_7$ interhalogen. It is prepared by the reaction of IF_5 and F_2 at elevated temperatures, and like other halogen fl uorides it is a strong fl uorinating agent. When it reacts with water, HF and HIO_4 are produced.

$$IF_7 + 4\,H_2O \rightarrow HIO_4 + 7\,HF$$

Interhalogens such as ClF_3, BrF_3, and BrF_5 will make fluorides out of almost anything. Metals, nonmetals, and many compounds are converted to the fluorides, and in order to make the reactions less vigorous, diluents are frequently added. If the reaction is being carried out in the gas phase, a relatively unreactive gas such as nitrogen or a noble gas is added. In most cases, the fl uorination of a nonmetal produces the fluoride of the nonmetal in its highest oxidation state. For example, phosphorus is converted to PF_5 and sulfur is converted to SF_6, although it is sometimes possible to limit the fluorination somewhat by controlling the ratio of the reactants. When ClF_3 or BrF_3 is the fluorinating agent, the halogen monofluoride is frequently a product. For example, with boron the reaction is

$$B_{12} + 18\,ClF_3 \rightarrow 12\,BF_3 + 18\,ClF$$

Halogen fluorides convert oxides to fluorides or oxyfluorides. The following equations illustrate reactions of this type:

$$P_4O_{10} + 4\ ClF_3 \rightarrow 4\ POF_3 + 2\ Cl_2 + 3\ O_2$$

$$SiO_2 + 2\ IF_7 \rightarrow 2\ IOF_5 + SiF_4$$

$$6\ NiO + 4\ ClF_3 \rightarrow 6\ NiF_2 + 2\ Cl_2 + 3\ O_2$$

Hydrogen Halides

Among the most useful compounds of the halogens are the hydrogen halides. Hydrogen iodide is not very stable, and it cannot be efficiently prepared by direct combination of the elements. Rather, as is the case with Te, P, and some other elements, it is easier to prepare compound containing iodine, and then carry out a hydrolysis reaction. For example,

$$P_4 + 6\ I_2 \rightarrow 4\ PI_3$$

$$PI_3 + 3\ H_2O \rightarrow 3\ HI + H_3PO_3$$

The properties of HF reflect the strong hydrogen bonding that persists even in the vapor state. As a result of its high polarity and dielectric constant, liquid HF dissolves many ionic compounds.

All of the hydrogen halides are very soluble in water, and acidic solutions result. Although HF is a weak acid, the others are strong and are almost completely dissociated in dilute solutions. HCl, HBr, and HI form constant-boiling mixtures with water that contain 20.2%, 47.6%, and 53% of the acid, respectively.

There are many ways in which the hydrogen halides can be prepared. Heating a salt containing the halide ion with a nonvolatile acid is the usual way in which HF, HCl, and HBr are obtained in laboratory experiments.

$$2\ NaCl + H_2SO_4 \rightarrow 2\ HCl + Na_2SO_4$$

$$KBr + H_3PO_4 \rightarrow HBr + KH_2PO_4$$

$$CaF_2 + H_2SO_4 \rightarrow 2\ HF + CaSO_4$$

Because HI is relatively unstable, this method is not appropriate for producing the compound. Instead, as mentioned earlier, HI is better prepared by hydrolysis reactions. If a halide salt is treated with an acid that is an oxidizing agent, a redox reaction occurs in which I_2 is produced.

$$8\ HI + H_2SO_4 \rightarrow H_2S + 4\ H_2O + 4\ I_2$$

$$2\ HI + HNO_3 \rightarrow HNO_2 + I_2 + H_2O$$

Other than the iodine compound, the hydrogen halides can be obtained by reactions of the elements, but the reactions can be explosive. A mixture of H_2 and Cl_2 will explode if the reaction is initiated with a burst of light, which separates some Cl_2 molecules to produce Cl^- . The reaction between H_2 and Cl_2 is a classic case of a free radical reaction. A vast number of hydrolysis reactions yield hydrogen halides.

$$SOBr_2 + 2\ H_2O \rightarrow 2\ HBr + H_2SO_3$$

$$SiCl_4 + 3\ H_2O \rightarrow 4\ HCl + H_2SiO_3$$

$$PI_3 + 3\ H_2O \rightarrow 3\ HI + H_3PO_3$$

On an industrial scale, HCl is also obtained in the chlorination of hydrocarbons.

Halogen Oxides

Compounds containing fluorine and oxygen are actually fluorides, and although O_2F_2 and OF_2 will be described, neither has significant uses. The compounds of chlorine and oxygen include some that are commercially important, but they are explosive in nature. Consequently, the discussion of halogen oxides will deal more heavily with the compounds containing chlorine.

Oxygen difluoride, OF_2 (m.p. - 223.8°C, b.p.- 145°C) is a pale yellow, poisonous gas. The molecule has a bent structure (C_{2v}), and the bond angle is 103.2°. OF_2 can be prepared by the reaction of fluorine with dilute NaOH or the electrolysis of aqueous solutions containing HF and KF. The reaction of OF_2 with water can be shown as

$$H_2O + OF_2 \rightarrow O_2 + 2\ HF$$

The reaction of OF_2 with HCl produces HF and liberates chlorine.

$$4\ HCl + OF_2 \rightarrow 2\ HF + H_2O + 2\ Cl_2$$

When heated to a temperature above 250°C, OF_2 decomposes into O_2 and F_2.

Dioxygen difl uoride, O_2F_2, is a yellow-orange solid with a melting point of -163°C.

Oxides of chlorine are both more numerous and more useful than those of fluorine. These oxides are the anhydrides of several important acids, and the oxyanions of those acids constitute the hypochlorites, chlorites, chlorates, and perchlorates. The fi rst oxide of chlorine, Cl_2O (m.p. -20°C, b.p.- 2°C), contains chlorine in the +1 oxidation state. It can be prepared by the reaction

$$2\ HgO + 2\ Cl_2 \rightarrow HgCl_2 \cdot HgO + Cl_2O$$

This oxide is soluble in water, with which it reacts to produce hypochlorous acid.

$$H_2O + Cl_2O \rightarrow 2\ HOCl$$

The bond angle in the bent (C_{2v}) molecule is 110.8°. One interesting reaction of Cl_2O is that with N_2O_5, which can be shown as

$$Cl_2O + N_2O_5 \rightarrow 2\ ClNO_3$$

but other products that depend on the reaction conditions are also produced. Cl_2O converts SbF_5 into an oxychloride.

$$Cl_2O + SbCl_5 \rightarrow SbOCl_3 + 2\ Cl_2$$

Chlorine dioxide, ClO_2 (m.p. -60°C, b.p. 11°C) is an explosive gas ($\Delta H_f°$ = +105 kJ/mol) that has been used industrially as a gaseous bleach for wood pulp, textiles, and fl our, and as a bactericide in water treatment. For most industrial applications, ClO_2 is produced on site. The molecule has a bent structure with a bond angle of 118°. Among the reactions that can be used to prepare ClO_2 is the following:

$$2\ HClO_3 + H_2C_2O_4 \rightarrow 2\ ClO_2 + 2\ CO_2 + 2\ H_2O$$

SO_2 acts as a reducing agent toward ClO_3^- , with the reduction product being ClO_2.

$$2\ NaClO_3 + H_2SO_4 + SO_2 \rightarrow 2\ ClO_2 + 2\ NaHSO_4$$

In industrial settings, ClO_2 is prepared from a solution of $NaClO_3$.

$$2\ ClO_3^- + 4\ HCl \rightarrow 2\ ClO_2 + Cl_2 + 2\ H_2O + 2\ Cl^-$$

When solid $NaClO_3$ reacts with gaseous chlorine, ClO_2 is produced.

$$4\ NaClO_3 + 3\ Cl_2 \rightarrow 6\ ClO_2 + 4\ NaCl$$

When chlorosulfonic acid reacts with a perchlorate, the product is Cl_2O_4, which is more accurately written as $ClOClO_3$, chlorine perchlorate.

$$CsClO_4 + ClOSO_2F \xrightarrow{-45\ °C} CsSO_3F + ClOClO_3$$

Although technically a chlorine oxide, chlorine perchlorate is of little importance. When ozone reacts with chlorine dioxide, the reaction produces Cl_2O_6 dichlorine hexoxide.

$$2\ O_3 + 2\ ClO_2 \rightarrow Cl_2O_6 + 2\ O_2$$

This compound is sometimes represented as $[ClO_2^+][ClO_4^-]$ on the basis of its reactions. One additional oxide of chlorine needs to be described. That compound is dichlorine heptoxide, Cl_2O_7 (m.p. -91.5°C and b.p. 82°C), which is produced when HClO 4 is dehydrated with P_4O_{10}.

$$4\ HClO_4 + P_4O_{10} \rightarrow 2\ Cl_2O_7 + 4\ HPO_3$$

The oxides of bromine and iodine are not numerous, and they are not particularly important. Bromine monoxide, Br_2O is obtained by the reaction

$$2\ HgO + 2\ Br_2 \rightarrow HgO \cdot HgBr_2 + Br_2O$$

and this oxide can be oxidized by ozone to produce BrO_2.

$$4\ O_3 + 2\ Br_2O \xrightarrow{-78\,°C} 3\ O_2 + 4\ BrO_2$$

Of the oxides of iodine that have been reported, the most common is I_2O_5, which is prepared by the dehydration of HIO_3.

$$2\ HIO_3 \xrightarrow{>170\,°C} I_2O_5 + H_2O$$

I_2O_5 is relatively stable but decomposes at 300°C. It is a strong oxidizing agent that is used in quantitative determination of CO, which it oxidizes to CO_2.

$$I_2O_5 + 5\ CO \rightarrow I_2 + 5\ CO_2$$

The reaction of I_2O_5 with water produces iodic acid, HIO_3. Although I_2O_4, I_4O_9, and I_2O_7 have been reported, they are of minor importance.

Oxyhalides

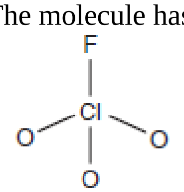
Oxyhalides of sulfur, nitrogen, and phosphorus are important compounds that have been described earlier. A considerable number of compounds of this type are known for the group VIIA elements, with most of them being fluorides. The known compounds include $FClO_2$, F_3ClO, $FClO_3$, F_3ClO_2, ClO_3OF, $FBrO_2$, $FBrO_3$, FIO_2, F_3IO, FIO_3, F3IO$_2$, and F_5IO. Most of these compounds react as strong oxidizing or fluorinating agents. They also undergo hydrolysis reactions such as

$$FClO_2 + H_2O \rightarrow HClO_3 + HF$$

As we have seen in other instances, strong Lewis acids can remove F^- from many types of molecules to generate cations. That is also the case with oxyf uorides.

$$FClO_2 + AsF_5 \rightarrow ClO_2^+AsF_6^-$$

Perchloryl fluoride, $FClO_3$, is stable at temperatures up to about 500°C, and it hydrolyzes only slowly. The molecule has the pyramidal (C_{3v}) structure



Hypohalous Acids and Hypohalites

When halogens dissolve in water, a reaction takes place to a slight extent that can be shown as

$$H_2O + X_2 \rightleftharpoons H^+ + X^- + HOX$$

The weak acids having the formula HOX are the hypohalous acids (they have K_a values of 3 x10^{-8}, 2 x 10^{-9}, and 1 x 10^{-11} for HOCl, HOBr, and HOI, respectively). They are oxidizing agents, as are the salts of the acids. HOF (m.p. -117°C) has been prepared, but it is unstable and has no serious uses. It does react with fluorine to produce OF_2 and HF.

$$HOF + F_2 \xrightarrow{H_2O} HF + OF_2$$

Although solutions containing the hypohalous acids are produced by dissolving the halogens in water, the process must be carried out at low temperature. Hypohalites are also produced when halogens react in basic solution.

$$X_2 + OH^- \rightarrow HOX + X^-$$

However, if the solutions are hot, the hypohalites undergo disproportionation to produce XO_3^- and X^-.

$$3\ HOX \rightarrow HXO_3 + 2\ HX$$

As a result, the products of reactions of halogens in basic solutions depend on pH, temperature, and the concentrations of the reactants. The acids are unstable compounds that decompose in two ways:

$$3\ HOX \rightarrow HXO_3 + 2\ HX$$

Although solutions of sodium hypochlorite are useful oxidizing agents, the solid is not very stable.

Calcium hypochlorite is used in bleaches, swimming pool treatments, and so forth. The decomposition of OCl^- is catalyzed by compounds containing transition metals.

$$2\ OCl^- \rightarrow 2\ Cl^- + O_2$$

The reaction has been studied kinetically, and it is found that when a solid catalyst (CoO and/or Co_2O_3) formed by adding $Co(NO_3)_2$ to a solution of NaOCl is present, the rate of the reaction is determined by the surface area of the catalyst. Hypobromites and hypoiodites are good oxidizing agents, though less commonly used than hypochlorites. One use of NaOBr is in analysis where it is used to oxidize urea and NH_4^+ to produce N_2.

$$2\ NH_4Cl + 3\ NaOBr + 2\ NaOH \rightarrow 3\ NaBr + 2\ NaCl + 5\ H_2O + N_2$$

$$CO(NH_2)_2 + 3\ NaOBr + 2\ NaOH \rightarrow 3\ NaBr + Na_2CO_3 + 3\ H_2O + N_2$$

Halous Acids and Halites

Of the acids having the formula HXO_2, only the chlorine compound is stable enough to be of much use. Sodium chlorite is a useful oxidizing agent that results when chlorine dioxide reacts with sodium hydroxide.

$$2\ NaOH + 2\ ClO_2 \rightarrow NaClO_2 + NaClO_3 + H_2O$$

Acidifying a solution containing a chlorite gives a solution of $HClO_2$. Halite salts are used primarily as bleaching agents.

Halic Acids and Halates

The *halic* acids may not be industrially important, but their salts certainly are. Sodium chlorate is produced in enormous quantities and used in processes in which its oxidizing strength makes it a versatile bleach. One such use is in making paper, and potassium chlorate is used as the oxidizing agent in matches.

The halic acids are strong, as would be expected on the basis of the formula $(HO)_a XO_b$ with $b = 2$, but they are not stable as pure compounds. The anions are isoelectronic with SO_3^{2-}, and they have the C_{3v} structure. In the anions, the observed bond angles are 106° in ClO_3^- and BrO_3^- and 98 to 100° in IO_3^- depending on the nature of the cation.

Numerous reactions are useful for producing the acids and salts containing the halogens in the +5 oxidation state. As was shown earlier, the disproportionation reaction

$$3\ HOX \rightarrow HXO_3 + 2\ HX$$

is one such reaction. Another important type of disproportionation reaction of the halogens is illustrated by the general equation

$$3\ X_2 + 6\ OH^- \rightarrow XO_3^- + 5\ X^- + 3\ H_2O$$

The acids are frequently prepared only as aqueous solutions by acidifying a solution of a metal halate. The following reaction is useful because the other product, $BaSO_4$, is insoluble:

$$Ba(ClO_3)_2 + H_2SO_4 \rightarrow BaSO_4 + 2\ HClO_3$$

The industrial production of sodium chlorate is by electrolysis of an aqueous solution of NaCl, which can be shown as

$$NaCl + 3\ H_2O \xrightarrow{\text{Elect.}} NaClO_3 + 3\ H_2$$

but this equation is an over-simplification because the process involves several steps.

At temperatures near or above the melting points, chlorates disproportionate.

$$4\ ClO_3^- \rightarrow 3\ ClO_4^- + Cl^-$$

However, decomposition of the acids takes place as shown in the equations

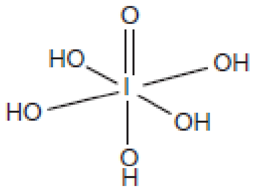
$$4\ HBrO_3 \rightarrow 2\ H_2O + 5\ O_2 + 2\ Br_2$$

$$8\ HClO_3 \rightarrow 4\ HClO_4 + 2\ H_2O + 3\ O_2 + 2\ Cl_2$$

Chlorates, bromates, and iodates are strong oxidizing agents that are useful in many industrial and synthetic processes.

Perhalic Acids and Perhalates

The oxyacids containing chlorine and bromine in the +7 oxidation state are very strong acids. This can be seen when the formulas are written as $(HO)XO_3$, which corresponds to $b = 3$ in the general formula $(HO)_a XO_b$. The acids have as their anhydrides the oxides X_2O_7, but the situation with iodine is different from that of the other halogens. The reaction of I_2O_7 with excess water produced an acid having the formula H_5IO_6, and the (C_{4v}) structure of the molecule can be shown as



However, when I_2O_7 reacts with water a series of reactions take place that can be shown as follows:

$$I_2O_7 \xrightarrow{H_2O} HIO_4 \xrightarrow{H_2O} H_4I_2O_9 \xrightarrow{H_2O} H_3IO_5 \xrightarrow{H_2O} H_5IO_6$$

RADIOACTIVITY

Radioactivity is the spontaneous disintegration of atomic nuclei. It is the spontaneous emission of radiation from an element.

Nucleons are particles comprising the nucleus; protons and neutrons.

Radioisotope is a radioactive isotope of an element.

Radionuclide is radioactive nuclide (the different atomic forms of all elements - in contrast to isotopes, which are different atomic forms of a single element).

Chemical properties are determined by electron distributions and are only indirectly influenced by atomic nuclei. Nuclear reactions involve changes in the composition of nuclei. These extraordinary processes are often accompanied by the release of tremendous amounts of energy and by transmutations of elements.

Some differences between nuclear reactions and ordinary chemical reactions follow.

Nuclear Reaction	Ordinary Chemical Reaction
1. Elements may be converted from one to another.	1. No new elements can be produced.
2. Particles within the nucleus are involved.	2. Only the electrons participate.
3. Tremendous amounts of energy are released or absorbed.	3. Relatively small amounts of energy are released or absorbed.
4. Rate of reaction is not influenced by external factors.	4. Rate of reaction depends on factors such as concentration, temperature, catalyst, and pressure.

Medieval alchemists spent years trying to convert other metals into gold without success. Years of failure and the acceptance of Dalton’s atomic theory early in the nineteenth century convinced scientists that one element could not be converted into another. Then, in 1896 Henri Becquerel discovered “radioactive rays” **(natural radioactivity)** coming from a uranium compound. Ernest Rutherford’s study of these rays showed that atoms of one element may indeed be converted into atoms of other elements by spontaneous nuclear disintegrations. Many years later it was shown that nuclear reactions initiated by bombardment of nuclei with accelerated subatomic particles or other nuclei can also transform one element into another—accompanied by the release of radiation **(induced radioactivity).**

Becquerel's discovery led other researchers, including Marie and Pierre Curie, to discover and study new radioactive elements.

THE NUCLEUS

Recall that the neutrons and protons together constitute the nucleus, with the electrons occupying essentially empty space around the nucleus. The nucleus is only a minute fraction of the total volume of an atom, yet nearly all the mass of an atom resides in the nucleus. Thus, nuclei are extremely dense. It has been shown experimentally that nuclei of all elements have approximately the same density, 2.4×10^{14} g/cm^3.

TABLE A: FUNDAMENTAL PARTICLES OF MATTER

Particle	Mass	Charge
Electron (e^-)	0.00054858 amu	1−
Proton (p or p^+)	1.0073 amu	1+
Neutron (n or n^0)	1.0087 amu	none

NEUTRON–PROTON RATIO AND NUCLEAR STABILITY

The term **"nuclide"** is used to refer to different atomic forms of all elements. The term "isotope" applies only to different forms of the same element. Most naturally occurring nuclides have even numbers of protons and even numbers of neutrons; 157 nuclides fall into this category. Nuclides with odd numbers of both protons and neutrons are least common (there are only four), and those with odd–even combinations are intermediate in abundance. Furthermore, nuclides with certain "magic numbers" of protons and neutrons seem to be especially stable. Nuclides with a number of protons *or* a number of neutrons *or* a sum of the two equal to 2, 8, 20, 28, 50, 82, or 126 have unusual stability.

Examples are $^{4}_{2}He$, $^{16}_{8}O$, $^{42}_{20}Ca$, $^{88}_{38}Sr$, and $^{208}_{82}Pb$.

This suggests an energy level (shell) model for the nucleus similar to the shell model of electron configurations. For low atomic numbers, the most stable nuclides have equal numbers of protons and neutrons ($N = Z$). Above atomic number 20, the most stable nuclides have more neutrons than protons.

NUCLEAR STABILITY AND BINDING ENERGY

Experimentally, we observe that the masses of atoms other than $^{1}_{1}H$ are always *less* than the sum of the masses of their constituent particles. We now know why this *mass deficiency* occurs. We also know that the mass deficiency is in the nucleus of the atom and has nothing to do with the electrons; however, *because tables of masses of isotopes include the electrons, we shall also include them.*

The **mass deficiency, Δm,** for a nucleus is the difference between the sum of the masses of electrons, protons, and neutrons in the atom (calculated mass) and the actual measured mass of the atom.

Δm = (sum of masses of all e^-, p^+ and n^0) - (actual mass of atom)
For most naturally occurring isotopes, the mass deficiency is only about 0.15 % or less of the calculated mass of an atom.

EXAMPLE
Mass Deficiency
Calculate the mass deficiency for chlorine-35 atoms in amu/atom and in g/mol atoms. The actual mass of a chlorine-35 atom is 34.9689 amu.
Plan
We first find the numbers of protons, electrons, and neutrons in one atom. Then we determine the "calculated" mass as the sum of the masses of these particles. The mass deficiency is the actual mass subtracted from the calculated mass. This deficiency is commonly expressed either as mass per atom or as mass per mole of atoms.
Solution
Each atom of $^{35}_{17}Cl$contains 17 protons, 17 electrons, and (35 -17) = 18 neutrons. First we sum the masses of these particles.

$$\begin{array}{rll} \text{protons:} & 17 \times 1.0073 \text{ amu} & = 17.124 \text{ amu} \qquad \text{(masses from Table} \\ \text{electrons:} & 17 \times 0.00054858 \text{ amu} & = 0.0093 \text{ amu} \\ \text{neutrons:} & 18 \times 1.0087 \text{ amu} & = 18.157 \text{ amu} \\ \hline & \text{sum} & = 35.290 \text{ amu} \longleftarrow \text{calculated mass} \end{array}$$

Then we subtract the actual mass from the "calculated" mass to obtain Δm.
Δm = 35.290 amu - 34.9689 amu = 0.321 amu(mass deficiency in one atom)
We have calculated the mass deficiency in amu/atom.

Recall that 1 gram is 6.022 x 1023 amu. We can show that a number expressed in amu/atom is equal to the same number in g/mol of atoms.

$$\frac{?\text{ g}}{\text{mol}} = \frac{0.321 \text{ amu}}{\text{atom}} \times \frac{1 \text{ g}}{6.022 \times 10^{23} \text{ amu}} \times \frac{6.022 \times 10^{23} \text{ atoms}}{1 \text{ mol } ^{35}\text{Cl atoms}}$$

$$= \boxed{0.321 \text{ g/mol of } ^{35}\text{Cl atoms}} \longleftarrow \text{(mass } \textit{deficiency} \text{ in a mole of Cl atoms)}$$

What has happened to the mass represented by the mass deficiency? In 1905, Einstein set forth the Theory of Relativity. He stated that matter and energy are equivalent. An obvious corollary is that matter can be transformed into energy and energy into matter. The transformation of matter into energy occurs in the sun and other stars. It happened on earth when controlled nuclear fission was achieved in 1939 . The reverse transformation, energy into matter, has not yet been accomplished on a large scale. Einstein's equation, is $E = mc^2$. E represents the amount of energy released, m the mass of matter transformed into energy, and c the speed of light in a vacuum, 2.997925 x 10^8 m/s (usually rounded off to 3.00 x 10^8 m/s).

A mass deficiency represents the amount of matter that would be converted into energy and released if the nucleus were formed from initially separate protons and neutrons. This energy is the **nuclear binding energy, *BE*.** It provides the powerful short-range force that holds the nuclear particles (protons and neutrons) together in a very small volume. We can rewrite the Einstein relationship as:

$BE = (\Delta m)c^2$

Specifically, if 1 mole of ^{35}Cl nuclei were to be formed from 17 moles of protons and 18 moles of neutrons, the resulting mole of nuclei would weigh 0.321 gram less than the original collection of protons and neutrons in the above example. Nuclear binding energies may be expressed in many different units, including kilojoules/ mole of atoms, kilojoules/gram of atoms, and megaelectron volts/nucleon. Some useful equivalence are 1 megaelectron volt (MeV) = 1.60×10^{-13} J and 1 joule (J) = 1 $kg.m^2/s^2$

Let us use the value of Δm for ^{35}Cl (0.321 g/mol) from the previous example atoms to calculate their nuclear binding energy.

EXAMPLE

Nuclear Binding Energy

Calculate the nuclear binding energy of ^{35}Cl in (a) kilojoules per mole of Cl atoms, (b) kilojoules per gram of Cl atoms, and (c) megaelectron volts per nucleon.

Solution

The mass deficiency is 0.321 g/mol = 3.21×10^{-4} kg/mol.

$$\text{(a)}\ BE = (\Delta m)c^2 = \frac{3.21 \times 10^{-4}\ \text{kg}}{\text{mol }^{35}\text{Cl atoms}} \times (3.00 \times 10^8\ \text{m/s})^2 = 2.89 \times 10^{13}\ \frac{\text{kg}\cdot\text{m}^2/\text{s}^2}{\text{mol }^{35}\text{Cl atoms}}$$

$$= 2.89 \times 10^{13}\ \text{J/mol }^{35}\text{Cl atoms} = \boxed{2.89 \times 10^{10}\ \text{kJ/mol of }^{35}\text{Cl atoms}}$$

(b) From Example 26-1, the actual mass of ^{35}Cl is

$$\frac{34.9689\ \text{amu}}{^{35}\text{Cl atom}} \quad \text{or} \quad \frac{34.9689\ \text{g}}{\text{mol }^{35}\text{Cl atoms}}$$

We use this mass to set up the needed conversion factor.

$$BE = \frac{2.89 \times 10^{10}\ \text{kJ}}{\text{mol of }^{35}\text{Cl atoms}} \times \frac{1\ \text{mol }^{35}\text{Cl atoms}}{34.9689\ \text{g }^{35}\text{Cl atoms}} = \boxed{8.26 \times 10^8\ \text{kJ/g }^{35}\text{Cl atoms}}$$

(c) The number of nucleons in *one* atom of ^{35}Cl is 17 protons + 18 neutrons = 35 nucleons.

$$BE = \frac{2.89 \times 10^{10}\ \text{kJ}}{\text{mol of }^{35}\text{Cl atoms}} \times \frac{1000\ \text{J}}{\text{kJ}} \times \frac{1\ \text{MeV}}{1.60 \times 10^{-13}\ \text{J}} \times \frac{1\ \text{mol }^{35}\text{Cl atoms}}{6.022 \times 10^{23}\ ^{35}\text{Cl atoms}} \times \frac{1\ ^{35}\text{Cl atom}}{35\ \text{nucleons}}$$

$$= \boxed{8.57\ \text{MeV/nucleon}}$$

RADIOACTIVE DECAY

Nuclei whose neutron-to-proton ratios lie outside the stable region undergo spontaneous radioactive decay by emitting one or more particles or electromagnetic rays or both. Common types of radiation emitted in decay processes are summarized in the Table below.

TABLE B Common Types of Radioactive Emissions(or radiations)

Type and Symbol[a]	Identity	Mass (amu)	Charge	Velocity	Penetration
beta (β^-, ${}_{-1}^{0}\beta$, ${}_{-1}^{0}e$)	electron	0.00055	1−	≤90% speed of light	low to moderate, depending on energy
positron[b] (${}_{+1}^{0}\beta$, ${}_{+1}^{0}e$)	positively charged electron	0.00055	1+	≤90% speed of light	low to moderate, depending on energy
alpha (α, ${}_{2}^{4}\alpha$, ${}_{2}^{4}He$)	helium nucleus	4.0026	2+	≤10% speed of light	low
proton (${}_{1}^{1}p$, ${}_{1}^{1}H$)	proton, hydrogen nucleus	1.0073	1+	≤10% speed of light	low to moderate, depending on energy
neutron (${}_{0}^{1}n$)	neutron	1.0087	0	≤10% speed of light	very high
gamma (${}_{0}^{0}\gamma$) ray	high-energy electromagnetic radiation such as X-rays	0	0	speed of light	high

[a]*The number at the upper left of the symbol is the number of nucleons, and the number at the lower left is the number of positive charges.*

[b]*On the average, a positron exists for only about a nanosecond (1×10^{-9} second) before colliding with an electron and being converted into the corresponding amount of energy.*

The particles can be emitted at different kinetic energies. In addition, radioactive decay often leaves a nucleus in an excited (high-energy) state. Then the decay is followed by gamma ray emission.

$$\text{(excited nucleus)} \longrightarrow {}_{Z}^{M}E^{*} \longrightarrow {}_{Z}^{M}E + {}_{0}^{0}\gamma$$

The energy of the gamma ray (*hv*) is equal to the energy difference between the ground and excited nuclear states. This is like the emission of lower energy electromagnetic radiation that occurs as an atom in its excited electronic state returns to its ground state. Studies of gamma ray energies strongly suggest that nuclear energy levels are quantized just as are electronic energy levels. This adds further support for a shell model for the nucleus. The penetrating abilities of the particles and rays are proportional to their energies.

Beta particles and positrons are about 100 times more penetrating than the heavier and slower-moving alpha particles. They can be stopped by 1/8-inch-thick (0.3 cm) aluminum plate. They can burn skin severely but cannot reach internal organs. Alpha particles have low penetrating ability and cannot damage or penetrate skin. They can damage sensitive internal tissue if inhaled, however. The high-energy gamma rays have great penetrating power and severely damage both skin and internal organs. They travel at the speed of light and can be stopped by thick layers of concrete or lead.

EQUATIONS FOR NUCLEAR REACTIONS

In *chemical* reactions, atoms in molecules and ions are rearranged, but matter is neither created nor destroyed, and atoms are not changed into other atoms. In earlier chapters, we learned to write balanced chemical equations to represent chemical reactions. Such equations must show the same total number of atoms of each kind on both sides of the equation and the same total charge on both sides of the equation. In a nuclear reaction, a different kind of transformation occurs, one in which a proton can change into a neutron, or a neutron can change into a proton, but the total number of nucleons remains the same.

This leads to two requirements for the equation for a nuclear reaction:

1. The sum of the mass numbers (the left superscript in the nuclide symbol) of the reactants must equal the sum of the mass numbers of the products.

2. The sum of the atomic numbers (the left subscript in the nuclide symbol) of the reactants must equal the sum of the atomic numbers of the products; this maintains charge balance. Because such equations are intended to describe only the changes in the nucleus, they do not ordinarily include ionic charges (which are due to changes in the arrangements of electrons).

NEUTRON-RICH NUCLEI (ABOVE THE BAND OF STABILITY)

Nuclei in this region have too high a ratio of neutrons to protons. They undergo decays that *decrease* the ratio. The most common such decay is **beta emission.** A beta particle is an electron ejected *from the nucleus* when a neutron is converted into a proton.

$${}^{1}_{0}n \longrightarrow {}^{1}_{1}p + {}^{0}_{-1}\beta$$

Thus, beta emission results in an increase of one in the number of protons (the atomic number) and a decrease of one in the number of neutrons, with no change in mass number.

Examples of beta particle emission are

$${}^{228}_{88}\mathrm{Ra} \longrightarrow {}^{228}_{89}\mathrm{Ac} + {}^{0}_{-1}\beta \qquad \text{and} \qquad {}^{14}_{6}\mathrm{C} \longrightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\beta$$

The sum of the mass numbers on each side of the first equation is 228, and the sum of the atomic numbers on each side is 88. The corresponding sums for the second equation are 14 and 6, respectively.

NEUTRON-POOR NUCLEI (BELOW THE BAND OF STABILITY)

Two types of decay for nuclei below the band of stability are **positron emission** or **electron capture** (*K* capture). Positron emission is most commonly encountered with artificially radioactive nuclei of the lighter elements. Electron capture occurs most often with heavier elements. A positron has the mass of an electron but a positive charge. Positrons are emitted when protons are converted to neutrons.

$${}^{1}_{1}p \longrightarrow {}^{1}_{0}n + {}^{0}_{+1}\beta$$

Thus, positron emission results in a *decrease* by one in atomic number and an *increase* by one in the number of neutrons, with *no change* in mass number.

$${}^{38}_{19}\mathrm{K} \longrightarrow {}^{38}_{18}\mathrm{Ar} + {}^{0}_{+1}\beta \qquad \text{and} \qquad {}^{15}_{8}\mathrm{O} \longrightarrow {}^{15}_{7}\mathrm{N} + {}^{0}_{+1}\beta$$

The same effect can be accomplished by electron capture (*K* capture), in which an electron from the *K* shell (n= 1) is captured by the nucleus.

$${}^{106}_{47}\mathrm{Ag} + {}^{0}_{-1}e \longrightarrow {}^{106}_{46}\mathrm{Pd} \qquad \text{and} \qquad {}^{37}_{18}\mathrm{Ar} + {}^{0}_{-1}e \longrightarrow {}^{37}_{17}\mathrm{Cl}$$

Some nuclides, such as ${}^{22}_{11}Na$, undergo both electron capture and positron emission.

$${}^{22}_{11}\mathrm{Na} + {}^{0}_{-1}e \longrightarrow {}^{22}_{10}\mathrm{Ne}\ (3\%) \qquad \text{and} \qquad {}^{22}_{11}\mathrm{Na} \longrightarrow {}^{22}_{10}\mathrm{Ne} + {}^{0}_{+1}\beta\ (97\%)$$

Some of the neutron-poor nuclei, especially the heavier ones, *increase* their neutron-to proton ratios by undergoing **alpha emission.** Alpha particles are helium nuclei,${}^{4}_{2}He$ consisting of two protons and two neutrons. Alpha emission also results in an increase of the neutron-to-proton ratio. An example is the alpha emission of lead-204.

$${}^{204}_{82}\mathrm{Pb} \longrightarrow {}^{200}_{80}\mathrm{Hg} + {}^{4}_{2}\alpha$$

NUCLEI WITH ATOMIC NUMBER GREATER THAN 83

All nuclides with atomic number greater than 83 are beyond the band of stability and are radioactive. Many of these decay by alpha emission.

$${}^{226}_{88}\mathrm{Ra} \longrightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\alpha \qquad \text{and} \qquad {}^{210}_{84}\mathrm{Po} \longrightarrow {}^{206}_{82}\mathrm{Pb} + {}^{4}_{2}\alpha$$

The decay of radium-226 was originally reported in 1902 by Rutherford and Soddy. It was the first transmutation of an element ever observed. A few heavy nuclides also decay by beta emission, positron emission, and electron capture. Some isotopes of uranium ($Z = 92$) and elements of higher atomic number, the **transuranium elements,** also decay by spontaneous nuclear fission. In this process a heavy nuclide splits into nuclides of intermediate mass and neutrons.

$$^{252}_{98}\mathrm{Cf} \longrightarrow {}^{142}_{56}\mathrm{Ba} + {}^{106}_{42}\mathrm{Mo} + 4\,{}^{1}_{0}n$$

DETECTION OF RADIATION

Radiation can be detected using the following methods:

1. Photographic Detection
2. Detection by Fluorescence
3. Fluorescent
4. Scintillation counter.
5. Cloud Chambers
6. Gas Ionization Counters

RATES OF DECAY AND HALF-LIFE

Radionuclides have different stabilities and decay at different rates. Some decay nearly completely in a fraction of a second and others only after millions of years. The rates of all radioactive decays are independent of temperature and obey *first-order kinetics.*

$$\text{rate of decay} = k[\mathrm{A}] \qquad \text{and} \qquad \ln\left(\frac{\mathrm{A}_0}{\mathrm{A}}\right) = akt$$

Here A represents the amount of decaying radionuclide of interest remaining after some time *t*, and A_0 is the amount present at the beginning of the observation. The *k* is the rate constant, which is different for each radionuclide.

We can therefore drop it from the calculations and write the integrated rate equation as

$$\ln\left(\frac{\mathrm{A}_0}{\mathrm{A}}\right) = kt$$

Because A_0/A is a ratio, A_0 and A can represent either molar concentrations of a reactant or masses of a reactant. The rate of radioactive disintegrations follows first-order kinetics, so it is proportional to the amount of A present; we can write the integrated rate equation in terms of *N*, the number of disintegrations per unit time:

$$\ln\left(\frac{N_0}{N}\right) = kt$$

In nuclear chemistry, the decay rate is usually expressed in terms of the half-life,$t_{1/2}$ of the process. This is the amount of time required for half of the original sample to react. For a first-order process; $t_{1/2}$ is given by the equation:

$$t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k}$$

EXAMPLE Rate *of Radioactive Decay*

The "cobalt treatments" used in medicine to arrest certain types of cancer rely on the ability of gamma rays to destroy cancerous tissues. Cobalt-60 decays with the emission of beta particles and gamma rays, with a half-life of 5.27 years.

$$^{60}_{27}Co \longrightarrow ^{60}_{28}Ni + ^{0}_{-1}\beta + ^{0}_{0}\gamma$$

How much of a 3.42 μg sample of cobalt-60 remains after 30.0 years?

Plan

We determine the value of the specific rate constant, *k*, from the given half-life. This value is then used in the first-order integrated rate equation to calculate the amount of cobalt-60 remaining after the specified time.

Solution

We first determine the value of the specific rate constant.

$$t_{1/2} = \frac{0.693}{k} \qquad \text{so} \qquad k = \frac{0.693}{t_{1/2}} = \frac{0.693}{5.27\ \text{y}} = 0.131\ \text{y}^{-1}$$

This value can now be used to determine the ratio of A_0 to A after 30.0 years.

$$\ln\left(\frac{A_0}{A}\right) = kt = 0.131\ \text{y}^{-1}(30.0\ \text{y}) = 3.93$$

Taking the inverse ln of both sides, $\frac{A_0}{A} = 51$.

$$A_0 = 3.42\ \mu g, \text{ so}$$

$$A = \frac{A_0}{51} = \frac{3.42\ \mu g}{51} = 0.067\ \mu g\ ^{60}_{27}Co \text{ remains after 30.0 years.}$$

USES OF RADIONUCLIDES

Radionuclides have practical uses because they decay at known rates.

(1) **Radioactive Dating**

The ages of articles of organic origin can be estimated by **radiocarbon dating.** The radioisotope carbon-14 is produced continuously in the upper atmosphere as nitrogen atoms capture cosmic ray neutrons.

$$^{14}_{7}N + ^{1}_{0}n \longrightarrow ^{14}_{6}C + ^{1}_{1}H$$

The carbon-14 atoms react with oxygen molecules to form $^{14}CO_2$. This process continually supplies the atmosphere with radioactive $^{14}CO_2$, which is removed from the atmosphere by photosynthesis. The intensity of cosmic rays is related to the sun's activity. As long as this remains constant, the amount of $^{14}CO_2$in the atmosphere remains constant. $^{14}CO_2$ is incorporated into living organisms just as ordinary $^{12}CO_2$ is, so a certain fraction of all carbon atoms in living substances is carbon-14. This decays with a half-life of 5730 years.

$$^{14}_{6}C \longrightarrow ^{14}_{7}N + ^{0}_{-1}\beta$$

After death, the plant no longer carries out photosynthesis, so it no longer takes up $^{14}CO_2$. Other organisms that consume plants for food stop doing so at death. The emissions from the ^{14}C in dead tissue then decrease with the passage of time. The activity per gram of carbon is a measure of the length of time elapsed since death. Comparison of ages of ancient trees calculated from ^{14}C activity with those determined by counting rings indicates that cosmic ray intensity has varied somewhat throughout history. The calculated ages can be corrected for these variations. The carbon-14

technique is useful only for dating objects less than 50,000 years old. Older objects have too little activity to be dated accurately.
The **potassium–argon** and **uranium–lead methods** are used for dating older objects.
Potassium-40 decays to argon-40 with a half-life of 1.3 billion years.

$$^{40}_{19}K + ^{0}_{-1}e \longrightarrow ^{40}_{18}Ar$$

Because of its long half-life, potassium-40 can be used to date objects up to 1 million years old by determination of the ratio of $^{40}_{19}K$ to $^{40}_{18}Ar$ in the sample. The uranium–lead method is based on the natural uranium-238 decay series, which ends with the production of stable lead-206. This method is used for dating uranium-containing minerals several billion years old because this series has an even longer half-life. All the ^{206}Pb in such minerals is assumed to have come from 238U. Because of the very long half-life of $^{238}_{92}U$, 4.5 billion years, the amounts of intermediate nuclei can be neglected. A meteorite that was 4.6 billion years old fell in Mexico in 1969. Results of 238U/206Pb studies on such materials of extraterrestrial origin suggest that our solar system was formed several billion years ago.

EXAMPLE *Radiocarbon Dating*
A piece of wood taken from a cave dwelling in New Mexico is found to have a carbon-14 activity (per gram of carbon) only 0.636 times that of wood cut today. Estimate the age of the wood. The half-life of carbon-14 is 5730 years.
Solution

First we find the first-order specific rate constant for ^{14}C.

$$t_{1/2} = \frac{0.693}{k} \qquad \text{or} \qquad k = \frac{0.693}{t_{1/2}} = \frac{0.693}{5730\ \text{y}} = 1.21 \times 10^{-4}\ \text{y}^{-1}$$

The present ^{14}C activity, N (disintegrations per unit time), is 0.636 times the original activity, N_0.

$$N = 0.636\ N_0$$

We substitute into the first-order decay equation

$$\ln\left(\frac{N_0}{N}\right) = kt$$

$$\ln\left(\frac{N_0}{0.636\ N_0}\right) = (1.21 \times 10^{-4}\ \text{y}^{-1})t$$

We cancel N_0 and solve for t.

$$\ln\left(\frac{1}{0.636}\right) = (1.21 \times 10^{-4}\ \text{y}^{-1})t$$

$$0.452 = (1.21 \times 10^{-4}\ \text{y}^{-1})t \qquad \text{or} \qquad t = 3.74 \times 10^{3}\text{y (or 3740 y)}$$

EXAMPLE Uranium–*Lead Dating*
A sample of uranium ore is found to contain 4.64 mg of ^{238}U and 1.22 mg of ^{206}Pb. Estimate the age of the ore. The half-life of ^{238}U is 4.51 x 10^9 years.
Plan
The original mass of ^{238}U is equal to the mass of ^{238}U remaining plus the mass of ^{238}U that decayed to produce the present mass of ^{206}Pb. We obtain the specific rate constant, *k*, from the known half-

life. Then we use the ratio of original 238U to remaining ^{238}U to calculate the time elapsed, with the aid of the first-order integrated rate equation.

Solution

First we calculate the amount of ^{238}U that must have decayed to produce 1.22 mg of ^{206}Pb, using the isotopic masses.

$$\underline{?}\ \text{mg}\ ^{238}\text{U} = 1.22\ \text{mg}\ ^{206}\text{Pb} \times \frac{238\ \text{mg}\ ^{238}\text{U}}{206\ \text{mg}\ ^{206}\text{Pb}} = 1.41\ \text{mg}\ ^{238}\text{U}$$

Thus, the sample originally contained 4.64 mg + 1.41 mg = 6.05 mg of ^{238}U.

We next evaluate the specific rate (disintegration) constant, k.

$$t_{1/2} = \frac{0.693}{k} \quad \text{so} \quad k = \frac{0.693}{t_{1/2}} = \frac{0.693}{4.51 \times 10^{9}\ \text{y}} = 1.54 \times 10^{-10}\ \text{y}^{-1}$$

$$\ln\left(\frac{A_0}{A}\right) = kt$$

$$\ln\left(\frac{6.05\ \text{mg}}{4.64\ \text{mg}}\right) = (1.54 \times 10^{-10}\ \text{y}^{-1})t$$

$$\ln 1.30 = (1.54 \times 10^{-10}\ \text{y}^{-1})t$$

$$\frac{0.262}{1.54 \times 10^{-10}\ \text{y}^{-1}} = t \quad \text{or} \quad t = 1.70 \times 10^{9}\ \text{years}$$

2.Medical Uses

Radionuclide	Uses
Cobalt (^{60}Co)	Radiation is used for the treatments of cancerous tumors. Sterilization of medical equipment.
^{24}Na	Salt solutions containing ^{24}Na can be injected into thebloodstream to follow the flow of blood and locate obstructions in the circulatory system.
Thallium-201 and technetium-99	Thallium-201 tends to concentrate in healthy heart tissue, whereas technetium-99 concentrates in abnormal heart tissue. The two can be used together to survey damage from heart disease.
plutonium-238	The energy produced by the decay of plutonium-238 is converted into electrical energy in heart pacemakers-which makes the heart beat strong or regular. The relatively long half-life of the isotope allows the device to be used for 10 years before replacement.

^{18}F	1. Brain imaging 2. Bone scan
^{32}P	Location of ocular, brain and skin tumors
^{43}K	Myocardial scan
^{47}Ca	Study of calcium metabolism
^{51}Cr	1. Determination of red blood cell volume 2. Spleen imaging 3. Placenta localization
^{99m}Tc	Imaging of various organs, bones and placenta location
Iodine-123	Iodine-123 concentrates in the thyroid gland, liver, and certain parts of the brain. This radioisotope is used to monitor goiter and other thyroid problems, as well as liver and brain tumors.
Iodine-125	1. Study of pancreatic function 2. Study of liver function 3. Thyroid imaging
Iodine-131	1. Study of thyroid activity 2. Study of liver function 3. Brain imaging

(3) Agricultural Uses

The pesticide DDT is toxic to humans and animals repeatedly exposed to it. DDT persists in the environment for a long time. It concentrates in fatty tissues. The DDT once used to control the screwworm fly was replaced by a radiological technique. Irradiating the male flies with gamma rays alters their reproductive cells, sterilizing them. When great numbers of sterilized males are released in an infested area, they mate with females, who, of course, produce no offspring. This results in the reduction and eventual disappearance of the population. Labeled fertilizers can also be used to study nutrient uptake by plants and to study the growth of crops. Gamma irradiation of some foods allows them to be stored for longer periods without spoiling. For example, it retards the sprouting of potatoes and onions. In 1999, the FDA approved gamma irradiation of red meat as a way to curb food-borne illnesses. In addition to significantly reducing levels of *Listeria*, *Salmonella*, and other bacteria, such irradiation is currently the only known way to completely eliminate the dangerous strain of *Escherichia coli* bacteria in red meat. Absorption of gamma rays by matter produces no radioactive nuclides, so foods preserved in this way are *not* radioactive.

(4) Industrial Uses

There are many applications of radiochemistry in industry and engineering. When great precision is required in the manufacture of strips or sheets of metal of definite thicknesses, the penetrating powers of various kinds of radioactive emissions are utilized. The thickness of the metal is correlated with the intensity of radiation passing through it. The flow Moderate irradiation with gamma rays from radioactive isotopes has kept the strawberries at the right fresh for 15 days, while those at the left are moldy. Such irradiation kills mold spores but does no damage to the food. The food does *not* become radioactive. The flow of a liquid or gas through a pipeline can be monitored by injecting a sample containing a radioactive substance. Leaks in pipelines can also be detected in this way. In addition to the ^{238}Pu-based heart pacemaker already mentioned, lightweight, portable power packs that use radioactive isotopes as fuel have been developed for other uses. Polonium-210, californium-242, and californium-244 have been used in such generators to power instruments for space vehicles and in Polar Regions. These generators can operate for years with only a small loss of power.

(5) Research Applications

The pathways of chemical reactions can be investigated using radioactive tracers. When radioactive 35S2- ions are added to a saturated solution of cobalt sulfide in equilibrium with solid cobalt sulfide, the solid becomes radioactive. This shows that sulfide ion exchange occurs between solid and solution in the solubility equilibrium.

$$CoS(s) \rightleftharpoons Co^{2+}(aq) + S^{2-}(aq) \qquad K_{sp} = 8.7 \times 10^{-23}$$

Photosynthesis is the process by which the carbon atoms in CO2 are incorporated into glucose, $C_6H_{12}O_6$, in green plants.

$$6CO_2 + 6H_2O \xrightarrow[\text{chlorophyll}]{\text{sunlight}} C_6H_{12}O_6 + 6O_2$$

The process is more complex than the net equation implies; it actually occurs in many steps and produces a number of intermediate products. By using labeled $^{14}CO_2$, we can identify the intermediate molecules. They contain the radioactive 14C atoms.

(6) Sterilization Germs are killed when an object is irradiated with γ-radiation, leaving it perfectly sterile with no trace of radioactivity. This is particularly very useful for sterilizing surgical equipment which can be irradiated after it is sealed so that there is no risk of further contamination.

ARTIFICIAL TRANSMUTATIONS OF ELEMENTS

Artificial transmutation is a process whereby a nuclear reaction is artificially induced by bombardment of a nucleus with subatomic particles or small nuclei. The first artificially induced nuclear reaction was carried out by Rutherford in 1915. He bombarded nitrogen- 14 with alpha particles to produce an isotope of oxygen and a proton.

$$^{14}_{7}N + ^{4}_{2}\alpha \longrightarrow ^{1}_{1}H + ^{17}_{8}O$$

Such reactions are often indicated in abbreviated form, with the bombarding particle and emitted subsidiary particles shown parenthetically between the parent and daughter nuclei.

$$^{14}_{7}N\ (^{4}_{2}\alpha,\ ^{1}_{1}p)\ ^{17}_{8}O$$

Several thousand different artificially induced reactions have been carried out with bombarding particles such as neutrons, protons, deuterons ($^{2}_{1}H$), alpha particles, and other small nuclei.

Bombardment with Positive Ions

A problem arises with the use of positively charged nuclei as projectiles. For a nuclear reaction to occur, the bombarding nuclei must actually collide with the target nuclei, which are also positively charged. Collisions cannot occur unless the projectiles have sufficient kinetic energy to overcome coulombic repulsion. The required kinetic energies increase with increasing atomic numbers of the target and of the bombarding particle. Particle accelerators called **cyclotrons** (atom smashers) and **linear accelerators** have overcome the problem of repulsion. Particle accelerators were used between 1937 and 1941 to synthesize three of the four “missing” elements: numbers 43 (technetium), 85 (astatine), and 87 (francium).

$$^{96}_{42}Mo + ^{2}_{1}H \longrightarrow ^{97}_{43}Tc + ^{1}_{0}n$$
$$^{209}_{83}Bi + ^{4}_{2}\alpha \longrightarrow ^{210}_{85}At + 3\ ^{1}_{0}n$$
$$^{230}_{90}Th + ^{1}_{1}p \longrightarrow ^{223}_{87}Fr + 2\ ^{4}_{2}\alpha$$

Many hitherto unknown, unstable, artificial isotopes of known elements have also been synthesized so that their nuclear structures and behavior could be studied.

Neutron Bombardment

Neutrons bear no charge, so they are not repelled by nuclei as positively charged projectiles are. They do not need to be accelerated to produce bombardment reactions. Neutron beams can be generated in several ways. A frequently used method involves bombardment of beryllium-9 with alpha particles.

$$^{9}_{4}Be + ^{4}_{2}\alpha \longrightarrow ^{12}_{6}C + ^{1}_{0}n$$

Nuclear reactors are also used as neutron sources. Neutrons ejected in nuclear reactions usually possess high kinetic energies and are called **fast neutrons.** When they are used as projectiles they cause reactions, such as (n, p) or (n, α) reactions, in which subsidiary particles are ejected. The fourth "missing" element, number 61 (promethium), was synthesized by fast neutron bombardment of neodymium-142.

$$^{142}_{60}Nd + ^{1}_{0}n \longrightarrow ^{143}_{61}Pm + ^{0}_{-1}\beta$$

Slow neutrons ("thermal" neutrons) are produced when fast neutrons collide with **moderators** such as hydrogen, deuterium, oxygen, or the carbon atoms in paraffin. These neutrons are more likely to be captured by target nuclei. Bombardments with slow neutrons can cause neutron-capture (n, Y) reactions.

$$^{200}_{80}Hg + ^{1}_{0}n \longrightarrow ^{201}_{80}Hg + ^{0}_{0}\gamma$$

Slow neutron bombardment also produces the 3H isotope (tritium).

$$^{6}_{3}Li + ^{1}_{0}n \longrightarrow ^{3}_{1}H + ^{4}_{2}\alpha \qquad (n,\ \alpha)\ \text{reaction}$$

E. M. McMillan (1907–1991) discovered the first transuranium element, neptunium, in 1940 by bombarding uranium-238 with slow neutrons.

$$^{238}_{92}U + ^{1}_{0}n \longrightarrow ^{239}_{92}U + ^{0}_{0}\gamma$$

$$^{239}_{92}U \longrightarrow ^{239}_{93}Np + ^{0}_{-1}\beta$$

Several additional elements have been prepared by neutron bombardment or by bombardment of the nuclei so produced with positively charged particles. Some examples are:

$$\left.\begin{aligned} ^{238}_{92}U + ^{1}_{0}n &\longrightarrow ^{239}_{92}U + ^{0}_{0}\gamma \\ ^{239}_{92}U &\longrightarrow ^{239}_{93}Np + ^{0}_{-1}\beta \\ ^{239}_{93}Np &\longrightarrow ^{239}_{94}Pu + ^{0}_{-1}\beta \end{aligned}\right\}\quad \text{plutonium}$$
$$^{239}_{94}Pu + ^{4}_{2}\alpha \longrightarrow ^{242}_{96}Cm + ^{1}_{0}n \qquad \text{curium}$$
$$^{246}_{96}Cm + ^{12}_{6}C \longrightarrow ^{254}_{102}No + 4\ ^{1}_{0}n \qquad \text{nobelium}$$
$$^{252}_{98}Cf + ^{10}_{5}B \longrightarrow ^{257}_{103}Lr + 5\ ^{1}_{0}n \qquad \text{lawrencium}$$

NUCLEAR FISSION

Nuclear fission is the process in which a heavy nucleus splits into nuclei of intermediate masses and one or more neutrons are emitted. Isotopes of some elements with atomic numbers above 80 are capable of undergoing fission in which they split into nuclei of intermediate masses and emit one or more neutrons. Some fission reactions are spontaneous; others require that the activation energy be supplied by bombardment. A given nucleus can split in many different ways, liberating enormous amounts of energy. Some of the possible fissions that can result from bombardment of fissionable uranium-235 with fast neutrons follow. The uranium-236 is a short-lived intermediate.

$$
{}^{235}_{92}\mathrm{U} + {}^{1}_{0}n \longrightarrow {}^{236}_{92}\mathrm{U} \longrightarrow
\begin{cases}
{}^{160}_{62}\mathrm{Sm} + {}^{72}_{30}\mathrm{Zn} + 4\,{}^{1}_{0}n + \text{energy} \\
{}^{146}_{57}\mathrm{La} + {}^{87}_{35}\mathrm{Br} + 3\,{}^{1}_{0}n + \text{energy} \\
{}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 3\,{}^{1}_{0}n + \text{energy} \\
{}^{144}_{55}\mathrm{Cs} + {}^{90}_{37}\mathrm{Rb} + 2\,{}^{1}_{0}n + \text{energy} \\
{}^{144}_{54}\mathrm{Xe} + {}^{90}_{38}\mathrm{Sr} + 2\,{}^{1}_{0}n + \text{energy}
\end{cases}
$$

Experiments with particle accelerators have shown that every element with an atomic number of 80 or more has one or more isotopes capable of undergoing fission provided they are bombarded at the right energy. Nuclei with atomic numbers between 89 and 98 fission spontaneously with long half-lives of 104 to 1017 years. Nuclei with atomic numbers of 98 or more fission spontaneously with shorter half-lives of a few milliseconds to 60.5 days.

Nuclear Power: Hazards and Benefits

Controlled fission reactions in nuclear reactors are of great use and even greater potential. The fuel elements of a nuclear reactor have neither the composition nor the extremely compact arrangement of the critical mass of a bomb. Thus, no possibility of nuclear explosion exists. However, various dangers are associated with nuclear energy generation. The possibility of "meltdown" has been discussed with respect to cooling systems in light water reactors. Proper shielding precautions must be taken to ensure that the radionuclides produced are always contained within vessels from which neither they nor their radiations can escape. Long-lived radionuclides from spent fuel must be stored underground in heavy, shock-resistant containers until they have decayed to the point that they are no longer biologically harmful. As examples, strontium-90 ($t_{1/2}$ = 28 years) and plutonium-239 ($t_{1/2}$ = 24,000 years) must be stored for 280 years and 240,000 years, respectively, before they lose 99.9% of their activities. Critics of nuclear energy contend that the containers could corrode over such long periods, or burst as a result of earth tremors, and that transportation and reprocessing accidents could cause environmental contamination with radionuclides. They claim that river water used for cooling is returned to the rivers with too much heat (thermal pollution), thus disrupting marine life. (It should be noted, though, that fossil fuel electric power plants cause the same thermal pollution for the same amount of electricity generated.) The potential for theft also exists. Plutonium-239, a fissionable material, could be stolen from reprocessing plants and used to construct atomic weapons. Proponents of the development of nuclear energy argue that the advantages far outweigh the risks. Nuclear energy plants do not pollute the air with oxides of sulfur, nitrogen, carbon, and particulate matter, as fossil fuel electric power plants do. The big advantage of nuclear fuels is the enormous amount of energy liberated per unit mass of fuel. At present, nuclear reactors provide about 22% of the electrical energy consumed in the United States. In some parts of Europe, where natural resources of fossil fuels are scarcer, the utilization of nuclear energy is higher. For instance, in France and Belgium, more than 80% of electrical energy is produced from nuclear reactors. With rapidly declining fossil fuel reserves, it appears likely that nuclear energy and solar energy will become increasingly important. Intensifying public concerns about nuclear power, however, may mean that further growth in energy production using nuclear power in the United States must await technological developments to overcome the remaining hazards.

NUCLEAR FUSION

Fusion, the joining of light nuclei to form heavier nuclei, is favourable for the very light atoms. In both fission and fusion, the energy liberated is equivalent to the loss of mass that accompanies the reactions. Much greater amounts of energy per unit mass of *reacting atoms* are produced in fusion than in fission. Spectroscopic evidence indicates that the sun is a tremendous fusion reactor consisting of 73 % H, 26 % He, and 1 % other elements. Its major fusion reaction is thought to involve the combination of a deuteron, ${}^{2}_{1}H$, and a triton ${}^{3}_{1}H$, at tremendously high temperatures to form a helium nucleus and a neutron.

$${}^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \longrightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}n + \text{energy}$$

Thus, solar energy is actually a form of fusion energy. Fusion reactions are accompanied by even greater energy production per unit mass of reacting atoms than are fission reactions. They can be initiated only by extremely high temperatures, however. The fusion of ${}^{2}_{1}H$, and ${}^{3}_{1}H$occurs at the lowest temperature of any fusion reaction known, but even this is 40,000,000 K! Such temperatures exist in the sun and other stars, but they are nearly impossible to achieve and contain on earth. **Thermonuclear** bombs (called fusion bombs or hydrogen bombs) of incredible energy have been detonated in tests but, thankfully, never in war. In them the necessary activation energy is supplied by the explosion of a fission bomb.

It is hoped that fusion reactions can be harnessed to generate energy for domestic power. Because of the tremendously high temperatures required, no currently known structural material can confine these reactions. At such high temperatures all molecules dissociate and most atoms ionize, resulting in the formation of a new state of matter called plasma**.** Very high temperature plasma is so hot that it melts and decomposes anything it touches, including structural components of a reactor. The technological innovation required to build a workable fusion reactor probably represents the greatest challenge ever faced by the scientific and engineering community.

Recent attempts at the containment of lower temperature plasmas by external magnetic fields have been successful, and they encourage our hopes. Fusion as a practical energy source, however, lies far in the future at best. The biggest advantages of its use would be that (1) the deuterium fuel can be found in virtually inexhaustible supply in the oceans; and (2) fusion reactions would produce only radionuclides of very short half-life, primarily tritium ($t_{1/2}$ =12.3 years), so there would be no long-term waste disposal problem. If controlled fusion could be brought about, it could liberate us from dependence on uranium and fossil fuels. Our sun supplies energy to the earth from a distance of 93,000,000 miles. Like other stars, it is a giant nuclear fusion reactor. Much of its energy comes from the fusion of deuterium, ${}^{2}_{1}H$, producing helium, ${}^{4}_{2}He$. The explosion of a thermonuclear (hydrogen) bomb releases tremendous amounts of energy. If we could learn how to control this process, we would have nearly limitless amounts of energy. Nuclear fusion provides the energy of our sun and other stars.

SOLVED EXAMPLES

1. Distinguish between nuclear fission and nuclear fusion.

NUCLEAR FISSION	NUCLEAR FUSION
(1) Favourable for heavy atoms	- Favourable for very light atoms
(2) Less amount of energy per unit mass of reacting atoms are produced	- Much greater amount of energy per unit mass of reacting atoms are produced
(3) Reactions are accompanied by lesser energy production per unit mass of reacting atoms	- Reactions are accompanied by greater energy production per unit mass of reacting atoms
(4) Can be initiated by lesser amounts of temperatures	- Can only be initiated by extremely high temperatures

2. Potassium, Z =19, has a series of naturally occurring isotopes: ^{39}K, ^{40}K, ^{41}K. Identify the isotope(s) of potassium that is/are most likely to be stable and which tends to decay.

isotope(s) of potassium that is/are most likely to be stable	**isotope(s) of potassium that is/are most likely to be unstable or tends to decay**
^{39}K Neutron = 20 are stable	^{40}K, ^{41}K Neutron = 21 and 22 respectively are not stable

3. Calculate the following for $^{49}_{22}Ti$ (actual mass = 48.94787 amu). (a) mass deficiency in amu/atom; (b) mass deficiency in g/mol; (c) binding energy in J/mol; (d) binding energy in kJ/mol; (e) binding energy in MeV/nucleon.

Particle	**Mass**
Electron (e^-)	0.00054858 amu
Proton (p)	1.0073 amu
Neutron (n)	1.0087 amu

Electron (e^-): 22 X 0.00054858 amu = 0.01207

Proton ($p$or p^+): 22 X 1.0073 amu = 22.1606

Neutron (n or n^0): 27 X 1.0087 amu = $\underline{27.2349}$

49.4076

(a) Δm = *calculated mass* – *actual mass* = 49.4076 – 48.94787 = 0.45973 ≈ 0.4597 amu/atom

(b) Δm in g/mol: $\frac{0.4597 \quad amu}{atom} \; X \; \frac{1\,g}{6.022\,X \;\; 10^{23} amu} \; X \; \frac{6.022\,X\;\;10^{23}\,atom}{1\,mol\;62_{Ni}\;\;atom}$ = 0.4597 g/mol ^{49}Ti

atom

BE = (Δm)C^2

Δm *in kg/mol* = 0.4597 x 10^{-3} kg/mol

(c) BE = 0.4597 X10^{-3} X (3 X $10^{8})^{2}$ =4.1373 X 10^{13} J/mol

(d) = 4.1373X 10^{13} X 10^{-3} = 4.1373 x 10^{10}kJ/mol

(e) $\frac{4.1373 \text{ x } 10^{10} \times 1000}{\mathbf{1.6 \times 10^{-13} \times 6.022 \times 10^{23} \times 49}} = \mathbf{8.76\ MeV}$

4. With the help of a Table, compare the behaviours of α, β and γ radiation (a) in an electrostatic field, and (b) with respect to ability to penetrate various shielding materials, such as a piece of paper and concrete. (c) **What is the composition of each type of radiation?**

Radiation	(i) In an electrostatic field	(ii) Ability to penetrate various shielding materials	(iii)The composition
α	deflected slightly towards the negative plate in an electrostatic	Alpha particles have low penetrating ability and cannot damage or penetrate skin	($^{4}_{2}\alpha$ or $^{4}_{2}$He) helium nucleus
β	highly deflected towards the positive plate in an electrostatic field	Beta particles are about 100 times more penetrating than the heavier and slower-moving alpha particles. They can be stopped by aluminum plate	$^{0}_{-1}$e, $^{0}_{-1}\beta$, *electron*
γ	not affected by an electrostatic field	have greatest penetrating power and severely damage both skin and internal organs. They travel at the speed of light and can be stopped by thick layers of concrete or lead.	($^{0}_{0}\gamma$), high-energy electromagnetic radiation (such as X-ray).

5. A nuclide of element rutherfordium, $^{257}_{104}Rf$, is formed by the nuclear reaction of californium-249 and carbon-12, with the emission of four neutrons. This new nuclide

rapidly decays by emitting an α-particle and forming a nuclide X. Write the equation for each of these nuclear reactions.

$^{249}_{98}Cf + ^{12}_{6}C \rightarrow ^{257}_{104}Rf + 4^{1}_{0}n$**.**

$^{257}_{104}Rf \rightarrow ^{253}_{102}X + ^{4}_{2}\alpha$ or $^{4}_{2}He$**.**

X = Nobelium.

6. The $^{14}_{6}C$ activity of an artifact from a burial site was 8.6/min mg C. The half-life of $^{14}_{6}C$ is 5730 years, and the current $^{14}_{6}C$ activity is 15.3/min. g C (that is, 15.3 disintegrations per minute per gram of carbon). How old is the artifact?

$$t_{1/2} = \frac{ln2}{k} = \frac{0.693}{k}$$

so $k = \frac{0.693}{t_{1/2}} = \frac{0.693}{5730\text{y}}$

$= 1.21 \times 10^{-4}\ y^{-1}$

$$ln\frac{N_0}{N} = kt$$

$$ln\frac{15.3}{8.6} = 1.21 \times 10^{-4}\ y^{-1}\ t$$

$ln1.7791 = 1.21 \times 10^{-4}\ y^{-1}\ t$

$0.5761 = 1.21 \times 10^{-4}\ y^{-1}\ t$

$t = 0.5761/(1.21 \times 10^{-4}) = 4761.2$

$t = 4761\ years$(OR yr)

7. Fill in the missing symbols in the following nuclear bombardment reactions.

ANSWERS ARE IN SQUARE BRACKETS AND BOLDED.

$$^{23}_{11}Na + \left[\boldsymbol{^{1}_{1}p\ OR\ ^{1}_{1}H}\right] \rightarrow ^{23}_{12}Mg + ^{1}_{0}n$$

$$^{59}_{27}Co + ^{1}_{0}n \rightarrow ^{56}_{25}Mn + \left[\boldsymbol{^{4}_{2}\alpha\ OR\ ^{4}_{2}He}\right]$$

$$^{96}_{42}Mo + ^{4}_{2}He \rightarrow ^{100}_{43}Tc + \left[\boldsymbol{^{0}_{+1}\beta}\right]$$

$$^{209}_{83}Bi + \left[^{2}_{1}\boldsymbol{H}\ \boldsymbol{OR}\ ^{2}_{1}\boldsymbol{D}\right] \rightarrow ^{210}_{84}Po + ^{1}_{0}n$$

8. Write the nuclear equation for each of the following bombardment processes:

   (i) $^{4}_{7}N$ (α, p) $^{17}_{8}O$; (ii) $^{23}_{11}Na$ (n, α) X. Identify X;

i. $^{14}_{7}N + ^{4}_{2}\alpha \rightarrow ^{17}_{8}O + ^{1}_{1}p$

ii. $^{23}_{11}Na + ^{1}_{0}n \rightarrow ^{20}_{9}X + ^{4}_{2}\alpha$

X = Fluorine (F)

9. Write the nuclear equation for each of the following processes:

i. $^{63}_{28}Ni$ undergoing β - emission;

ii. Two deuterium ions undergoing fusion to give $^{3}_{2}He$ and a neutron;

iii. A nuclide being bombarded by a neutron to form $^{7}_{3}Li$ and an α –particle.

(i) $^{63}_{28}Ni \rightarrow ^{63}_{29}Cu + ^{0}_{-1}e$

(ii) $^{2}_{1}H^{+} + ^{2}_{1}H^{+} \rightarrow ^{3}_{2}He + ^{1}_{0}n$

(iii) $^{10}_{5}X + ^{1}_{0}n \rightarrow ^{7}_{3}Li + ^{4}_{2}\alpha$

EXERCISES

1. Define and compare nuclear fission and nuclear fusion.

2. Differentiate between natural and induced radioactivity. Use the periodic table to identify the locations of those elements that are the result of induced radioactivity.

3. How do nuclear reactions differ from ordinary chemical reactions?

4. What is the equation that relates the equivalence of matter and energy? What does each term in this equation represent?

5. What is mass deficiency? What is binding energy? How are the two related?

6. What are nucleons? What is the relationship between the number of protons and the atomic mass?

7. What are nuclear bombardment reactions? Explain the shorthand notation used to describe bombardment.